

SPCI107

POSTGRADUATE COURSE

M.Sc. - Cyber Forensics & Information Security

FIRST YEAR

SECOND SEMESTER

CORE PAPER - VI

BASICS OF CYBER FORENSICS



**INSTITUTE OF DISTANCE EDUCATION
UNIVERSITY OF MADRAS**

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With best wishes from mind and heart,

DIRECTOR i/c

M.Sc. Cyber Forensics & Information Security
FIRST YEAR - SECOND SEMESTER

CORE PAPER - VI
BASICS OF
CYBER FORENSICS

COURSE COMPILED, WRITTEN & EDITED BY
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M.Sc. DEGREE COURSE
FIRST YEAR
SECOND SEMESTER
Core Paper - VI
BASICS OF CYBER FORENSICS
SYLLABUS

Unit 1: Digital Investigation - Digital Evidence and Computer Crime - History and Terminology of Computer Crime Investigation - Technology and Law - The Investigative Process -Investigative Reconstruction - Modus Operandi, Motive and Technology -Digital Evidence in the Courtroom.

Unit 2 :Understanding information - Methods of storing data: number systems, character codes, record structures, file formats and file signatures - Word processing and graphic file formats - Structure and Analysis of Optical Media Disk Formats - Recognition of file formats and internal buffers - Extraction of forensic artifacts– understanding the dimensions of other latest storage devices – SSD Devices.

Unit 3: Computer Basics for Digital Investigators - Computer Forensic Fundamentals - Applying Forensic Science to computers - Computer Forensic Services - Benefits of Professional Forensic Methodology -Steps taken by computer forensic specialists.

Unit 4:Standards, Guidelines and Best Practices- Handling the Digital Crime Scene - Digital Evidence Examination Guidelines –ACPO – IOCE – SWGDE -DFRWS – IACIS – HTCIA - ISO 27037

Unit 5: Types of Computer Forensics Tools and Technology -Tools and Types of Military Computer Forensics Technology -Tools and Types of Law Enforcement Computer Forensic Technology -Tools and Types of Business Computer Forensic Technology

M.Sc. DEGREE COURSE
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LESSON - 1

DIGITAL INVESTIGATION

Learning Objectives

After reading this lesson you will be able to understand the following:

- Digital Evidence and Computer Crime
- History and Terminology of Computer Crime Investigation
- Technology and Law
- The Investigative Process
- Investigative Reconstruction
- Modus Operandi
- Motive and Technology
- Digital Evidence in the Courtroom

Structure

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1.1.13. Digital Evidence**1.2. History and Terminology of Computer Crime Investigation****1.2.1. Emergence of Cyber Forensic Investigating Agencies****1.2.2. Development of Cyber Forensics****1.2.3. Digital Data****1.2.3.1. Active Data****1.2.3.2. Latent Data****1.2.3.3. Archival data****1.2.4. Cardinal Rules in Cyber Forensics****1.2.5. Cyber Forensic Lab****1.2.6. Operational Perspective****1.2.7. Technological Perspective****1.2.8. Scientific Perspective****1.2.9. An Artistic Perspective****1.2.10. Budget****1.2.11. Core Missions and Services****1.2.12. Increasing awareness of Digital Evidence****1.2.13. Challenging aspects of Digital Evidence****1.2.14. Following the cyber trail****1.2.15. Challenging aspects of the cyber trail****1.1. Digital Evidence and Computer Crime**

Crime is as old as mankind. Crime is as old as humankind. Most crimes leave behind traces of “silent evidence”. The traces of materials, known as physical evidence, are found at the scene of crime and acts as potent clues that become the most eloquent witness subsequently. Evidence that is conspicuously found in the crime scene be it dust, pollens, bloodstain, bullet, glass, paint, hair, ash and the like serves as silent speakers of truth. These materials as such can become valuable pieces of evidence in the hands of forensic scientists who seek truth and nothing but truth.

The term forensics roots from the Latin word, “forensics” which generally means forum for discussion. In modern days Forensic investigation uses various facets of Science coupled with Technology in the process of establishing facts in criminal or civil courts of law. Forensic science is a multidisciplinary subject, drawing principally from chemistry, biology, physics, geology, psychology, social science and the like. Forensic Science is the science that deals with analysis of physical evidence collected from all possible sources in which criminal and victim are associated with. The latest entrant in this field is cyber forensics. It is digital and it includes advanced computer science technology and its main goal is submission of the proven claims of scientific methods and strategies to recover any significant digital traces. Cyber crimes are staggering. As computers become more advanced, so do criminal activities. Therefore, computer forensics niche is in constant progression along with the technological advancements of computers.

Although technology in general and computers specifically, since their introduction and dissemination into mainstream society, have benefited society, there is also a sinister, dark side to this technology when it is abused. In recent years, society has seen the rise in abuse of

- Fraud audits
- Identity theft
- Hacking
- Embezzlement
- Instances of homicide
- Drug and embezzlement record keeping
- Child pornography
- Civil litigation in cases of divorce, age or race discrimination, sexual harassment, wrongful dismissal, termination.
- Compromise customer privacy data stored electronically
- Peer-to-peer file sharing
- Leak or unauthorized disclosure of internal and confidential information
- Theft of trade secrets, intellectual property
- Unlawful access to company computers
- Use of company computers or technology for personal gain (running auction sites, shopping, E-bay, fantasy sports leagues, etc.)

- Violation of company acceptable use policies (downloading music and movies, accessing adult Web site, etc.)
- Launching denial of service attacks against a competitor

1.1.1. Crime and the Internet

Internet, has gone beyond imagination since its inception in 1989, the most significant technological advancement conceived by TIM Berner Lee “The world wide web” has now caused the world into a global village. Digital communication and interaction has grown to nearly two thirds of the world population. This has contributed to a change in society, significant impact on business and critical infrastructure, and fundamental aspects of modern society. Cyber space has transformed the global economy in a phenomenal way. The scale of cyber space usage and its magnitude and increased acquiescence of internet demonstrates a global obsession which leads internet to an intrinsic and compelling part of society’s day to day activities. Increased number of communications via email, chat, social networking, huge volume of online shopping, internet banking, gaming, purchase of online gifts, travel tickets, online purchase of property and so on only reflects that the society has a paradigm shift from citizens to netizens. A recent survey was conducted to find out the dependence of youth on the internet. The outcome of the survey was that 75% of the youth could not survive without the internet and they belong to the age group between 16 to 24 years. Despite its useful advancement of technology, internet provides ample opportunity for unscrupulous individuals performing undesirable activities caused by the ease of access, open nature, and increased anonymity facilitated in the virtual world. E-governance is becoming more and more pervasive and this has begun to touch the citizens life in many ways. Driving license, Vehicle registration certificate, Filing of income tax returns, Land records, Banking transactions, Credit Cards, Railway reservation, Passport issuance, Immigration control and many more have become the way of life and many of these could be categorized as mission critical rolled out on all these projects. The nature of risks and vulnerabilities are becoming more and more transformational today and there is a need to create new knowledge and understanding new risks, interactions and probabilities and costs. Citizens are gradually becoming dependent on computers, networks and the like and transforming into Netizens. Any deviant behavior, with malicious intent leads to crime in the cyber space. Studying cyber crime from e-governance perspective is the need of the hour. Cyber spans not only State but National boundaries as well. Cyber crime is going to be a major problem to any country’s law enforcement. These crimes have no geographic boundaries that are clearly definable, transcends the border

of States and Nations' physical area that is accessible over computer and Telecommunication networks across the globe with a time frame of milliseconds. It is difficult to implement regulations because laws are different in different countries.

The growth of Internet and its proliferation worldwide now increased the need for conducting cyber crime investigations through efficient and meticulous Cyber Forensics. Extensive knowledge and skill in computer technology is required for Police, Judiciary, Lawyers, Forensic experts, Private Investigators and Network Administrators to counter this fraud/menace through effective training, creation of specialized units, necessary legislations, and international co-operation. Those are some of the steps that require immediate attention of all Governments in the world. Computers can be used to commit crimes, and crimes can be recorded on computers, including violation of company policies, records of embezzlement, email harassment, leaks of proprietary information, murder and even terrorism.

The global internet usage has risen over 2 billion people over a span of ten years. Unclassified and classified networks and foreign intelligence organization are working to acquire the capacity to disrupt elements of critical information infrastructure. The tools and techniques used by cyber criminals are increasing in sophistication at an incredible rate. Beyond government activities, cyber criminals can control botnets with millions of infected hosts, whether the goal is monetary, access to intellectual property, or disruption of critical infrastructure systems.

The pursuit of cyber crime detection has two main objectives. One to prevent the occurrence cyber crime in vulnerable institutions requiring security from loss, pilferage and mishandling by accidental or intentional manipulations and the other detection and documenting cyber crimes through a disciplined methodology. With advancement of technology, computer is both the instrument of crime, as well as the location of evidence.

Forensic Science in its broadest definition is the application of science to Law, especially to those of criminal and civil laws that are enforced by Police agencies in Criminal justice system. Cyber Forensics presents the investigation and analysis techniques on computers to obtain potential legal evidence. It involves the preservation, identification, extraction and documentation of computer evidence stored as data or magnetically encoded information. The computer operating system invariably leaves behind the computer evidences transparently without the knowledge of computer user which may be hidden from view. Special Forensic software tools and techniques are required in order to recognize and retrieve such evidences. Computer Forensics involves obtaining and analyzing such digital information for use in civil/

criminal or administrative cases. Digital evidence was not considered as tangible evidence in courts until recently but now they are gaining importance.

1.1.2. Cyber Forensics /Digital Forensics

Cyber Forensics is simply application of computer investigation and analysis techniques in the interest of determining potential legal evidence. Forensic computing is the process of identifying preserving, analyzing and presenting the digital evidence in a manner that is legally acceptable. It is the study of evidence from attacks on computer system in order to learn what has occurred, how to prevent it from recurring and the extent of the damage

Cyber Forensics is one of the emerging professions of 21st century. It can be thought of as an investigation of computer based evidence of criminal activity, using scientifically developed methods that attempts to discover and reconstruct event sequences from such activity. The fascinating part of the science is that the computer operating system invariably leaves behind the computer evidences transparently without the knowledge of computer operator. The information may actually be hidden from view. Any enterprise that uses computer networks should have concern for both security and forensic capabilities (Yasinsac and Manzano, 2001). They suggest that forensic tools should be developed to scan continually computers and networks within an enterprise for illegal activities.

When misuse is detected these tools should record sequence of events and store relevant data for further investigation. Special Forensic software tools and techniques are required in order to recognize and retrieve such evidences. Cyber Forensics involves obtaining and analyzing such digital information for use in civil/criminal or administrative cases. Digital evidence was not considered as tangible evidence in courts until recently but now they are gaining importance.

1.1.3. Definition of Cyber Forensics

- In a Computer Crime, the process of identifying, preserving, analyzing and presenting digital evidence to prove the crime in a way that is legally acceptable in the court of law.
- According to A.Lunn, cyber forensics is employment of a set of predefined procedures to examine a Computer system using software or hardware tools to extract and preserve the evidence of criminal activity.

1.1.4. A Military Cyber Forensics Definition

The exploration and application of scientifically proven methods to gather, process, interpret, and utilize digital evidence in order to:

- Provide a conclusive description of all cyber-attack activities for the purpose of complete post-attack enterprise and critical infrastructure information restoration
- Correlate, interpret, and predict adversarial actions and their impact on planned military operations
- Make digital data suitable and persuasive for introduction into a criminal investigative process

While the last goal of the definition directly addresses law enforcement needs in cyber forensics, it is also required for the military commander who will be making decisions on how to engage aggressors in cyberspace, who may be civilian enemy combatants rather than state-sponsored attackers. The commander will need to know for sure who is attacking prior to taking any action that would be viewed as a violation of a Treaty or other international agreement. Forensic results that meet criminal investigative criteria can help justify a commander's actions in retaliation.

1.1.5. Terminologies

- Disk Forensics: deals with extracting data/information from storage media by searching active, deleted files and also from unallocated and slack space.
- Network Forensics: It is a sub branch of digital forensics relating to monitoring and analysis of computer network traffic for the purpose of information gathering, legal evidence or intrusion detection. Unlike other areas of digital forensics, network investigation deal with volatile and dynamic information. It is also called Pro-active forensics.
- Wireless Forensics: It is a sub part of network forensics. The main goal of wireless forensics is to provide the tools required to collect and analyze the data from wireless network traffic. The data collected can correspond to plain data or with the broad usage of Voice over Internet Protocol(VoIP) technologies especially over wireless technology.

- Database Forensics: is a branch of digital forensics relating to study and examine databases and their related metadata. A forensic examination of a database may relate to the timestamps that apply to the row (update time) in a relational table being inspected and tested for validity in order to verify the actions of a database user.
- Malware Forensics: deals with analysis and identification of a malicious code, to study their payload, viruses, worms, Trojans, Keyloggers etc.
- Mobile Phone Forensics deals with examination and analysis of mobile devices, to retrieve phone and SIM contacts, call logs(Dialled, Missed & Received), incoming and outgoing SMS/MMS, Audio, videos, paired device history and in some smart phones, geolocation and calendar information etc.
- GPS Forensics is also called SatNav Forensics, is a relatively new discipline with the fast paced world of Mobile Device Forensics. It is used for examining and analysing GPS devices to retrieve information such as TrackLogs, TrackPoints, WayPoints, Routes, Photos, audio etc.
- Email Forensics: Deals with recovery and analysis of emails including deleted emails, calendars and contacts.
- Memory Forensics deals with collecting data from system memory (system registers, cache, RAM) in raw form and then carving the data from Raw dump.
- E-Discovery: E-Discovery is the process of evaluating solutions for organization. A defensible e-Discovery process is repeatable, systemized and meets legal requirements for proper handling and admissibility of computer evidence. Email archiving can be a useful complement e-Discovery. A defensible e-discovery process is repeatable, systematized and meets legal requirements for proper handling and admissibility of computer evidence. An ideal e-discovery process identifies, collects, preserves, processes, reviews and produces relevant electronically stored information. Relevant information may be found in unmanaged, unstructured, semi-structured or structured data sources dispersed across networks on desktops, laptops, servers, share drives, removable storage media and other devices. As the name implies, email archiving is limited to the contents of email system since they work only with the set of emails and do not extend to data on the network. An effective repeatable and defensible eDiscovery response plan requires an

organization to proactively anticipate the type of discovery that could be initiated and develop an offensive strategy that employs both technology and human resources (Scott Carlson, 2009). The same is illustrated in the figure 1.1.

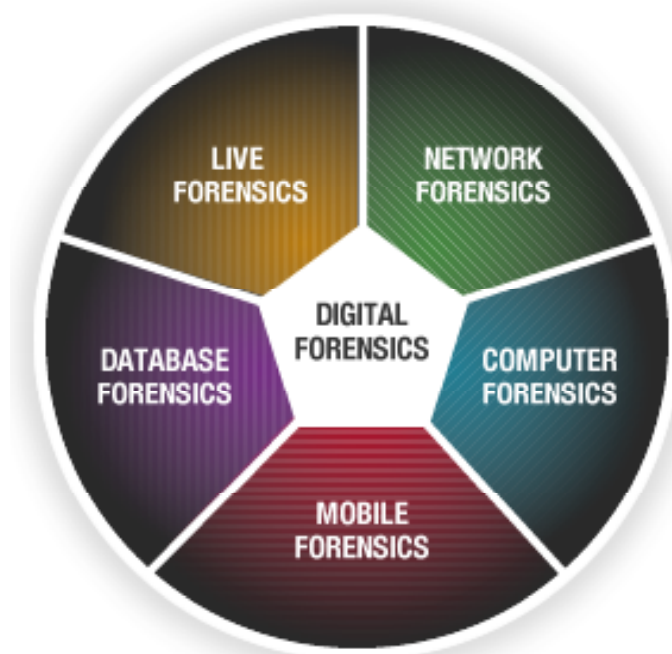


Figure 1.1: Digital Forensics

1.1.6. Digital Spectrum

With the advent of new forms of criminality associated with growth of digital technologies, numbers of terms are used within the forensic community. These include cyber crime, high tech crime, e-crime, new technology crime to indicate new and digitized versions of existing crime. Some crimes can be placed on the spectrum depending upon the extent of digital environment. Consider a street thief who first observes an unwary user input the PIN in an ATM and then steals the card and later withdraws cash; this is not a crime that appears to be particularly digital and it is therefore placed at the less digital end of the spectrum. On the other hand, skimming the magnetic strip, cloning a card and then using it to make transactions is clearly a crime unique to digital era and would appear to be placed at the more digital end of the spectrum. Similarly, some crimes are more likely to have a digital aspect rather than uniquely exploit digital technologies. For example, a fraud enacted via an e-bay, a fraudster may have had

planning for the fraud in the internet, setting up temporary and difficult to trace email accounts, surfing the internet for images, descriptions and prices. These are activities which exploit the advantages of digital technologies but nonetheless arise from a conventional and classic form of crime. This is illustrated in Figure 1.2.

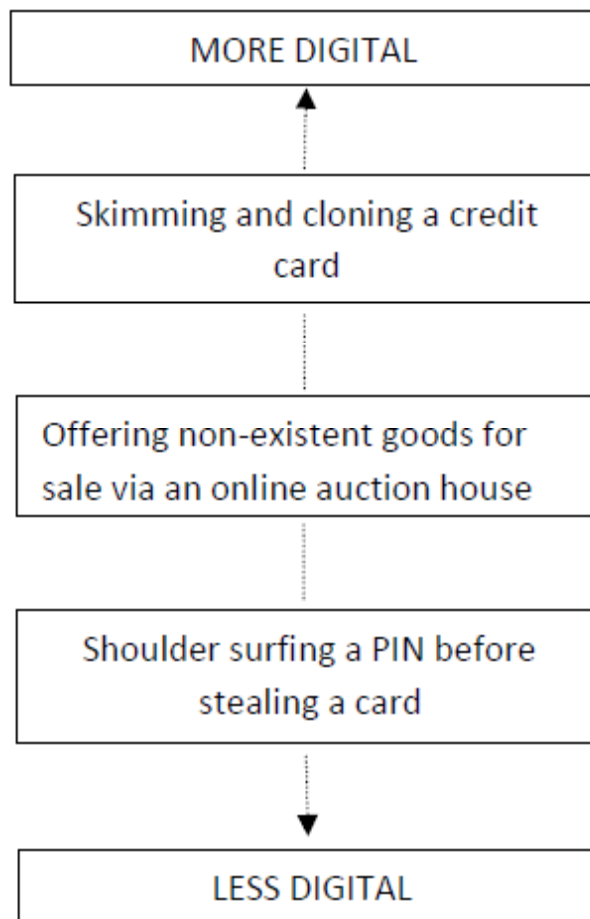


Figure 1.2: The Digital Spectrum
(Source: Bryant -Investigating Digital Crimes)

1.1.7. Good Field Practice in Processing a Crime Scene

Crime Scene: It is crucial to understand the definition of crime scene. For practical purpose, a Crime scene is the aftermath of an event that is considered, by law, to be illegal. For basic understanding the crime scene can be considered the apex of an Inverted pyramid that expands to encompass the five phases,

- Investigation of crime,
- the recognition,
- analysis,
- interpretation of evidence, and
- finally court trial.

Crime scene should be processed with due diligence, utmost care and by the application of technology because any mistake made in processing the crime scene are impossible to rectify. Both errors of omission and commission made in processing a crime scene can confound the final resolution in two ways to make thing worse. The investigators use general guidelines for processing crime scene and exercise the use of check sheets, forms lists as templates for search and examination to be counterproductive. Each crime scene is unique and must be approached with knowledge, education and experience of the investigator. Crime scene is the apex of an inverted pyramid. This is illustrated in the Figure.1.3.

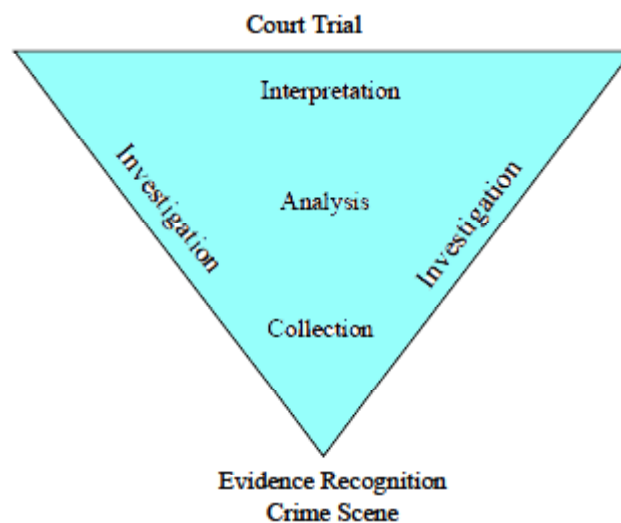


Figure 1.3: Cyber Crime Scene Pyramid

1.1.8. Digital Event and Case Relevance

Digital Event is an occurrence that changes the state of one or more objects. If the state of an object changes as a result of an event, then it is an effect of the event. Some types of

objects have the ability to cause the events and these called causes (Carrier and Spafford, 2004). The property of any piece of information,, which is used to measure its ability to answer the investigative “who, what, where, when, why and how” questions in criminal investigation (Rubin and Garrtner, 2005). The ultimate purpose of crime scene investigation is to seek to solve the commission of crime inevitably that fall under the umbrella of the six “W” questions:

1. What happened?
2. When did it happen?
3. Where did it happen?
4. Who was involved?
5. How was it done?
6. Why was it done?

In the examination of physical evidence the first five questions are relevant. The question “Why” is irrelevant for laboratory analysis and it is left for the establishment of motive by” “Profilers, “Criminologist” and “Courts”. Degrees of case relevance is illustrated in Figure 1.4.

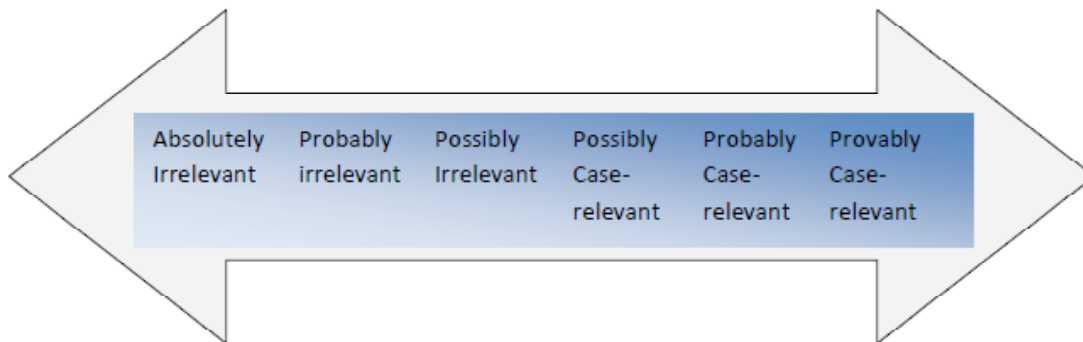


Figure.1. 4: Degrees of Case Relevance

1.1.9. Locard’s Principle: Traditional Forensics Vs. Cyber Forensics

Locard’s Exchange Principle is often cited in forensics publications, “Every contact leaves a trace.” Essentially Locard’s Exchange Principle is applied to crime scenes in which the perpetrator(s) of a crime comes into contact with the scene. The perpetrator(s) will both bring something into the scene, and leave with something from the scene. In the cyber world, the perpetrator may or may not come in physical contact with the crime scene, thus, this brings a new facet to crime scene analysis. According to the World of Forensic Science,

Locard's publications make no mention of an "exchange principle," although he did make the observation *"Il est impossible au malfaiteur d'agir avec l'intensité que suppose l'action criminelle sans laisser des traces de son passage."* (It is impossible for a criminal to act, especially considering the intensity of a crime, without leaving traces of this presence).

1.1.10.Cyber Exchange Principle

"Artifacts of electronic activity in digital devices are detectable through forensic examination, although such examination might require access to computer and network resources involving expanded scope that may involve more than one venue and geolocation." (Zatyko and Bay, 2012). Locard's Exchange Principle does apply to cyber crimes involving computer networks, such as identity theft, electronic bank fraud, or denial of service attacks, even if the perpetrator does not physically come in contact with the crime scene. Although the perpetrator may make virtual contact with the crime scene through the use of a proxy machine, we believe he will still "leave a trace" and digital evidence will exist.

Breaking apart the principle into its parts and analyzing the application of Locard's Exchange Principle, one has to determine whether or not the following occurs:

- Are there two items?
- Is there contact?
- Is there an exchange of material?

To illustrate the application of Locard's Exchange Principle to a cyber crime, we take the example of identity theft where someone's identity is stolen and the perpetrator intends to use the stolen information for criminal gain. Let us further suppose the perpetrator steals the identity through the use of a Trojan horse and keyboard logger on the victim's computer. One could contend that during this type of cyber crime Locard's Exchange Principle does not apply. The rationale is that because a human is not at the crime scene there is no trace evidence from the human on the computer or digital media at the scene.

However, in actuality there may be lots of digital evidence such as the Trojan horse itself, changed passwords, digital logs, and so on. Thus, in this example, there is a trace at and from the scene. It may involve finding the trace evidence at other physical locations than just the one scene of the crime. The key logger could be added software or hardware or both, but in both cases it remains behind for an investigator to discover. This examination typically involves bits and bytes of information.

1.1.11 Goals of Cyber Forensics

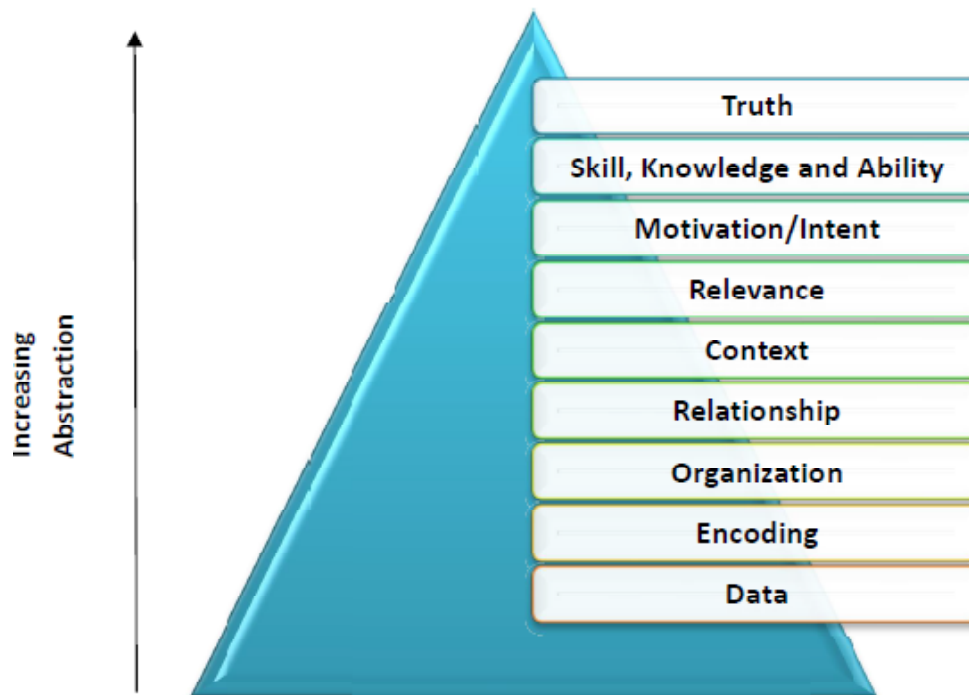


Figure 1.5: Goals of Cyber Forensics

In pursuit of finding the truth, the goal of digital forensics moves from specific to abstract. This can be from acquiring the evidence from the storage media in the form of data or information. This information could be represented inside the media as an encoding (i.e) ASCII or binary. The encoded data in the form of information may be involving incidents that transpired in an organization and event reconstruction needs to be made on timeline basis. The correlation or the relationship between events must be established specific to the context for e.g.: an unauthorized exploit. After acquiring the data must be relevant or coincidental to the questioned occurrence. This incident may involve the motivation or the malicious intent of the attacker. The intent may be *information component* or *human component*. It requires the skill, knowledge and ability of the investigator in the process of unraveling the truth. Incident and response time have inverse relationship. Severity and number of attacks/penetrations are high when compared to earlier years as against the time to respond is very low. A graphical representation of the goal from specific to abstract is represented in the Figure 1.5.

1.1.12.Digital Evidence

Digital evidence or electronic evidence is any probative information stored or transmitted in digital form that a party to a court case may use at trial. Before accepting digital evidence a court will determine if the evidence is relevant, whether it is authentic, if it is hearsay and whether a copy is acceptable or the original is required.

Any data stored or transmitted using a computer that support or refute a theory of how an offense occurred or that address critical elements of the offense such as intent or alibi – Chisum(1988).

Digital evidence is defines as any data that can establish that a crime has been committed or can provide a link between crime and its perpetrator (Casey, 2000)

Definition provided by Scientific Working Group on Digital Evidence (SWGDE) – any information o probative value that is either stored or transmitted in digital form.

International Organization on Computer Evidence (IOCE) defines DE as information stored or transmitted in binary form that may be relied in court.

The term digital evidence and electronic evidence are sometimes used interchangeably. However, an effort should be made to distinguish between electronic devices such as mobile phones and the digital data it can contain. When considering the many sources of digital evidence it is useful to categorize the computer systems into three groups. (Henseler, 2000).

- **Open computer systems:** Open computer systems are what most people think of as computers-systems comprised of hard drives, keyboards, monitors – devices such as laptop, desktop and servers that obey the standards. These systems with their ever in creating amounts of storage space can be rich sources of digital evidence. A simpe like can contain incriminating information and can have associated properties that are useful in an investigation. For example, details such as when a file was created, who created, or that it was created on another computer can all be important.
- **Communication systems** – Tradition telephone system, wireless telecommunication systems, the internet and networks in general can be a source of digital evidence. For instance, the internet carries email messages around the world. The time a message was sent, who sent it, or what the message contained can all be important

in an investigation. To verify when a message was sent, it may be necessary to examine the log files from intermediate servers and routers that handled a given message. To verify the contents of a message, it may be necessary to eavesdrop on the communications as it occurs.

- **Embedded systems** - Mobile phones, PDA, smart cards and many other systems with embedded computers may contain digital evidence. For example, navigation system can be used to determine where a vehicle has been and sensing and diagnostic modules in many vehicles hold data that can be useful for understanding vehicle accidents, including the vehicle speed, brake status, throttle position during the last five seconds before impact.

The use of digital evidence has increased in the past few decades as courts have allowed the use of e-mails, digital photographs, ATM transaction logs, word processing documents, instant message histories, files saved from accounting programs, spreadsheets, internet browser histories, databases, the contents of computer memory, computer backups, computer printouts, Global Positioning System tracks, logs from a hotel's electronic door locks, and digital video or audio files.

The same is illustrated in Table 1.1.

Table 1.1 Sources of Digital Evidence

Open Computer Systems	Communication Systems	Embedded Computer System
Comprise of Hard Drives, Keyboards and Monitors that obey standards.	Traditional Telephone System, Wireless Telecom System, Internet and Networks. Examine Log Files from Intermediate Servers and Routers.	Mobile, Personal Digital Assistants, Smart Cards, Vehicle Modules, IoT (Internet of Things).

1.2. History and Terminology of Computer Crime Investigation

The field of computer forensics began in the 1980s, shortly after personal computers became a viable option for consumers. In 1984, an FBI program was created. Known for a time

as the Magnetic Media Program, it is now known as the Computer Analysis and Response Team (CART). Shortly thereafter, the man who is credited with being “the father of computer forensics” began work in this field. His name was Michael Anderson, and he was a special agent with the criminal investigation division. Anderson worked for the American government until the mid 1990s, after which he founded New Technologies, Inc., a leading computer forensics firm.

1.2.1. Emergence of Cyber Forensic Investigating Agencies

A meeting held in 1988 in Oregon led to the formation of the International Association of Computer Investigative Specialists (IACIS). Shortly after that, the first classes were held to train SCERS (Seized Computer Evidence Recovery Specialists). Computer Forensic Timeline is illustrated in Figure.1.6 and it represents the evolution of digital forensics domain as such.



Figure 1.6: Computer Forensics Timeline

Different Phases of Cyber Forensics are:

1. Ad-hoc phase which was characterized by lack of structure, lack of clear goals, lack of adequate tools, processes and procedures. Further huge legal issues on how to proceed with digital evidence were seen.

2. Structured Phase is complex solution for computer forensic from accepted procedures, special tools developed and what is more important enabling criminal legislation to wide use of digital evidences.

3. Enterprise phase –Three areas of this phase are real-time collection of evidence, developing field collection tools and forensic becoming a service in companies.

1.2.2. Development of Cyber Forensics

Computer forensics is the study of extracting, analyzing and documenting evidence from a computer system or network. It is often used by law enforcement officials to seek out evidence for a criminal trial. Government officials and business professionals may also have need of a specialist familiar with computer forensic techniques. The discipline of computers forensics is relatively new, having been founded in the 1980s. This digital evidence consists of data from storage media like hard disk, floppy disks, zip disks, Compact discs, DVDs, emails, data transmitted over communication links (wired and wireless), log files generated by the operating system, logs from perimeter devices like router and switch, PDA, Mobile Phones, MP3 players, USB devices, and even plethora of devices which do not fit into the original concept of computers like the washing machines, engine management system in cars, GPS devices. In order for evaluation and acceptance of Digital evidence – a new type of evidence, previously not considered by courts, certain basic principles are suggested by research scholars and these are given below:

- **Authenticity** – the evidence should be authentic that is “it should specifically linked to the circumstances and persons alleged – and produced by someone who can answer questions about such links”
- **Accuracy** – the evidence should be free from any reasonable doubt about the quality of procedures used to collect the material, analyze the material if that is appropriate and necessary and finally to introduce it into court and produced by someone who can explain what has been done. In the case of exhibits which themselves contain statements – a letter of other document, for example – ‘accuracy’ must also encompass accuracy of content and that normally requires the document’s originator to make a witness statement and be available for cross examination”. Accuracy – when presented evidence contains statements created in a computer “accuracy must encompass the accuracy of the process which produced the statement as well as the accuracy of the contents
- **Completeness** - “tells within its own terms a complete story of a particular set of circumstances or events”.
- **Chain of custody**, transparency and explainable are the other basic principles the needs to be followed among the other principles. Unlike the physical evidence, the digital evidence, by itself has no informational value. It requires skill and talent in

interpreting the latent information which is dependent on the process by which it is unraveled and the process in turn depends on the basic principles of computer science. In order to be legally acceptable in the court of law, it requires a motivated skilled expert who will apply appropriate tools to achieve, efficiency and reliability.

1.2.3. Digital Data

Digital data are all around us and should be collected in any investigation routinely. More likely than not someone involved in the crime used a computer, personal digital assistant, mobile phone or accessed the internet. Therefore every corporate investigation should consider relevant information stored on the computer systems by their employees both at work and at home. Every search warrant should include digital evidence to avoid the need for the second warrant and the associated time lost. Even if digital data do not provide a link between a crime and its victim or a crime and its perpetrator they can be useful in a investigation,, digital evidence can reveal how a crime was committed, provide investigative leads, disprove or support witness statements and identify likely suspects. Cyber Forensics focuses on three kinds of data namely active data, latent data and archival data.

1.2.3.1 Active Data

An active data is one that is currently available, visible and that which can be understood using an application within a computer. Active data might also be protected using passwords or some means of encryption. Some of the active data are: word processor files, spread sheets, files and directories, email content, database programs, system files, history files, temporary internet files, cookies, recycle bin and the like.

1.2.3.2. Latent Data

Latent data also called as ambient data, volatile data may be in the form of deleted files, memory dumps and similar data which reside in swap files, temporary files, printer spool files, metadata, shadow file and so on. It requires an expert talent to bring to light the latent data using specialized tools and techniques.

1.2.3.3. Archival data

Data that has been stored or backed up to external storage media such as tapes, CDs, DVDs, external hard disks, pen drive, zip disks, network servers or the internet is archival data. Necessary precautions have to be taken while performing forensic examination since the backup

peripheral devices do not have all the information. Hence, it is always better to perform forensic examination on original source media because backups do not store latent data.

1.2.4. Cardinal Rules in Cyber Forensics

The cardinal rules of computer forensics can be expressed as the five 'A's

- Admissibility must guide actions: document everything that is done
- Acquire the evidence without altering or damaging the original
- Authenticate your copy to be certain it is identical to the source data
- Analyze the data while retaining its integrity and
- Anticipate the unexpected.
- The cardinal rules are designed to facilitate a forensically sound examination of computer media and enable a forensic examiner to testify in court as to their handling of a particular piece of evidence. A forensically sound examination is conducted under controlled conditions, such that it is fully documented, replicable, and verifiable.

A forensically sound methodology changes no data on the original evidence, preserving it in pristine condition. The results must be replicable such that any qualified expert who completes and examination of the media employing the same tools and methods employed will secure the same results.

1.2.5. Cyber Forensic Lab

The role of cyber forensics will be increasing importance to the legal system as information continues to evolve into purely a digital form and the systems on which such digital information is stored becomes more technologically advanced. Strategic planning for setting a laboratory involves in developing a forensic practice: Operational Perspective, Technological perspective/venue, Scientific Perspectives and an artistic perspective. Other areas of significance are Core Mission and Services, Budget and Standard operating procedures.

Cyber Forensics Lab requires adequate infrastructure for examination and analysis of Digital Evidence. Adequate Infrastructure not only includes technical infrastructure but also assets such as workspace, dedicated communication lines, 24/7 internet connection among others which should be made available to the cyber forensic experts.

1.2.6. Operational Perspective

All business venue must have sound business management, financial profitability, core service etc. a police cyber forensics lab may not have profit per se, but the lab has to demonstrate value of service and return of Investment in order to acquire annual budget allocations and training in new technologies to continue fighting crime.

1.2.7. Technological Perspective

Technological advancement calls for sophisticated knowledge in data and data storage technologies. More complex techniques are used by criminals to hide their criminal activity. The commercial market is rolling out a new wave of newest and shiniest technologies available to upkeep the demand for progress; again forensics community is at the front of the line, dismantling and investigating every new gadget that hits the shelves in order to reveal its secrets.

1.2.8. Scientific Perspective

In order reveal facts objectively through empirical observation, deductive reasoning and conversion of hypothesis to demonstrable proof of the fact, examiners have to perform their duties according to reliable, repeatable, valid, objective, consistent and accurate methodologies, thereby enabling the presentation of the findings as acceptable in court of law.

1.2.9. An Artistic Perspective

A great degree of technological prowess or expertise and competency is required in fact finding. Although the investigative process involves a rigid set of procedures, intuitive and creative skills of the forensic examiner is also required. Raw technological skill does not empower an examiner to understand the man and the machine. More artistry and creativity is required to enable a better understanding of how the tools of technology and human nature and thought process interact.

1.2.10. Budget

Every forensic facility be it business organization or government, require funds to function. It has to spend on building, staff, and stock. It has to operate, maintain and grow a facility. Every operation needs to demonstrate return of investment in order to prove viability of the venture.

1.2.11. Core Missions and Services

Primary consideration of forensics facility is the design plan and what services are to be rendered and the scope at which it is to be provided. A firm grasps of a prospective lab core mission and range of service will provide guidance on every aspect of building, functional forensic facility, touching on everything from annual budget to furniture ergonomics. The technical aspects would include the requisite hardware and software tools for examination and analysis. The cyber lab should possess robust operating system software such as Microsoft Windows, MAC OS, Linux, SOLARIS. It must also have powerful computer workstation with standard peripherals.

Another vital requisite would be Uninterruptible Power management software is also required for extensive control and monitoring capabilities. In general, the infrastructure is layered and it should be built, frequently updated as per the requirements and needs and the same is illustrated in Figure.1.7.

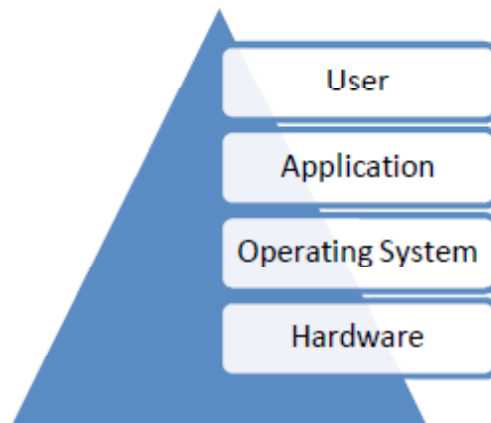


Figure.1.7: Layered Infrastructure

1.2.12 Increasing awareness of Digital Evidence

In the recent years, judiciary and law enforcement are encountering progressively more digital evidence in their work. Less obviously, cyber security professionals and military decision makers are concerned with digital evidence. An increasing number of organizations are faced with necessity of collecting evidence on their networks in response to incidents such as computer intrusions, fraud, intellectual property theft, child pornography, stalking, sexual exploitation and

harassment, and more violent crimes. Organizations are forced to consider legal remedies when criminals target them. The need is that the organizations giving more attention to handling digital evidence in a way that it will hold up in court. In addition to collection of evidence, organization by processing digital evidence properly is protecting their organizations liabilities such as invasion of privacy and unfair dismissal claims. As a result, there is an increasing training and knowledge in digital evidence handling. Organization must also respond to such incidents and recover from such incidents in a rapid manner in order to minimize the loss/damage to the organization. Hence, cyber security professionals hundreds of security incidents on a daily basis as a result there is no enough time to open an investigation for each incident. There are three significant drawbacks to this. They are:

- Each unreported incident deprives defense council an opportunity to learn basics about computer related crimes. They are only involved only when the stakes are high and cases are complicated.
- Due to poor evidence collections policies, lawyers and law enforcement officers job difficult to prosecute an perpetrator
- This approach results in under reporting of criminal activity, deflating the statistics that are used to allocate the corporate and government spending on computer related crimes.

Government agencies are also interested in using digital evidence to detect terrorist activities and prevent future attack. As a result data mining techniques are used to detect and investigate criminal activity. Understandably the possibility of the government freely sifting through every citizens personal data for anything that looks suspicious is privacy advocate's worst nightmare. In certainty the risk associated with these pre-crime systems will do more harm that the problems they aim to address. The reason for this is

- Inadequate training data sets
- Inaccurate data
- High number of false positives and
- Information overloaded

1.2.13. Challenging aspects of Digital Evidence

Digital evidence is a form of physical evidence creates several challenges for digital forensicators.

- Slippery from of evidence makes it difficult to handle – Example: Hard Disk platter contains amalgam of data – pieces of information scattered across the disk and are layered on top of each other over a period of time. Only a small portion of this may be relevant to a case, making it necessary to extract useful pieces and translate them to meaningful form that can be interpreted.
- Radio waves and microwaves travelling through the air contain a tangle of data making it necessary to find the desired signal amongst the noise and translate it into the data that can be understood
- Volatile nature of the evidence – transient nature of data and packets makes it difficult to collection of evidence a difficult task
- Digital evidence is generally an abstraction of some event or digital object. When a person instructs a computer to perform a task such as composing a email and sending it, the activities generate data remnants that give only a partial view of what occurred. Unless some surveillance equipments are installed – individual mouse clicks, keystrokes, internal system commands, and minutiae are not recorded or retained. Only certain results of the activity such as email message and server logs remain to give a partial view of what transpired. Email message and server logs are several layers of abstraction to magnetic fields on the disk to letters and numbers seen on the screen. Each layer can introduce error
- Digital evidence can be manipulated – it can be altered either maliciously by perpetrators or accidentally during collection
- Fortunately there are several features that can be used to mitigate this problem
 - o Digital evidence can be duplicated exactly and a forensic copy can be examined as it is original. It is a common practice when dealing with digital evidence to examine a copy, thus allowing the risk of damaging the original.
 - o With right tools it is very easy to determine if digital evidence has been modified or tampered with, by comparing it with an original copy.

- o Digital evidence is difficult to destroy. Even when a file is deleted or a hard disk is formatted, still the digital evidence can be retrieved.
- o When criminals attempt to destroy the evidence – copies and associated remnants can remain in places that they were not aware of.
- o Digital evidence is circumstantial making it difficult to attribute computer activity to an individual. Therefore digital evidence can be only one component of a solid investigation. If a case hinges upon a single form of digital evidence such as a timestamp on computer files, then the case is unacceptably weak. Without additional information it could be argued that someone else used the computer at that time. If the defendant argues that some exonerating digital evidence was not collected from one system, this would impact a weak case that does not have supporting evidence of guilt from other sources.

1.2.14 Following the cyber trail

It is a common belief that cyber space is separate from the physical space. This is simply not the case – crime on cyber space mirrors crime in physical space. There are several reasons to this:

- Crime on internet reflects crime in physical space with human perpetrators, victims and should be treated with same gravity
- To neglect the very real and direct link between people and the online activities that involve them limits one's capability to investigate and understand crimes with an online component – example – auction fraud provides a simple demonstration of how a combination from the virtual world to physical world is used to apprehend a criminal
- Criminals feel that they are safe in the internet but they don't realize the fact that they are observable, and hence vulnerable. This opportunity can be utilized to uncover the crimes in physical world that would have been possible without investigation in the cyber space
- Some of the cases that have been solve by observing the criminals include:
 - o Murder
 - o Child pornography

- o Child abusers
 - o Drug dealers
 - o Weapon traffickers
 - o Human traffickers
- When a crime is committed in physical world, the cyber space has related digital evidence and should be considered as an extension of crime scene.

The crimes of today and future requires the digital forensicators to become skilled at finding connections between the cyber criminal and the cyber crime which can be extended to the physical world by following the cyber trail. The cyber trail should be considered even when there is no sufficient internet activity. Criminals are learning to conceal their internet activities. With the rise of wireless the difficulty rises. There are growing numbers of sensors on the internet such as cameras showing highway traffic on the web. These cameras may inadvertently capture evidence relating to crime. In addition to the internet, the cyber trail can be found on commercial systems.(e.g: ATMs, credit cards, debit cards) and on privately owned networks. The rich source of digital evidence on privately owned networks include:

- Internal emails
- Chats
- News groups
- Web servers
- Databases
- Document management systems
- Time clock systems
- Individuals information

1.2.15.Challenging aspects of the cyber trail

The challenging aspects of the cyber trail include:

- The dynamic and distributed nature of networks makes it difficult to collect digital evidence
- Data can be spread over beyond geographic boundaries

- It is not feasible to take a snapshot of a given network at a given instant
- Network traffic is transient and must be captured while it is in transit. Once a network traffic is captured – only copies remain and the original data is unavailable
- The amount of data lost during evidence collection process can be documented but the lost evidence cannot be retrieved
- Huge volume of data travels across the network and it is difficult to sniff – useful information can be like a needle in a haystack
- Even when the digital evidence is obtained from the network, a great deal of anonymity makes it difficult for attributing to a criminal

Summary

- Crime is as old as humankind. Most crimes leave behind traces of “silent evidence”. The traces of materials, known as physical evidence, are found at the scene of crime and acts as potent clues that become the most eloquent witness subsequently. Evidence that is conspicuously found in the crime scene is dust, pollens, bloodstain, bullet, glass, paint, hair, ash and the like serves as silent speakers of truth.
- The pursuit of cyber crime detection has two main objectives. One to prevent the occurrence cyber crime in vulnerable institutions requiring security from loss, pilferage and mishandling by accidental or intentional manipulations and the other detection and documenting cyber crimes through a disciplined methodology. With advancement of technology, computer is both the instrument of crime, as well as the location of evidence.
- Cyber Forensics is simply application of computer investigation and analysis techniques in the interest of determining potential legal evidence. Forensic computing is the process of identifying preserving, analyzing and presenting the digital evidence in a manner that is legally acceptable. It is the study of evidence from attacks on computer system in order to learn what has occurred, how to prevent it from recurring and the extent of the damage.
- In a Computer Crime, the process of identifying, preserving, analyzing and presenting digital evidence to prove the crime in a way that is legally acceptable in the court of law.
- Cyber forensics is employment of a set of predefined procedures to examine a

Computer system using software or hardware tools to extract and preserve the evidence of criminal activity.

- Disk Forensics: deals with extracting data/information from storage media by searching active, deleted files and also from unallocated and slack space.
- Network Forensics: It is a sub branch of digital forensics relating to monitoring and analysis of computer network traffic for the purpose of information gathering, legal evidence or intrusion detection. Unlike other areas of digital forensics, network investigation deal with volatile and dynamic information. It is also called Pro-active forensics.
- Wireless Forensics: It is a sub part of network forensics. The main goal of wireless forensics is to provide the tools required to collect and analyze the data from wireless network traffic. The data collected can correspond to plain data or with the broad usage of Voice over Internet Protocol (VoIP) technologies especially over wireless technology.
- Database Forensics: is a branch of digital forensics relating to study and examine databases and their related metadata. A forensic examination of a database may relate to the timestamps that apply to the row (update time) in a relational table being inspected and tested for validity in order to verify the actions of a database user.
- Malware Forensics: deals with analysis and identification of a malicious code, to study their payload, viruses, worms, Trojans, Keyloggers etc.
- Mobile Phone Forensics deals with examination and analysis of mobile devices, to retrieve phone and SIM contacts, call logs (Dialled, Missed & Received), incoming and outgoing SMS/MMS, Audio, videos, paired device history and in some smart phones, geolocation and calendar information etc.
- GPS Forensics is also called SatNav Forensics, is a relatively new discipline with the fast paced world of Mobile Device Forensics. It is used for examining and analysing GPS devices to retrieve information such as TrackLogs, TrackPoints, WayPoints, Routes, Photos, audio etc.
- Email Forensics: Deals with recovery and analysis of emails including deleted emails, calendars and contacts.

- Memory Forensics deals with collecting data from system memory (system registers, cache, RAM) in raw form and then carving the data from Raw dump.
- E-Discovery: E-Discovery is the process of evaluating solutions for organization. A defensible e-Discovery process is repeatable, systemized and meets legal requirements for proper handling and admissibility of computer evidence.
- Crime Scene: It is crucial to understand the definition of crime scene. For practical purpose, a Crime scene is the aftermath of an event that is considered, by law, to be illegal. For basic understanding the crime scene can be considered the apex of an Inverted pyramid that expands to encompass the five phases, Investigation of crime, the recognition, analysis, interpretation of evidence, and finally court trial.
- Locard's Exchange Principle is often cited in forensics publications, "Every contact leaves a trace." Essentially Locard's Exchange Principle is applied to crime scenes in which the perpetrator(s) of a crime comes into contact with the scene. The perpetrator(s) will both bring something into the scene, and leave with something from the scene. In the cyber world, the perpetrator may or may not come in physical contact with the crime scene, thus, this brings a new facet to crime scene analysis.
- Digital evidence or electronic evidence is any probative information stored or transmitted in digital form that a party to a court case may use at trial. Before accepting digital evidence a court will determine if the evidence is relevant, whether it is authentic, if it is hearsay and whether a copy is acceptable or the original is required.
- **Authenticity** – the evidence should be authentic that is "it should specifically linked to the circumstances and persons alleged – and produced by someone who can answer questions about such links"
- **Accuracy** – the evidence should be free from any reasonable doubt about the quality of procedures used to collect the material, analyze the material if that is appropriate and necessary and finally to introduce it into court and produced by someone who can explain what has been done. In the case of exhibits which themselves contain statements – a letter or other document, for example – 'accuracy' must also encompass accuracy of content and that normally requires the document's originator to make a witness statement and be available for cross examination". Accuracy – when presented evidence contains statements created in a computer "accuracy must encompass the accuracy of the process which produced the statement as well as the accuracy of the contents"

- **Completeness** - “ tells within its own terms a complete story of a particular set of circumstances or events “In addition to these considerations, forensic evidence must exhibit the following properties:
- **Chain of custody**, transparency and explainable are the other basic principles the needs to be followed among the other principles. Unlike the physical evidence, the digital evidence, by itself has no informational value. It requires skill and talent in interpreting the latent information which is dependent on the process by which it is unraveled and the process in turn depends on the basic principles of computer science. In order to be legally acceptable in the court of law, it requires a motivated skilled expert who will apply appropriate tools to achieve, efficiency and reliability.
- Digital data are all around us and should be collected in any investigation routinely. More likely than not someone involved in the crime used a computer, personal digital assistant, mobile phone or accessed the internet. Therefore every corporate investigation should consider relevant information stored on the computer systems by their employees both at work and at home.
- Cyber Forensics focuses on three kinds of data namely active data, latent data and archival data.
- The cardinal rules of computer forensics can be expressed as the five ‘A’s namely: Admissibility; Acquisition; Authenticate; Analyze; Anticipate.

Check your answers

- is employment of a set of predefined procedures to examine a Computer system using software or hardware tools to extract and preserve the evidence of criminal activity.
- deals with extracting data/information from storage media by searching active, deleted files and also from unallocated and slack space.
- is a sub branch of digital forensics relating to monitoring and analysis of computer network traffic for the purpose of information gathering, legal evidence or intrusion detection.
- is a sub part of network forensics. The main goal of wireless forensics is to provide the tools required to collect and analyze the data from wireless network traffic.

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- It is crucial to understand the definition of crime scene. For practical purpose, a Crime scene is the aftermath of an event that is considered, by law, to be illegal.
- "Every contact leaves a trace" is the dictum coined by
-is any probative information stored or transmitted in digital form that a party to a court case may use at trial.

LESSON - 2

TECHNOLOGY AND LAW

After reading this lesson you will be able to understand the following:

- Technology and Law
- Magnitude of Computer & Information Technology Crimes in India

Structure

2.1 Technology and Law

2.2 Magnitude of Computer & Information Technology Crimes in India

2.1 Technology and Law

Need for Law human beings are vulnerable to unlawful acts which are crimes, and therefore, rules of law are required to protect them against such acts. Computer systems are vulnerable. This technological progress can easily dupe, exploit a person by illegal or unauthorized access. The result is damage to the computer system. With the help of networks the cyber criminals indulge in criminal activity without any fear of being apprehended and tried for the offences committed by them.

The reasons for committing cybercrimes by criminal may briefly be stated as;

1. Wider accesses to information.
2. Complexity of computer system.
3. Negligence of network users.
4. Non- availability or loss of evidence.
5. Lack of Jurisdiction leading to fearless crimes.

Cybercrime Evolution or Neo-criminology: Computer Technology and Internet with its advancements has brought about new variety of crimes thereby providing ever increasing opportunities for the criminals to indulge in illegal activities unabated. The last quarter of twentieth century witnessed several sophisticated ways through which the perpetrators of cybercrime found it easy to penetrate into the system of software and internet to commit cybercrime which may be characterized as the new species of white collar crime. These crimes have global

ramifications sabotaging the national economy and business ventures. They are not restricted to any geographical area or territory and may be committed within a fraction of a second affecting victim who may be thousands of miles away. These new variety of crime have thrown a big challenge before the law enforcement agencies and they are required to evolve a neo-criminological approach to handle these internet crimes. Here the risk in such crimes is that the victims are totally unaware of the crime committed on their property or on person.

2.1.1 Groups of Cyber Criminals

1. Children and Adolescent age between 8-18 years
2. Professional Hackers / Crackers
3. Disgruntled Employees or vengeful persons
4. Cyber abuser.

By the use of internet the incidences of cybercrime have increased, therefore the need for such crimes to be channelized and regulated by effective cyber laws. Cyber laws are that branch of law which deals with the newly emerging issues out of the development of the internet and other communication mediums functioning over the cyberspace. These laws need to regulate all aspects of transactions of the internet and the cyberspace.

There are many issues faced by the development of the internet and the corresponding crimes arising there from-

1. Domain name
2. Intellectual Property Rights
3. E-commerce
4. Encryption
5. Electronic contracts
6. On-line banking

Cyber law may be defined as the law governing cyberspace which is a non-physical terrain created with two or more computers networked together. Online system create a cyber space. Cyberspace is not restricted to internet alone, but has computers, its networks and software. Therefore cyber law refers to the law relating to computer, computer networks and

includes all activities that take place in relation to information stored, exchanged or retrieved using the computer system.

2.1.2 Reasons to Pass Cyber Laws

1. To prevent computer abusers to carry on their illegal activities for personal gains.
2. To prevent political rivalry so that innocent people do not become a victim of their criminal act. E.g.) A person staying in one country can easily dupe a person having a bank account in any other nation, by transferring millions in the bank of a third nation, within no time with the help of lap-top and cell phone.

Like many other countries India to has developed independent laws, to tackle cybercrime. But still in some countries there exists traditional laws to deal with such offences.

2.1.3 Importance of Cyber Laws

1. To regulate all the features of contacts and actions with respect to the internet.
2. To tackle issues on e-commerce, encryption, electronic contracts, on-line banking.
3. To tackle multidimensional crimes.
4. To have laws as cybercrime recognizes no territorial boundaries

2.1.4 Provisions of the I.T. Act

Jurisdiction, Extent and Scope of the IT Act 2000

Short title, extent and applicability -

The Act is called The Information Technology Act 2000 and is applicable to entire India; it is applicable to all the crimes which is committed in and abroad.

2.1.5 Statement of Object of Reasons of The Information Technology Act,2000

Objects of the Act of 2000: The progress in communication and technology led to the need to have a law governing transaction in business. Today's business world is internet savvy and conducts most of their business transactions in electronic form instead of traditional papers. It was observed then that information stored in electronic form was much easier, cheaper, and retrieve and speedier to communicate. Despite the very many plus point of internet businessmen are reluctant to settle any transaction in electronic form as there is no proper framework of law.

Objects

1. To provide for legal outline.
2. Legal Sacrosanctity to all e-records.
3. To make in its ambit all business transaction legal and transfer of information from one computer to another.

2.1.6 Purpose and the Applicability of the IT Act 2000

The IT Act 2000 was introduced with the following intent:

The Act was conceived to regulate electronic commerce. As the preamble of the Act states that the Act was framed “to provide legal acknowledgement for transaction made over electronic means and any other means of electronic communication” The necessary changes were made to harmonize the existing laws.

The major challenges were the requirements of digital signatures for legal acknowledgement. The Information Technology Act provides for **legal recognition** for transactions made over the internet or any electronic mode of electronic communication, known as “electronic commerce” which is alternatives to paper based methods of communication and storage of information. Need was felt to amend the I.P.C, the Indian Evidence Act, 1872, the Bankers Book of Evidence Act, 1891 and the Reserve Bank of India Act, 1934. The reason for passing of the Act was to safeguard the interest of the internet users.

The I.T Act, 2000 does not apply to the following

- a. A Negotiable Instrument Act 1881.(Amended in IT Act 2008)
- b. A Power of attorney as defined in section 1-A of Powers of Attorney Act 1882.
- c. A Trust as defined in section 3 of The Indian Trusts Act 1882.
- d. A will as defined in clause (h) of the Indian Succession Act 1925.
- e. For agreements in the form of contracts with respect to sale of fixed assets or property.
- f. Any kind of document or transactions as stated in the official gazette.

2.1.7 The IT Act 2000 prior to amendment was divided into 13 chapters

- Chapter I – Preliminary information about Information Technology
- Chapter II –stated that a Digital Signature may be affixed to authenticate any electronic document of the subscriber. The public key of the subscriber verifies the digital signature.
- Chapter III – **Electronic Governance.**
- Chapter IV –**Attribution, Acknowledgement and Dispatch of ElectronicRecord.**
- Chapter V — **Protected Electronic Records and Digital Signatures**
- Chapter VI – **Regulation of Certifying Authorities-**
- Chapter VII – **Digital Signature Certificate**
- Chapter VIII – **Duties of Subscribers**
- Chapter IX –**Punishments.**
- Chapter X – The **procedure, powers**
- Chapter XI- **Offences.**
- Chapter XII – **Service Provides for Network and their liability.**
- Chapter XIII — Here the powers of the police officers are laid down with respect to enter any premises, appoint a controller along with his deputy. Powers, protection, offences by companies, Constitution of Advisory Committee, controller's powers.

Table 2.1 Offences

Sections	Description	Punishment	Amendment (A) / Inserted(I)/ Substituted(S)
65	Computer Documents and its Tampering	three years Custody, or with fine of 2,00,000 rupees, or both	
66	Offences related to Computer	Custody two to three years or fine of 5,00,000 rupees or both.	(S) ITAA 2008

66 A	Offensive messaging	Custody for a term two to three years and fine.	(I) ITAA 2008
66 B	Deceitfully receiving stolen computer or communication device	Custody for a term of three years or fine of 1,00,000 rupees. Or both.	(I) ITA 2008
66C	Identity theft	for a term which may extend to three years and shall also fine which may to rupees 1,00,000 rupees.	(I) ITA 2008
66D	cheating by Personation	Custody for a term of three years and fine to 1,00,000 rupees.	(I) ITA 2008
66E	Privacy and Violation	Custody to three years or fine of 2,00,000 rupees,	(I) ITA 2008
66F.	Punishment for cyber Terrorism	Punishable with Custody which may extend to Custody for life'.	(A) ITAA 2008
67	Punishment for publishing or transmitting obscene material in electronic Form	punished on first conviction with Custody for a term which may extend to two three years and with fine which may extend to five lakh rupees and in second or subsequent Custody for a term which to five years and fine to ten lakh rupees.	(A) ITAA 2008

67 A	Punishment for publishing or transmitting of material containing sexually explicit act in electronic form	Custody to five years and fine to 10,00,00 rupees,	(I) ITAA 2008
67 B	Punishment for publishing or transmitting of material depicting children in sexually explicit act in electronic form	on first conviction imprisonment for a term to five years and with a fine ten lakh rupees and in second or subsequent conviction with imprisonment for a term to seven years and with fine to 10,00,000 rupees.	
67 C	Protection, Keeping of information by Intermediaries	Custody to three years and fine.	
68	Power of Controller to give directions	Custody for two years or fine or both.	(I) ITAA 2008
69	Powers –for interception , monitoring, decryption of any Information	Custody to seven years and fine.	
69 A	Power - for blocking for public access Information	Custody to seven years and fine.	
69B	Power to authorize to monitor and collect traffic data or information through	Custody to three years and fine.	

	any computer resource for Cyber Security		
70 A	National nodal agency.	cognizance by an officer authorized by the agency	(I) ITAA 2008
71	misrepresentation, Breach of confidentiality, privacy		
72 A	Disclosure of information in breach of lawful contract		
73	Penalty - publishing false electronic Signature Certificate		
74	Publication for fraudulent purpose		
75	Act for offence outside India		
76	Confiscation		
77	Compensation, penalties , confiscation		
77 A	Compounding of Offences		
77 B	Offences imprisonment	three years	
78	Investigate and Powers		(A) ITAA 2008

2.1.8 Need of Today

The goal of any country is to codify a commercial law so that it can be used across multiple political states. This can be achieved through harmonization and unification of laws. This process is difficult as it involves balancing issues and choices revolving around different social cultures and legal systems having different historical traditions. Alternatively, the various other states like USA already have homogenous attitudes with some issues. The economic systems of the USA are highly developed, as compared to many parts of the world. The disparities existing in legal traditions, cultures, and economic development make harmonization project a complex task, most particularly at the international level. However, only through such harmonization can the uniformity of law that is so crucial to support efficient and fair commercial transactions be advanced. This is achieved to a certain extent through the UNCITRAL Model.

2.1.9 Lacunas in the initials stages of the Act

1. Problematic for courts to determine jurisdictional issues to bring criminals for trial.
2. Ambiguity in application of country's laws either at place of origin of crime or at place of commencement of crime/victim is located / both these laws.

In a landmark case, an internet news service was sued for trademark violation and misusing the plaintiff's domain name. The Court held that a passive website which only made information available to interested users was subject to Court's jurisdiction, unless it entered into specific contract knowingly and repeatedly transferred computer files to the plaintiff.

2.1.10 Shortcomings of I.T. Act 2008

1. Infringement of copyright excluded from this law.
2. Domain names lack protection
3. Power of attorney trusts and wills not covered under the Act.
4. Act is silent on taxation.
5. Electronic documents have no stamp duty.
6. No proper laws governing credit card frauds.
7. The Act did not cover all the offences.

2.1.11 Advantages of I.T. Act 2008

1. Promotes e-commerce

- Valid Email contacts
- Valid Digital signature.
- Valid credit card payment.
- Valid online communications

2. Improve business

Through issue of digital signature by the authorities authorized by the Act.

3. Filling online forms:-

Filling online forms for different purposes like promoting e-governance has made task easy for all the citizens of India.^{xi}

4. High penalty

High penalty is imposed for persons doing any cybercrime due to which the rate of cybercrime has reduced.

2.2 MAGNITUDE OF COMPUTER AND INFORMATION TECHNOLOGY CRIMES IN INDIA

Despite penal provisions and preventive measures provided in the Indian Penal Code and the Information Technology Act, the statistics of the preceding years clearly show that there has not been any decline in the crime rate and in fact there is a steady rise in the crimes. Many crimes exist which need improvised investigation and legal techniques and skills to handle them efficiently.^{xii} Crimes like Credit Card Fraud are much on the rise. Crime statistics have an important role in formulating preventive crime strategy as they contain relevant data on specific crimes and criminals, which helps the criminal law enforcement agencies to make best possible use of them for working out effective strategy to tackle such crimes efficiently.^{xiii}

2.2.1 Law relating to electronic signatures

Any subscriber of the contract on the internet, signs the electronic agreement by digital signature is treated as authenticate, by the presence of electronic signature.

2.2.2 Procedure for verification of electronic records:

1. Affixed Digital signature
2. Further these asymmetric crypto system and hash function are used to wrap up into another electronic record.
3. The hash results are generated.
 - a. An algorithm is generated to give results.
 - b. Any person can verify e-record.
 - c. Subscriber has unique private and public key.

2.2.3 Electronic Signature

The Electronic and Digital signatures are sometimes used interchangeable yet there exists a considerable difference between the two. The electronic signature adds data which may be text, sound, symbol, picture to the document.^{lxv} This identifies the signer. The major disadvantage of electronic signature is that they are forgeable as there is no proper format or standard and therefore vulnerable to copying and interfering are easy. In most of the cases they was a need to authenticate the e-signature, which are universally accepted.

2.2.4 The content of such system

The Digital signature is based on Public Key Infrastructure .This signature is based on the cryptographic operation. This of the signer is **authentic**. Digital Signatures are valid in India.

2.2.5 Advantage of such signatures

1. They cannot be copied, tampered or altered.
2. They contain three algorithms i.e. a Key generator, possible private keys, a signature verifying algorithm.
3. They are made in one program (e.g. Microsoft, Word, Adobe, PDF) so they can be authenticated by using the same application.

Electronic signatures are used in many ways

- a. Filing online tax returns
- b. E-contracts

- c. Online government orders.
- d. Issuing forms and licenses.
- e. Online file movement system
- f. Public information records
- g. E-voting
- h. E-ticketing
- i. Online money transfer
- j. Online transfer of data, e-filing

2.2.6 Need of Digital Certificate

- 1. It is a Certificate that gives validity to your credentials on web while doing business.
- 2. It is a method by which your identity is proved in electronic transactions. The need of such certificate could be equivalent to that of the driver's license or passport. Each Authority certifying, stores digital signature certificates issued by it online.
- 3. Digital Certificates help prevent someone from using a phony key to impersonate someone else.

2.2.7 Secure Digital Signature

A safe digital signature should have the following conditions:

- 1. It should be unique identity.
- 2. It should be capable of identifying the subscriber
- 3. The key is under the exclusive control of the subscriber.^{lxvii}
- 4. It should create a link so as to prevent alteration.

All software programs create digital signatures with its features.

The procedures for digital signatures are stated in the notification by the Central Government.

Section 87 of the **Information Technology Act, 2000** Central Government states theregulation for application and other guidelines for Certifying Authorities. This is passed in the 6th rule, which relate to the formation, loading of the digital signature.

2.2.8 Status in India

In India, the Ministry of Communication & Information Technology, Department of Electronics is the Controller of Certifying Authorities (C.C.A). It has been formed under Section 18, of the I.T. Act, 2000. They license and regulate the working of the Authorities for Certification. They issue Certificates for electronic verification of users. The Agency which are authorized under the CCA: A Certifying Authority (CA) is licensed to issues the digital signature.

The following organizations are authorized C.A. under CCA, G.O.I.

1. NIC (For Government Departments/ Undertakings only)
2. (n) Code Solutions CA (GNFC)
3. Safescript
4. TCS
5. MTNL
6. Customs & Central Exercise
7. e-Mudhra
8. IDRBT

From the above data we can conclude for the need for I.T. Act, 2000. The advent of the computers and trade being carried on the internet, gave rise to a number of crimes being conducted. The information technology tools were used to commit crimes. Therefore there was a need to have a law to govern trade in the virtual world. This gave the need to pass the Information Technology Act of 2000.

According to the National Crime records Bureau, the following table illustrates the offences that were listed in the year 2016 accross India.

Table 2.2 : Offences involving Computer as Medium / Target

S. No.	State/UT	SLL Offences (Involving Computer as Medium/Target)				Total Offences Under Cyber Crimes			
		Copy Right Act	Trade Marks Act, 1999	Other SLL Offences	Total Cyber Crimes under SLL	IT Act	IPC	SLL	Total Cyber Crimes
1	2	3	4	5	6	7	8	9	10
STATES:									
1	Andhra Pradesh	35	0	0	35	396	185	35	616
2	Arunachal Pradesh	0	0	0	0	3	1	0	4
3	Assam	0	0	0	0	696	0	0	696
4	Bihar	0	0	0	0	299	10	0	309
5	Chhattisgarh	0	0	0	0	55	35	0	90
6	Goa	0	0	0	0	21	10	0	31
7	Gujarat	16	0	1	17	98	247	17	362
8	Haryana	7	1	0	8	386	7	8	401
9	Himachal Pradesh	8	0	0	8	23	0	8	31
10	Jammu & Kashmir	7	0	0	7	20	1	7	28
11	Jharkhand	0	0	0	0	259	0	0	259
12	Karnataka	7	0	0	7	1091	3	7	1101
13	Kerala	15	0	0	15	218	50	15	283
14	Madhya Pradesh	9	0	0	9	194	55	9	258
15	Maharashtra	12	1	2	15	359	2006	15	2380
16	Manipur	3	0	0	3	2	6	3	11
17	Meghalaya	0	0	0	0	39	0	0	39
18	Mizoram	0	0	0	0	1	0	0	1

S. No.	State/UT	SLL Offences (Involving Computer as Medium/Target)				Total Offences Under Cyber Crimes			
		Copy Right Act	Trade Marks Act, 1999	Other SLL Offences	Total Cyber Crimes under SLL	IT Act	IPC	SLL	Total Cyber Crimes
1	2	3	4	5	6	7	8	9	10
19	Nagaland	0	0	0	0	2	0	0	2
20	Odisha	0	0	0	0	25	292	0	317
21	Punjab	5	0	0	5	95	2	5	102
22	Rajasthan	3	0	0	3	625	313	3	941
23	Sikkim	0	0	0	0	1	0	0	1
24	Tamil Nadu	14	0	0	14	104	26	14	144
25	Telangana	0	0	0	0	538	55	0	593
26	Tripura	0	0	0	0	3	5	0	8
27	Uttar Pradesh	2	0	0	2	2562	75	2	2639
28	Uttarakhand	0	0	0	0	58	4	0	62
29	West Bengal	38	0	0	38	349	91	38	478
TOTAL STATE(S)		181	2	3	186	8522	3479	186	12187
UNION TERRITORIES:									
30	A & N Islands	0	0	0	0	2	1	0	3
31	Chandigarh	0	0	0	0	17	9	0	26
32	D&N Haveli	0	0	0	0	1	0	0	1
33	Daman & Diu	0	0	0	0	0	0	0	0
34	Delhi UT	0	0	0	0	69	29	0	98
35	Lakshadweep	0	0	0	0	0	0	0	0
36	Puducherry	0	0	0	0	2	0	0	2
TOTAL UT(S)		0	0	0	0	91	39	0	130
TOTAL (ALL INDIA)		181	2	3	186	8613	3518	186	12317

Col.6 = Col.3 + Col.4 + Col.5
Col 10=Col.7 + Col.8 + Col.9

TABLE 9A.2(iii) PAGE 1 OF 1

Source : NCRB

Summary

- Cyber law is important because it touches almost all aspects of transactions and activities on and concerning the Internet, the World Wide Web and Cyberspace. Initially it may seem that Cyber laws are a very technical and that it does not have any bearing to most activities in Cyberspace. But the actual truth is that nothing could be further than the truth. Whether we realize it or not, every action and every reaction in Cyberspace has some legal and Cyber legal perspectives.
- Cybercrimes can involve criminal activities that are traditional in nature, such as fraud, forgery, theft, mischief and defamation all of which are subject to the Indian Penal Code. The abuse of computers has also given birth to a range of new age crimes that are addressed by the Information Technology Act, 2000
- Cyber law is vital because it touches almost all aspects of transactions and behaviour on and concerning the Internet, the World Wide Web and Cyberspace. Primarily it may seem that Cyber laws is a very technical field and that it does not have an attitude to most activities in Cyberspace. But the actual fact is that nothing could be further than the truth. Whether we realize it or not, every work and every reaction in Cyberspace has some legal and Cyber legal perspectives.
- Cyberlaws prevent or reduce large scale damage from Cybercriminal activities by protecting information access, privacy, communications, intellectual property (IP) and freedom of speech related to the use of the Internet, websites, email, computers, cell phones, software and hardware, such as data storage devices.
- Groups of cyber criminals include script kiddies, scammers, phishers, political/religious/commercial groups, insiders, advanced persistent threat agents, children and adolescent age between 8-18 years, professional hackers/crackers, disgruntled employees and cyber abuser.
- There are many issues faced by the development of the internet and corresponding crimes arising from there which includes:-domain name, intellectual property rights, e-commerce, encryption, electronic contracts and online banking.
- The IT Act 2000 was conceived to regulate electronic commerce. As the preamble of the act states that the act was framed to provide legal acknowledgement for transaction made over electronic means and any other means of electronic communication. The necessary changes were made to harmonize the existing laws.

- A digital signature is an electronic, encrypted, stamp of authentication on digital information such as email messages, macros, or electronic documents. A signature confirms that the information originated from the signer and has not been altered.
- Signing certificate is used to create a digital signature, you need a signing certificate, which proves identity. When you send a digitally-signed macro or document, you also send your certificate and public key. Certificate are issued by a certification authority, and like a driver's license, can be revoked. A certificate is usually valid for a year, after which, the signer must renew, or get a new, signing certificate to establish identity.
- Digital signature assures authenticity, integrity, non-repudiation and notarization and the content creator must digitally sign the content by using a signature by satisfying the criteria such as the valid digital signature; the certificate associated with the digital signature is current (non expired). The signing person or organization, known as the publisher; is trusted and the certificate associated with the digital signature is issued to the signing publisher by a reputable certificate authority (CA).

Check your Answers

Explain the following:

- a. The need for cyber law
- b. Digital signature
- c. Offences under IT Act
- d. Certifying Authority

LESSON - 3

THE INVESTIGATIVE PROCESS

After reading this lesson you will be able to understand the following:

- The Investigative Process
- Case Scenario
- The role of digital evidence
- Investigative methodology
- Investigative reconstruction
- Modus Operandi, Motive and Technology
- Digital Evidence in the courtroom

Structure

- 3.1 The Investigative Process**
 - 3.1.1 Case scenario**
 - 3.1.2 The role of digital evidence**
 - 3.1.3 Investigative Methodology**
- 3.2 Investigative Reconstruction**
- 3.3 Modus Operandi and Motive and Technology**
- 3.4 Digital Evidence in the Courtroom**

3.1 The Investigative Process

The goal of any investigation is to uncover and present the truth. The methodology involved in this lesson is designed to assist in development of case management tools, standard operating procedure (SOPs) and final investigative report. This methodology has grown out of experience, discussion in the field and is believed to be sufficient in scope. However, every investigation is unique and can bring unforeseen challenges. This methodology should not be viewed as end-point but rather a framework or a foundation upon which to build. The investigative process is part of a larger methodology most often associated with courts of law as illustrated in figure 3.1.

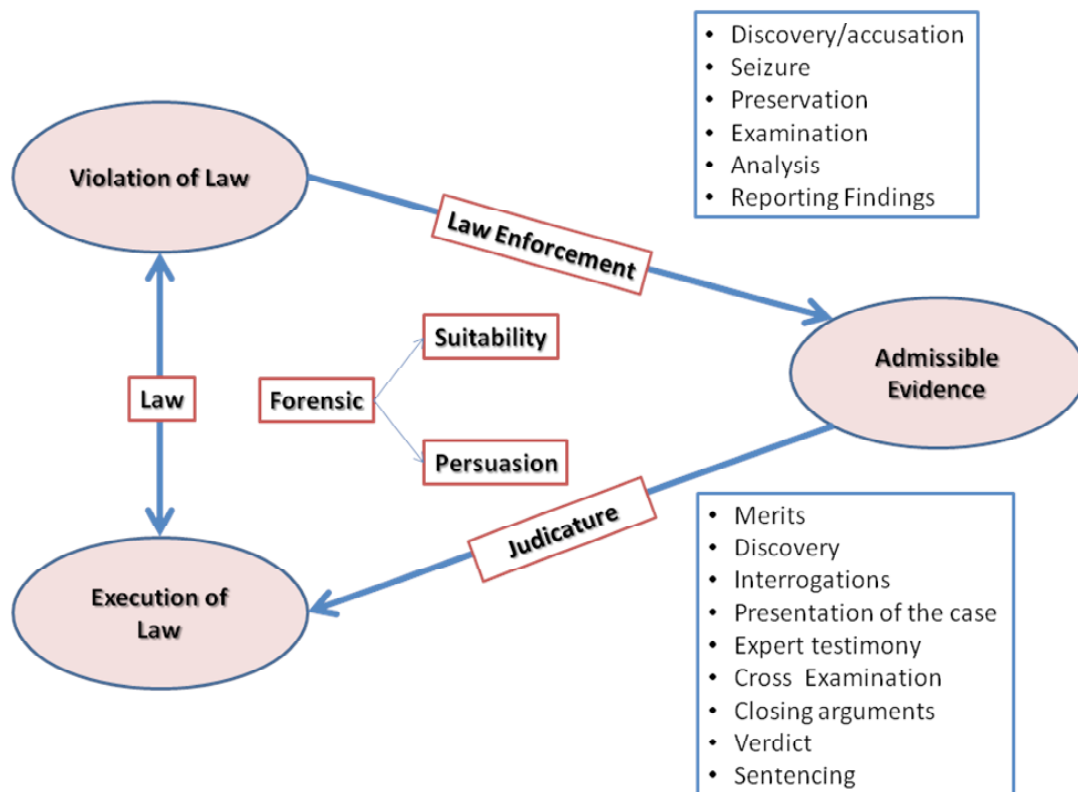


Figure 3.1: Overview of simplified investigative methodology and process involved

The process of determining the fraudulent activity has occurred and if punitive measures are warranted is complex and goes beyond investigative steps and is normally referred to a “Forensics”. Forensics means a characteristic of evidence that satisfies its suitability for admission as fact and its ability to persuade based upon proof.

The investigative process begins with an accusation and progresses through evidence handling to a clear and precise explanation of facts and techniques in expert testimony. This linear representation is useful for structuring procedures and final report that describes each step of an investigation to the decision makers.

Trained experienced investigators will begin asking themselves a series of questions aimed at deciding if a crime or infarction has actually occurred. The answer to these questions will help determine whether or not a full investigation will proceed or if valuable and limited resources are better applied to other matters.

Example:

- 1) A log file may indicate that an employee misused a system, but he repudiates adamantly, a digital forensicator should carefully examine the logs for signs of error.
- 2) When large amount of data are missing on a computer and an intruder is suspect, the digital forensicator should determine if the damage is more consistent with disk corruption than an intrusion.

3.1.1 Case scenario

In a suicide case, suicide note was computer. It raised a concern because it had a creation date after the victim's death. Upon analysis of the computer hard disk, the reason for the time mismatch was ascertained due to the incorrect clock setting and the note was indeed before the suicide.

When questions are answered affirmatively, the focus shifts towards determining what happened, where, when, how, who was involved and why. The process by which digital evidence is uncovered and applied to these issues is composed of several steps each employing a set of protocols, proven methods and in some cases trusted tools. More importantly the success of this process depends upon the experience and skill of the investigators, evidence examiners and crime scene technicians who must collaborate to fit the pieces of evidence together and develop a convincing account of the offense.

The effectiveness of the investigative process depends upon high level of objectivity applied to all stages. Some cases and the nature of evidence uncovered will take the digital forensicators to emotional limits, testing their resolve. Quite often they have to investigate cases that involve brutal abuse of innocent victims inciting distraught individuals and communities to strike out at first available suspect. A good investigator can remain objective in most trying situations.

As experience increases and methods employed are verified the accuracy of these predictions may improve. Conjecture based upon experience has its place in effective triage but should not be relied upon to the exclusion of rigorous investigative measures. The investigative process demands that each case be viewed as unique with its own set of circumstances and exhibits. Letting the evidence speak for itself is particularly important when offenders take steps to misdirect investigators by staging a crime scene or concealing evidence.

While preparing for final step of investigative process that is “the decision” or “the verdict” that the discrepancies it is important to keep in mind scientific and legal truth may arise out of lack of understanding on the part of the decision makers. This is different from scientific peer review, where reviewers are qualified to understand and comment on the relevant facts and methods with credibility. When technical evidence supporting a set of reviewers who are unfamiliar with methods used, misunderstandings and misconceptions may result. In order to avoid such misconceptions a clear presentation of the findings is absolutely necessary when the investigative process and evidence uncovered to support prosecution must be presented clearly to the courts to enable the decision makers to decide.

3.1.2 The role of digital evidence

One of the main goals in an investigation is to attribute the crime to its perpetrator by uncovering compelling links between the offender, victim and crime scene. Witness may identify a suspect but evidence of an individual's involvement is usually more compelling and reliable. According to Locard's Exchange principle – anyone, or anything, entering a crime scene takes something from the crime scene and leaves something of themselves behind when they leave. This is illustrated in figure 3.2. In the physical world the offender might inadvertently leave fingerprints or hair at the scene of crime and take a fiber from the scene. The same applies to electronic crime scene as well. This type of exchange principle produces evidence in either one of the two general categories:

- The evidence that fit in the group called class characteristics – are common traits in similar items
- The exhibits that fall under the category called individual characteristics – are more unique and be linked to a specific person or activity with greater certainty



Figure 3.2: Locard's Exchange Principle

The table 3.1 illustrates example of class and individual characteristics in physical realm and digital realm.

Table 3.1: Class and individual characteristics in physical realm and digital realm

Physical realm	Digital realm
Class Characteristics	
Shoe print left near the window at a crime scene – forensic analysis of the shoe print impressions reveal the make and model of the shoe placing it in the class of all shoes with same make and model. Therefore if a suspect possession of a pair with the same manufacturer and model, a tenuous circumstantial link can be made between the suspect and wrong doing.	Digital equipments and their attributes can be classified as class and individual characteristics. Scanners, printers, skimmers may exhibit discernable artifacts that lead to common class characteristics allowing the identification of an EPSON, CANON or a Mini123 device. More conclusive individual characteristics are rare but not impossible.
Individual Characteristics	
If forensic analysis uncovers detailed wear patterns in the shoe prints and finds identical wear patterns in one of the shoe's of the suspect, a much stronger link is possible. The margin of error has just been significantly reduced by the discovery of an individual characteristic making the link much circumstantial and harder to refute	In digital realm, it is more virtual and less tangible space. With the philosophy of openness and anonymity associated with the internet, the very notion of individual identity is almost at odds. However similar exchange principle occurs in digital realm. Example: (1) Data from offender computer recorded by a server or data from server stored on offenders computer. Such links have been used to demonstrate that an individual was involved. (2) Browsing of the web – if an individual sends a threatening message using email service – the browsing details, stored files, exchange of the threatening mails along with date and time can provide information (3) Unique marks on a digitized photo might be used to demonstrate that the suspect's scanner, digital camera was involved.(4) Logs from FTP servers confirm that the files were downloaded to the suspects IP address at the time in question.

These examples describe suspected offenses and allude to types of locations of potential evidentiary material.

3.1.3 Investigative Methodology

The investigative process depicted as a sequence of ascending stairs in Figure 3.3, is structured to encourage a complete, rigorous investigation ensure proper evidence handling and reduce the chance of mistakes created by preconceived theories and potential pitfalls. This process applies to criminal investigations as well as military and corporate inquiries dealing with policy violations or system compromise.

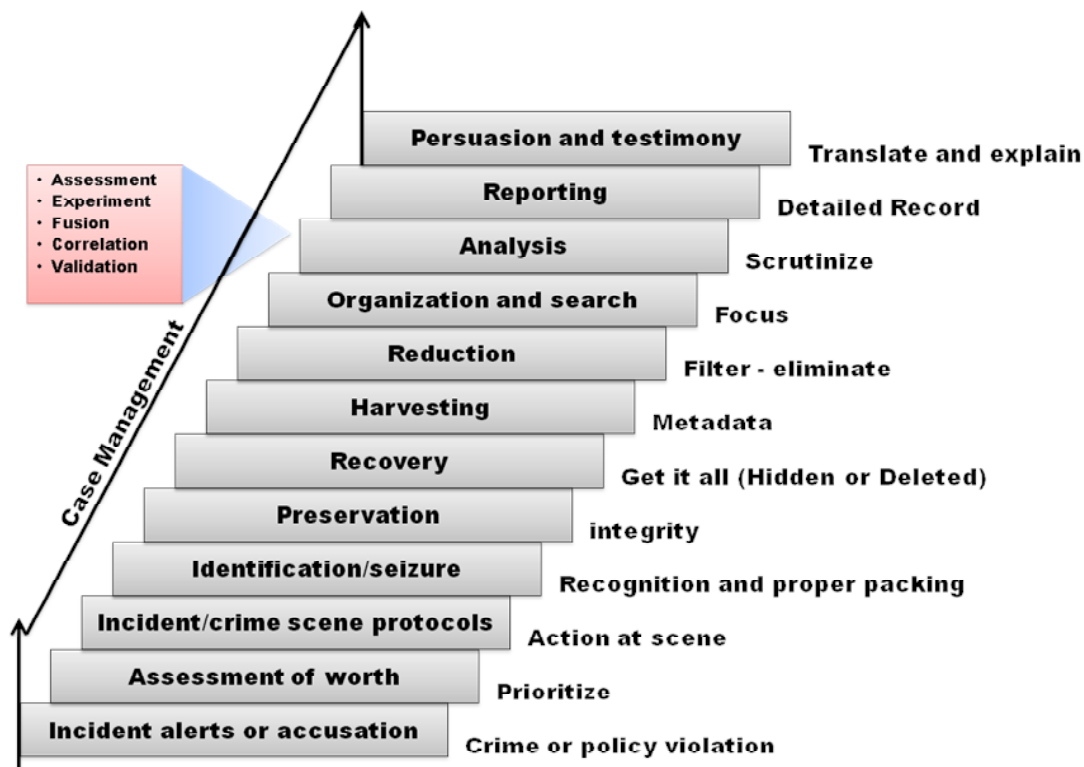


Figure 3.3: Categories of Investigative Process Model
(Source: *Digital Evidence and Computer Crime* by Eoghan Casey)

The investigators and forensic examiners work together to scale these steps from bottom to top in a systematic, determined manner in order to present the evidence to the trier-of-fact by a way of persuasion and testimony in order to make a decision. The judge, jury, military operation communities or resource managers, who will rely on the confidence and accuracy of the

information, based on the merits of the evidence, before taking decisions. Two items of particular note and special importance stand out and they are:

- (1) Case Management plays a vital role and spans across all steps in the process model. It provides stability and enables investigators effectively to tie all relevant information together in order to prosecute.
- (2) Analysis phase of investigative process - a scientific method beginning with fact gathering and validation, proceeding to hypothesis formation and testing the actively seeking evidence that disproves the hypothesis and revising conclusions as new evidence emerges.

In general this model affords investigators and examiners a logical flow of events that, taken together seek to provide:

- (1) Acceptance – the steps and methods have earned professional consensus
- (2) Reliability – the methods employed can be proven to support findings
- (3) Repeatability – the process can be applied by all independent of time and place
- (4) Integrity – the state of evidence is proven to be unaltered
- (5) Cause and effect – logical connection between suspected individuals, events and exhibits
- (6) Documentation- recordings essential for testimonial evidence individuals, and exhibits

All six tenets have a common purpose – to form the most persuasive arguments possible based on facts, not supposition, and to do so considering the legal criteria for admissibility.

3.2 Investigative Reconstruction

Reconstructing human behavior from physical evidence is a multi-dimensional jigsaw puzzle. Pieces of the puzzle are missing, damaged and some are even camouflaged. The puzzle pieces come in seemingly incompatible data types – some are visual, some are in such microscopic form that it takes days of specialized analysis to show their existence and in some cases evidence is intangible, such as oral testimony. But practitioners of these two disciplines, each for totally different reasons, sit at their desks and doggedly persist in completing these puzzles – archaeologist and forensic scientists. (Scott and Corner, 1997).

Crime is not always committed in a straight forward or easily decipherable manner. Nor it is always possible for the investigator to prove what they suspect occurred with the evidence left behind. A crime can involve multiple victims, multiple crime locations and offenders engaging in various degrees of planning, aggression, fantasy, concealment, victim response and a multitude of other behavioral interactions. Only the offender knows the full story or their involvement in a crime and it can be difficult to establish their associated motives, movements, interactions, sequences and timing using the fragmentary clues.

Reconstruction refers to the systematic process of piecing together evidence and information gathered during an investigation to gain a better understanding of what transpired between the victim and the offender during the crime. A core tenet of this process is that when they commit a crime, criminals leave an imprint of themselves at the scene. In conventional crime footprints, fingerprints and blood patterns are clear examples of imprints left by an offender at the scene. If Locard's principle is recalled, which states that when any two objects come in contact there is a transfer of material (cross transfer). Reconstruction involves taking physical imprints a step further, using them to infer offense related to behavior or behavioral imprints. Such behavioral imprints established as a particular crime scene can be used to provide who did what, when and how. Taken together, a series of connected behavioral imprints can be used to establish

- an offender's modus operandi
- their knowledge of crime
- their knowledge of victim
- even their motivation

This is true in digital crime scenes as it is in corporeal world ; digital crime scene contains behavioral imprints . For example, the words that the offender uses on the internet may disclose precious details.

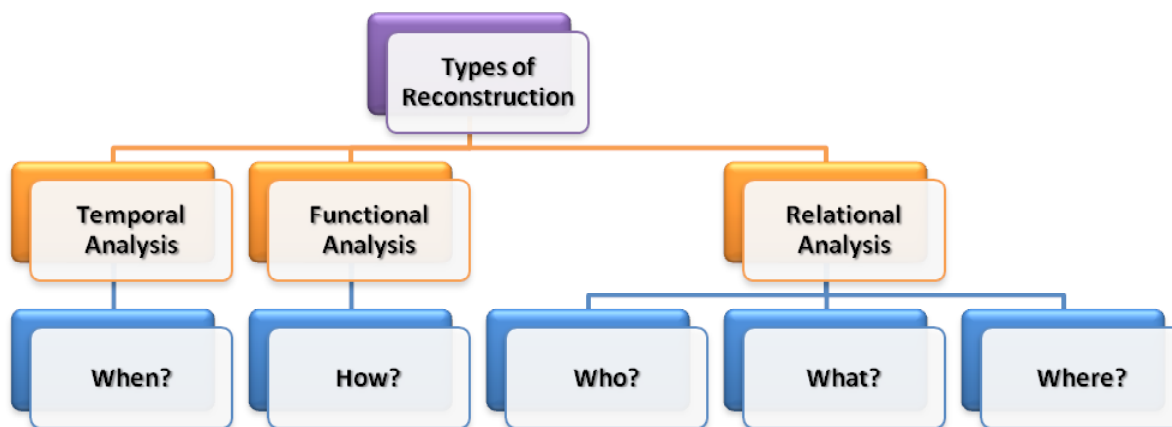
- the tools that the offender uses online can be significant
- how offender conceals their identity and
- criminal activity can be telling some intruder may use an online tool to automate certain aspects of their modus operandi. Questions may arise that require investigation.

- Did the offender erase all signs of tool kit?
- Was the offender a skilled enough not to need a toolkit?

Therefore creating as complete reconstruction of the crime as possible using the available evidence as a crucial stage in investigation. In addition helping investigative reconstruction it can be used to:

- Develop and understanding of case facts and how they relate
- Forms the investigation by exposing important features and fruitful avenues of inquiry
- Locate concealed evidence
- Develop suspects with motive, means and opportunity
- Establish evidence of insider or intruder knowledge
- Anticipate intruder actions and assess potential for escalation
- Link related crimes with the same behavioral imprints
- Give insight into offender fantasy, motives, intents and state of mind
- Case presentation in court

Figure 3.4 : Types of Reconstruction



Temporal Analysis

- Creating a chronological list of events to gain insight into what happened and who were involved in the crime.

- Such timeline of events can help an investigation to
- Identify patterns and anomalies,
- Shedding light on the crime and leading to other source of evidence.
- Reconstructing the sequence of events can be very useful for digital evidence investigation
- Creating a histogram of times can reveal a period of high activity that deserves closer inspection
- Repeated patterns and deviations from those regular events can be achieved by arranging times in a grid with days on the horizontal axis and hours on the vertical axis.

Example:

- The time a file was created
- The time a person logged on using private password
- A computer log file with a large gap or entries that are out of sequence may be an indication that the log was tampered with

Relational analysis

- Determining where an object or person was in relation to other objects or people is very useful when investigating crimes involving networked computers.
- In large computer fraud cases, thousands of people and computers can be involved, making it difficult to keep track of many relationships between objects.
- Creating a diagram depicting the associations between the people and the computers can clarify what had occurred.

Example:

- Link analysis may reveal how the offender is obtaining information about the victim in a cyber stalking case.
- Network traffic logs show connections between the compromised machine and the accounting server in a computer intrusion scenario.

Functional Analysis

- Functional testing of the original hardware to ensure that the system was capable of performing basic actions.
- Purpose is to reconsider all possible explanations for a given set of circumstances.
- Functional testing can be performed to determine if the suspect's computer was capable of downloading and displaying the graphic files that are presented as incriminating evidence.
- In Non-Technical cases determining how a given computer or application functions can shed light on available digital evidence can help investigators assess the reliability and meaning of the digital evidence.

3.3 Modus Operandi (MO): Motive and Technology

MO is Latin term that means, "a method of operating". It refers to the behaviors that are committed by a criminal for the purpose of successfully completing an offense for the purpose of successfully completing an offense. A criminal's MO reflects how they committed their crimes. A criminal's MO has traditionally been investigative, relevant for the case linkage, efforts of law enforcement.

A criminal's MO consists of learned behaviors that can evolve and develop over time. An offender becomes more experienced, sophisticated and confident over time.

Three purposes of an offender's MO behaviour:

- Protects offender's identity.
- Ensures the successful completion of the crime.
- Facilitates the offender's escape.

3.4 Digital Evidence in the Courtroom

3.4.1 Admissibility

The concept of admissibility is a simple one. Courts need to determine whether evidence is "safe" to put before a jury and will help provide a solid foundation for making a decision in the case.

In practice, admissibility is a set of legal tests carried out by a judge to assess an item of evidence.

3.4.2 Authentication of Digital Evidence

To demonstrate that digital evidence is authentic, it is generally necessary to satisfy the court that it was acquired from a specific computer and/or location, that a complete and accurate copy of digital evidence was acquired, and that it has remained unchanged since it was collected.

3.4.3 Reliability of Digital Evidence

There are two general approaches to assessing whether digital evidence can be relied upon in court.

The first approach is to focus on whether the computer that generated the evidence was functioning normally, and the other approach is to examine the actual digital evidence for evidence of tampering and other damage.

3.4.4 Best Evidence

When dealing with the contents of writing, recording, or photograph, courts sometimes require the original evidence.

The original purpose of this rule was to ensure that decisions made in court were based on the best available information.

3.4.5 Presenting Digital Evidence

Digital investigators are commonly asked to testify or produce a written summary of their findings in the form of an affidavit or expert report.

Testifying or writing a report is one of the most important stages of the investigative process because, unless findings are communicated clearly in writing, others are unlikely to understand or make use of them.

Summary

- Digital evidence can be used to answer fundamental questions relating to a crime, including what happened when (sequencing), who interacted with whom (linkage), the origination of a particular item (evaluation of source), and who was responsible (attribution).
- The forensic analysis process involves taking factual observations from available evidence, forming and testing possible explanations for what caused the evidence, and ultimately developing deeper understanding of a particular item of evidence or the crime as a whole.
- Forensic analysis of digital evidence depends on the case context and largely relies on the knowledge, experience, expertise, thoroughness, and in some cases the curiosity of the practitioner performing the work. Although every forensic analysis will have differing aspects based on the dataset, objectives, resources, and other factors, the underlying process remains fundamentally the same.
- Locard's Exchange Principle is often cited in forensics publications. "Every contact leaves a trace". In the cyber world, the perpetrator may or may not come in physical contact with the crime scene, thus, this brings a new facet to crime scene analysis.
- To illustrate the application of Locard's Exchange Principle to a cyber crime, we take the example to identity theft where someone's identity is stolen and the perpetrator steals the identity through the use of a Trojan horse and keyboard logger on the victim's computer. One could contend that during this type of cyber crime Locard's Exchange Principle does not apply. The rationale is that because a human is not at the crime scene there is no trace evidence from the human on the computer or digital media at the scene. However, in actually there may be lots of digital evidence such as the Trojan horse itself, changed passwords, digital logs, and so on. Thus, in this example, there is a trace at, to, and from the scene. It may involve finding the trace evidence at other physical locations than just the one scene of the crime. The key logger could be added software or hardware or both, but in both cases it remains behind for an investigator to discover.
- Reconstruction is systematic process of piecing together evidence and information gathered during investigation to gain better understanding of what transpired between the victim and offender during the crime. There are three types of reconstruction.

They are temporal (chronology/timeline), relational (in relation to) and functional (capability of functional)

- Modus operandi, (Latin: "operating method") abbreviation MO, it is a distinct pattern or manner of working that comes to be associated with a particular criminal. Criminologists have observed that, whatever his speciality-burglary, auto theft, or embezzling-the professional criminal is very likely to adhere to his particular way of operating. If, for example, a burglar begins his career by entering houses from the roof, he will, in all probability, continue this method for as long as he is able to work. Some burgains become so attached to their *modus operandi* that they burglarize the same places or people again and again.
- **The original evidence is the best evidence.**

Check your Answers

Explain the following:

1. Investigative process
2. Best evidence
3. Reconstruction
4. Digital Evidence in court room.

LESSON - 4

UNDERSTANDING INFORMATION

After reading this lesson you will be able to understand the following

- Methods of storing data
- Number systems
- Character codes
- Record structures
- File formats and file signatures
- Structure and Analysis of Optical Media Disk Formats
- Recognition of file formats and internal buffers
- Extraction of forensic artifacts
- Understanding SSD Devices

Structure

4.1 Methods of Storing Data

4.1.1. Primary Storage

4.1.2. Secondary Storage

4.1.3. Tertiary Storage

4.1.4. Storage Media

4.1.4.1. Paper

4.1.4.2. Semiconductor

4.1.4.3. Magnetic

4.1.4.4. Optical

4.2. Number systems

4.2.1. Decimal

4.2.2. Binary

4.2.3. Octal

4.2.4. Hexadecimal

4.3. Character codes

4.4. Record structures

4.4.1. Unordered

4.4.2. Ordered

4.4.3. Structured

4.5. File formats and file signatures

4.6. Structure and Analysis of Optical Media Disk Formats

4.6.1. Optical Media

4.7. Extraction of forensic artifacts

4.7.1. Recorder Identification of CD-R(W)

4.7.2. Unique ID Field of DVD-R(W)

4.7.3. File and MAC time analysis

4.7.4. The imaging process

4.7.5. Deleted media

4.7.6. RID (CD)

4.8. Understanding SSD Devices

4.1. Methods of storing data

4.1.1. Primary Storage

Main Memory (or) Internal Memory

Directly accessible to the CPU through Memory Bus (Address Bus and Data bus)

Memory Management Unit acts as interface between CPU and RAM

(recalculates actual memory address and provides abstraction of virtual memory)

Volatile Primary Storage: RAM

Non-Volatile Primary Storage: ROM, BIOS

Data accessed in nanoseconds

4.1.2. Secondary Storage**External Storage (or) Auxiliary Storage**

Not directly accessible to CPU

Non-Volatile

CPU uses Input/output channels to access data

Data accessed in milliseconds

Hard Disk, CD/DVD, Flash memory (e.g. USB flash drives or keys), floppy disks, magnetic tape, paper tape, punched cards, standalone RAM disks, and lomegaZip drives.

4.1.3. Tertiary Storage

Tertiary storage involves Robotic mechanism which will mount and dismount removable mass storage device according to system's demand near line. Storage Data is not immediately available, but can be made online quickly without human intervention.

4.1.4. Storage Media**4.1.4.1. Paper**

In form of Paper tape (or) Punch Cards. Information was recorded by punching holes into the paper or cardboard medium and was read mechanically to determine whether a particular location on the medium was solid or contained a hole.

4.1.4.2. Semiconductor

- Uses semiconductor based Integrated Circuits to store information
- Contains millions of tiny transistors and capacitors
- Primary Storage typically consists of volatile semiconductor memory (DRAM)
- Offline Storage typically consists of non-volatile semiconductor memory (Flash Drives, SSD)

4.1.4.3 Magnetic

- Uses different patterns of magnetization on a magnetically coated surface to store information
- Non-volatile, accessed using one or more read/write heads which has one or more recording transducers
- Head, Medium (or) both must move relative to other to access data
- Magnetic Disk (Floppy, HDD)
- Magnetic Tape
- Carousel memory (magnetic rolls)

4.1.4.4 Optical

Stores information in deformities on the surface of a circular disc and reads this information by illuminating the surface with a laser diode and observing the reflection.

Non-volatile

Deformities may be permanent (read only), formed once (write once), reversible (read/write) CD, DVD, BD, UDO

4.2 Number systems

4.2.1 Decimal

- Base 10
- Uses 0,1,2,3,4,5,6,7,8,9
- Each position represents a specific power of the base (10)
- 1234 is represented as
- $(1 \times 1000) + (2 \times 100) + (3 \times 10) + (4 \times 1)$
- $(1 \times 10^3) + (2 \times 10^2) + (3 \times 10^1) + (4 \times 10^0)$
- $(1000 + 200 + 30 + 4)_{10}$
- $(1234)_{10}$

4.2.2 Binary

- Base 2
- Uses 0, 1
- Each position represents a specific power of the base (2)

Step	Binary Number	Decimal Number
Step 1	10101 ₂	$((1 \times 2^4) + (0 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0))_{10}$
Step 2	10101 ₂	$(16 + 0 + 4 + 0 + 1)_{10}$
Step 3	10101 ₂	21 ₁₀

Fig 4.1: Binary

4.2.3. Octal

- Base 8
- Uses 0,1,2,3,4,5,6,7
- Each position represents a specific power of the base (8)

Step	Octal Number	Decimal Number
Step 1	12570 ₈	$((1 \times 8^4) + (2 \times 8^3) + (5 \times 8^2) + (7 \times 8^1) + (0 \times 8^0))_{10}$
Step 2	12570 ₈	$(4096 + 1024 + 320 + 56 + 0)_{10}$
Step 3	12570 ₈	5496 ₁₀

Fig 4.2 : Octal

- Taking the byte as the basic unit of memory interpret max & min – can be determined
- The pattern 00000000 clearly gives a value of 0
- The pattern 11111111 gives:

$$1 \times 2^7 + 1 \times 2^6 + 1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$
 this is equal to 255
- The range of whole numbers that can be represented in a single byte

4.2.4. Hexadecimal

- Base 16
- Uses 0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F
- Each position represents a specific power of the base (16)

Step	Hexadecimal Number	Decimal Number
Step 1	19FDE ₁₆	$((1 \times 16^4) + (9 \times 16^3) + (F \times 16^2) + (D \times 16^1) + (E \times 16^0))_{10}$
Step 2	19FDE ₁₆	$((1 \times 16^4) + (9 \times 16^3) + (15 \times 16^2) + (13 \times 16^1) + (14 \times 16^0))_{10}$
Step 3	19FDE ₁₆	$(65536 + 36864 + 3840 + 208 + 14)_{10}$
Step 4	19FDE ₁₆	106462 ₁₀

Fig 4.3: Hexadecimal

4.3. Character codes

Character encoding is used to represent a repertoire of characters by some kind of encoding system.

Depending on the abstraction level and context, corresponding code points and the resulting code space may be regarded as bit patterns, octets, natural numbers, electrical pulses, etc.

Character encoding using internationally accepted standards permits worldwide interchange of text in electronic form.

4.4. Record structures

4.4.1. Unordered

- Stores the records in the order they are inserted
- Offers good insertion efficiency, but inefficient retrieval times

4.4.2. Ordered

- Stores the records in order
- May have to rearrange or increase the file size when a new record is inserted, hence low insertion efficiency
- Provides more efficient retrieval as the records are pre-sorted

4.4.3 Structured

- Heap Files
- Lists of unordered records of variable size
- Insert efficient, Retrieval inefficient
- Hash Buckets
- Hash functions calculate the address of the page in which the record is to be stored based on one or more fields in the record
- Efficient for exact matches on key field, not suitable for range retrieval
- Collisions are possible, so collision detection and restoration is required
- • B+ Trees
- Most commonly used in practice
- Full index so data file does not have to be ordered
- Versatile data structure – sequential as well as random access
- Supports exact, range, part key and pattern matches efficiently

4.5. File formats and file signatures

A file format is a standard way that information is encoded for storage in a computer file. It specifies how bits are used to encode information in a digital storage medium. File formats may be either proprietary or free and may be either unpublished or open

Filename Extension

A filename extension is an identifier specified as a suffix to the name of a computer file. The extension indicates a characteristic of the file contents or its intended use.

that skipped over header data to the start of executable code”

“it is very desirable that files should generally present themselves as self-describing objects from which an application launcher or navigation tool can readily deduce both their uses and at least some of the semantics of their contents”

Graphic Formats

<

GIF Format- Graphic Interchange Format

1987 – GIF87a

1989 – GIF89a

- The file signature (or magic number)
- “47 49 46 38 39 61” or “GIF89a”
- followed immediately by the logical screen descriptor block, starting at address 06H, which specifies the width - of the logical screen in pixels as “40 01” (translated from little endian this becomes 0140H = 320 decimal)
- height of the logical screen in pixels as “d6 00” (again, translated from little endian, this is 00d6H = 214 decimal) [at address 08h]
- shows at address 0aH that there is a global colour table and that the colour resolution is 8 bits for each primary colour in the colour table
- The global colour table starts at address 0dH with “00 00 00” and consists of 256 triples,

- each triple representing the red, green, blue (RGB) intensity values of one of the 256 possible colours in the colour palette
- In this case, the colours in the palette are all greyscale colours
- We can tell this because the intensities of each of the R, G and B values for each colour triple are identical, thus resulting in a palette of grey/black tones.
- The second segment, starting with address 300H, shows the last entry of the colour table at address 300aH with R, G, B values “ff ff ff”
- The next value, at address 300dh, is “2c” and this signals the start of a GIF picture
- This is the first byte of the image descriptor block which defines at address 30eH the left coordinate of the picture in pixels, 0000H, at address 310H the top coordinate of the picture in pixels, 0000H, at address 312H the width of the picture in pixels (again, to be translated from little endian, thus 0140H = 320 decimal) and at address 314H the height of the picture in pixels (also, to be translated from little endian, 00d6H = 214 decimal).

Portable Network Graphic

Gander - I:\EXAMPLES\ZEON.PNG																															
File Edit Display Search Filters Help																															
	00	01	02	03	04	05	06	07	08	09	0a	0b	0c	0d	0e	0f	01	23	45	67	89	a	b	c	d	e	f				
00	89	50	4e	47	0d	0a	1a	0a	00	00	00	0d	49	48	44	52	-PNG----				----IHDR										
10	00	00	01	40	00	00	00	d6	08	00	00	00	00	85	b3	04	---@----				-----										
20	9c	00	00	ad	f2	49	44	41	54	78	da	2c	bd	09	d0	65	----IDATx,---				---e										
30	e9	59	1e	f6	ed	e7	9c	7b	ef	bf	74	f7	f4	f4	f4	f4	-Y--<--t---				-----										
40	cc	f4	ec	1a	ad	23	90	40	08	89	2d	08	59	12	92	90	----#@--Y---				-----										
50	1d	07	1c	5c	76	91	00	71	ca	49	39	81	4a	62	5c	65	--\u--q-19-Jb\				e										
60	5c	15	bb	62	53	76	95	5d	b6	29	9c	32	10	53	81	18	\--bSu-l->2-S--				--										
70	43	11	6c	e2	62	df	57	49	a0	05	69	24	66	46	1a	cd	C-l-b-WI-i\$FF--				--										
80	3e	d3	eb	74	ff	cb	bd	f7	9c	f3	ad	79	9e	f7	8e	2c	>--t-----y---				--										
90	e4	ee	bf	ff	7b	ee	77	be	ef	5d	9e	e7	dd	3e	fd	e3	----<-w-l-->--				--										

- This file is 44 kilobytes in length.
- The file signature is the first eight bytes and is always “89 50 4e 47 0d 0a 1a 0a”
- Followed by length of data in the chunk 8H- Value = 00 00 00 0d this is in big endian hence it is = 13

- Next follows signature for the chunk in character = IHDR@ ocH
- Next 13 data bytes starting @ 10H
- 00 00 01 40 – in big endian results in width of the image =320 decimal
- Followed by height 00 00 00 d6 = 214 decimal
- Finally, the last five bytes “08 00 00 00 00”, starting from address 18H, indicate an 8 bit per pixel greyscale colour depth.

JPEG Compression

Gander - I:\EXAMPLES\ZEON.JPG																															
File Edit Display Search Filters Help																															
	00	01	02	03	04	05	06	07	08	09	0a	0b	0c	0d	0e	0f	01	23	45	67	89	a	b	c	d	e	f				
00	ff	d8	ff	e0	00	10	4a	46	49	46	00	01	01	01	01	2c				JF	IF										
10	01	2c	00	00	ff	db	00	43	00	05	03	04	04	04	03	05				C											
20	04	04	04	05	05	05	06	07	0c	08	07	07	07	07	0f	0b															
30	0b	09	0c	11	0f	12	12	11	0f	11	11	13	16	1c	17	13															
40	14	1a	15	11	11	18	21	18	1a	1d	1d	1f	1f	1f	13	17				!											
50	22	24	22	1e	24	1c	1e	1f	1e	ff	c0	00	0b	08	00	d6				"\$"	\$										
60	01	40	01	01	11	00	ff	c4	00	1f	00	00	01	05	01	01				@											
130	e7	e8	e9	ea	f1	f2	f3	f4	f5	f6	f7	f8	f9	fa	ff	da															
140	00	08	01	01	00	00	3f	00	dc	b8	42	fb	b7	73	db	8a				?		B		s							
150	c7	9e	3d	b2	9c	fa	f7	a8	5b	8e	78	15	1a	3e	d2	d9				=		[	x		>						
160	39	c9	19	a5	91	f6	ee	38	e4	f6	23	18	a8	a3	b8	56				9		8		#		U					
170	cb	63	3d	fe	95	a9	e1	db	53	25	f7	da	a4	07	6a	03				c		S	%		j						

- The start of image (SOI) marker block, “ff d8”, is required by the JPEG standard to signal the start of a JPEG file and
- the “ff e0” indicates an application (APP0) marker block which, together with the characters “JFIF”, starting at address 06H identify this as a JFIF file
- the file ends with “ff d9”, the end of image (EOI) marker.
- the sequence that starts at address 59H of value “ff c0”
- This is a start of frame marker block which at address 5dH has the bits per pixel value of 08
- At address 5eH is a two-byte word giving the height of the image in pixels “00 d6” (big endian format of 00d6H results in 214 decimal) and at address 60H is a two byte word giving the width of the image in pixels “01 40”(big endian format of 0140H results in 320 decimal)

- The file archive program pkzip has already been mentioned in the context of the LZ compression algorithms.
- File Signature
- File Metadata

File Header

The metadata contained in a file header are usually stored at the start of the file, but might be present in other areas too

Character-based files usually have character-based headers, whereas binary formats usually have binary headers

Used by software for reading or interpreting the file during the loading process and afterwards

Used by an operating system to quickly gather information about a file without loading it all into memory

File Magic Number

A short sequence of bytes (most are 2-4 bytes long) placed at the beginning of the file, used to identify the format of the file

Well-designed magic number test is a pretty sure sign that the file is either corrupt or of the wrong type

A valid magic number does not guarantee that the file is not corrupt or is of a correct type

- **File Checksum**

The result of a hash function over the file contents

To verify the integrity of the file contents against transmission errors or malicious attacks

Included at the end of the file or in a separate file

- **File Structure**

Unstructured:

Raw Memory Dumps

Directly dumping the memory images of one or more structures into the file

Extending and improving this type of file is very difficult unless the memory images have reserved spaces

Specific to one platform or programming language

Chunk Based

- Each piece of data is embedded in a container (chunk) that identifies it
- Chunk's scope is identified by
 - o Start-end markers
 - o Explicit length field
- Fixed requirements of file formats definition
- Tools that do not know certain chunk identifiers simply skip those that they do not understand

Directory Based

- Extensible format, that closely resembles a file system
- File is composed of 'directory entries' that contain the location of the data within the file itself as well as its signatures
- Example – Disk Images (.iso, .dmg)

4.6. Structure and Analysis of Optical Media Disk Formats

4.6.1. Optical Media

An optical disc is designed to support one of three recording types: read-only (e.g.:CD and CD-ROM), recordable (write-once, e.g. CD-R), or re-recordable (rewritable, e.g. CD-RW). Write-once optical discs commonly have an organic dye recording layer between the substrate and the reflective layer.

Rewritable discs typically contain an alloy recording layer composed of a phase change material, most often AgInSbTe, an alloy of silver, indium, antimony, and tellurium.^[4]

Optical discs are most commonly used for storing music (e.g. for use in a CD player), video (e.g. for use in a Blu-ray player), or data and programs for personal computers (PC)

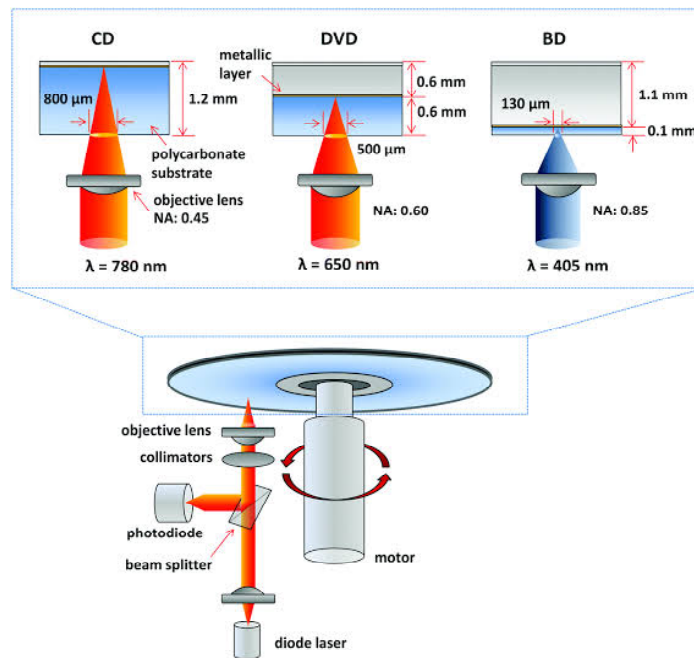


Figure 4.4 : Optical Media

4.7. Extraction of forensic artifacts

4.7.1. Recorder Identification of CD-R(W)

- In 1995, the company Philips gave under the pressure of the music industry, represented by IFPI[®], RIAA[†] and the RIAJ[‡], in their "Orange Book" standard for recordable CDs a possible identification for the tracking of pirated content.
- This Recorder Identification Code (RID) can be written in the Q-sectors of the individual songs or/and in the roll-in/out-blocks of the disc.
- The RID code consists in 3 groups, including the following information: the manufacturer's code, followed by the device type and unique serial number.

4.7.2. Unique ID Field of DVD-R(W)

- Identification for the drive which was used for burning a DVD is defined in ISO/IEC 23912 standard for recordable DVD media.
- This Unique ID Field is written in a specified area previous to the lead-in.
- The size of the Unique ID Field is 32 bytes for the manufacturer, 16 bytes for the serial number and 16 bytes for the model name.
- Disc Definition Structure Field with BD-R(E), moreover it is possible to identify the drive that was used for burning in the Blu-ray Disc standard.
- In a fixed area on the BD is the Disc Definition Structure Field (DDS) for identifying the burning hardware.
- The size is variable, but may not exceed 2048 bytes for storing manufacturer, model and the serial number.
- At rewritable BD-REs, the identifier is located in the Physical Access Control Field (PAC).
- Both fields are located outside the data area of the optical medium.

4.7.3. File and MAC time analysis

The programs X-Ways Forensics, EnCase and FTK are very comprehensive tools for inspecting the file contents of the medium. All these programs use file carving technologies, have filters for different file types and have possibilities for generating hash values. Interesting for a forensic investigation are the MAC times of the files. X-Ways, EnCase and FTK provide this information in tabular form to the individual files.

4.7.4 The imaging process

- The creation of forensic duplicates is a standard procedure in investigations of computer crime.
- It was found that in opposite to the imaging process on hard drives, where the data of the unpartitioned space is taken up in the duplicate, only the data that containing between the border areas is copied in the duplicate.
- The reason can be found in the fact that optical drives will be mounted just as logical drives in the system, however, as physical drives. The choice of the operating system was not relevant in this case.

- The image generation under the operating systems Windows XP, Windows 7 and Linux (with the programs dump device (dd), Nero Burning Rom, FTK Imager, IsoBuster PRO, X-Ways and EnCase) took place and showed identical results.

4.7.5 Deleted media

This section examines the possibilities to reconstruct the contents of a deleted CD-RW, DVD-RW or BD-RE. Numerous tests were performed on this. Erased discs were checked for access under Mac OS, Linux and Windows 7. None of the operating systems allowed direct access to the blank disc. The imaging process by using Windows and Linux broke with an error message. The programs X-Ways, EnCase, FTK, IsoBuster PRO and CD/DVD Diagnostic found no data on the disk.

4.7.6. RID (CD)

- Information can be examined about the used burning hardware on a CD.
- To determine the Recorder Identification Code the open-source tool
- The manufacturer's code, the device type and the serial number could be found.
- Often the serial number is 012345(hex). So only manufacturer and type are clearly ascertainable.

4.8. Understanding SSD Devices

A solid-state drive (SSD) is a solid-state storage device that uses integrated circuit assemblies as memory to store data persistently. It is also sometimes called a solid-state device or a solid-state disk, although SSDs do not have physical disks.

SSDs can use traditional hard disk drive (HDD) interfaces and form factors, or newer form factors and interfaces that have been developed to address specific advantages of the flash memory technology used in SSDs. Traditional interfaces (e.g., SATA and SAS), and standard HDD form factors allow such SSDs to be used as drop-in replacements for HDDs in computers and other devices.

Working of SSD

An SSD is a larger, more complex device that aggregates pools of NAND flash storage, the type of storage also found in MP3 players and digital cameras. Unlike RAM, which doesn't

retain data when the machine shuts off, SSD flash memory is non-volatile, which means data is retained whether the device is powered on or not.

With SSDs, every block of data is accessible at the same speed as every other block, no matter the location. This makes SSDs inherently faster than hard drives, where platters are spinning and drive heads are moving to the right location.

With HDDs, large files can be broken up and tucked into unused nooks and crannies of the drive, and data can be easily updated in place. This allows for efficient use of the total drive capacity. On the other hand, scattered data obviously takes longer to locate, which is why defragmenting a hard drive has become a standard part of device maintenance.

SSDs have a different and bigger problem – SSDs can only write to empty blocks. That's okay when the SSD is new and all the blocks are empty. But over time, as blocks get filled up, overwriting data becomes an issue, because the only way an SSD can update an existing page is to copy the contents of the entire block into memory, erase the block and then write the contents of the old block in addition to the new data. If there are no empty blocks available, the SSD must scan for blocks marked for deletion, but not yet deleted, erase them, and then write the data to the now-erased page. Over time, as the SSD fills up, writing to the drive becomes more complicated and slower.

SSDs use a combination of methods to overcome this problem, including overprovisioning; wear leveling, a type of load balancing for flash cells; and garbage collection, a process that works in the background to delete stale files and optimize the available space, similar to the defrag function in HDDs.

4.9 Summary

- Methods of storing data: Primary Storage, Secondary Storage, Tertiary Storage, Nearline Storage.
- The Storage Media are Paper, Semiconductor, Magnetic, Optical
- The Number Systems Decimal, Binary, Octal, Hexadecimal
- A filename extension is an identifier specified as a suffix to the name of a computer file.

- An optical disc is designed to support read-only, recordable or re-recordable
- The programs X-Ways Forensics, EnCase and FTK are very comprehensive tools for inspecting the file contents of the medium
- The creation of forensic duplicates is a standard procedure in investigations of computer crime.
- This section examines the possibilities to reconstruct the contents of a deleted CD-RW, DVD-RW or BD-RE

4.10 Check your answers

1. Volatile Primary Storage_____
2. Secondary Storage is also known as_____
3. Data accessed in primary storage_____
4. Convert Binary to Decimal :101011
5. Using a _____the RID code in some media could be found with PLScsi.
6. The size of the Unique ID Field is _____bytes for the manufacturer, _____ bytes for the serial number and _____bytes for the model name.
7. Name any four file extension

LESSON - 5

COMPUTER BASICS FOR DIGITAL INVESTIGATORS

After reading this lesson you will be able to understand the following:

- Computer Forensic Fundamentals
- Computer Forensic Services
- Benefits of Professional Forensic Methodology
- Steps taken by computer forensic specialists.

Structure

- 5.1 Computer Forensic Fundamentals**
- 5.2 Computer Forensic Services**
- 5.3 Benefits of Professional Forensic Methodology**
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5.1. Computer Forensic Fundamentals

What is Computer Forensics?

Computer forensics is the process of methodically examining computer media (hard disk, diskettes, taps, etc) for evidence. In other words computer forensics is the collection, preservation, analysis, and presentation of computer related evidence.

Computer forensics also referred to as computer forensic analysis, electronic discovery, electronic evidence discovery, digital discovery, data recovery, data discovery, and computer analysis and computer examination. Computer evidence can be useful in criminal cases, civil

disputes and human resource employment proceeding

Use of Computer Forensics in Law Enforcement

Computer forensics assists in Law Enforcement. This can include:

- Recovery deleted files such as documents, graphics and photos.
- Searching unallocated space on the hard drive places where an abundance of data often resides.
- Tracing artefacts, those tidbits of data left behind by the operating system. Our experts know how to find these artifacts and more importantly they know how to evaluate the value of the information they find.
- Processing hidden files that are not visible or accessible to the user that contain past usage information. Often this process requires reconstructing and analyzing the date codes for each file and determining when each file was created, last modified, last accessed and when deleted.
- Running a straight search for email when no email client is obvious.

5.1.1 Choosing a Computer Forensics Specialist for a Criminal Case

When the services of a computer forensics specialist are required, look very carefully at the level of experience of the individuals involved. There is far more to proper computer forensic analysis than the ability to retrieve data, especially when a criminal case is involved. The services of person who will be called to testify in court, who will be able to explain what was done to computer Services and its data.

The court will want to know that individual's own level of training and experience, not the experience of his or her employer. Make sure you find someone who not only has the expertise and experience, but also the ability to stand up to the scrutiny and pressure of cross-examination.

5.2. Computer Forensic Services

No matter how careful they are, when people attempt to steal electronic information (everything from customer databases to blueprints), they leave behind traces of their activities. Likewise, when people try to destroy incriminating evidence contained on a computer (from harassing memos to stolen technology), they leave behind vital clues. In both cases, those

traces can prove to be the smoking gun that successfully wins a court case. Thus, computer data evidence is quickly becoming a reliable and essential form of evidence that should not be overlooked. A computer forensics professional does more than turn on a computer, make a directory listing, and search through files. Forensics professionals should be able to successfully perform complex evidence recovery procedures with the skill and expertise that lends credibility to your case. For example, they should be able to perform the following services:

- Data seizure
- Data duplication and preservation
- Data recovery
- Document searches
- Media conversion
- Expert witness services
- Computer evidence service options
- Other miscellaneous services

5.2.1 Data Seizure

Federal rules of civil procedure let a party or their representative inspect and copy designated documents or data compilations that may contain evidence. Computer forensics experts, following federal guidelines, should act as this representative, using their knowledge of data storage technologies to track down evidence & experts should also be able to assist officials during the equipment seizure process.

5.2.2 Data Duplication and Preservation

When one party must seize data from another, two concerns must be addressed:

(1) the data must not be altered in any way,

(2) and the seizure must not put an undue burden on the responding party. Computer forensics experts should acknowledge both of these concerns by making an exact duplicate of the needed data. Because duplication is fast, the responding party can quickly resume its normal business functions, and, because experts work on the duplicated data, the integrity of the original data is maintained.

5.2.3 Data Recovery

Using proprietary tools, computer forensics experts should be able to safely recover and analyze otherwise inaccessible evidence. The ability to recover lost evidence is made possible by the expert's advanced understanding of storage technologies. For example, when a user deletes an email, traces of that message may still exist on the storage device. Although the message is inaccessible to the user, experts should be able to recover it and locate relevant evidence.

5.2.3 Document Searches

Computer forensics experts should also be able to search over 200,000 electronic documents in seconds rather than hours. The speed and efficiency of these searches make the discovery process less complicated and less intrusive to all parties involved.

5.2.4 Media Conversion

Some clients need to obtain and investigate computer data stored on old and unreadable devices. Computer forensics experts should extract the relevant data from these devices, convert it into readable formats, and place it onto new storage media for analysis.

5.2.5 Expert Witness Services

Computer forensics experts should be able to explain complex technical processes in an easy-to-understand fashion. This should help judges and juries comprehend how computer evidence is found, what it consists of, and how it is relevant to a specific situation.

5.2.6 Provide Expert Consultation and Expert Witness Services Computers

a) Expert Testimony

- Has regularly testified multiple times as an expert witness in computers and computer forensics in courts.

b) Computer Expertise

- Belongs to the Computer Crime Investigators Association
- Trained in the forensic examination of computers (PC & Mac), having conducted examinations in countless cases including child exploitation, homicide, militia, software piracy, and fraud

- Has testified in state and federal courts as an expert in computers, computer forensics, the Internet, and America Online; often as an expert witness.
- Is thoroughly familiar with both computer hardware and software, having written software and repaired & assembled computers
- Teaches computer crime investigation, including computer search and seizure, for the Institute of Police Technology and Management
- Regularly consults with law enforcement officers in the search and seizure of computers
- Has provided forensic training to numerous law enforcement officers and corporate security officers
- Regularly consulted by other forensic examiners for advice in difficult cases

c) Electronic Surveillance

- Theft by employees or others
 - o Time
 - o Property
 - o Propriety information and trade secrets
- Embezzlement
- Inappropriate employee actions
- Burglary

Computer forensics expert's experience should include installing cameras in every imaginable location (indoors and outdoors, offices, homes, warehouses, stores, schools, or vehicles) for every conceivable crime (theft, burglaries, homicides, gambling, narcotics, prostitution, extortion, or embezzlement) under every conceivable circumstance (controlled settings, hostage crisis, or court-ordered covert intrusion). Computer forensics experts should be able to covertly install video monitoring equipment to protect. This even includes situations where employees may be misusing company computers. By using video surveillance to document employees who are stealing time, property, or secrets to take appropriate action against the employees.

d) Child Exploitation

- Child sexual exploitation
- Child pornography
 - o Manufacture
 - o Use
 - o Sale
 - o Trading Collection
 - o Child erotica
- Use of computers in child exploitation
- Search and seizure
- Victim acquisition
- Behavior of preferential and situational offenders
- Investigation
 - o Proactive
 - o Reactive

5.2.7 Computer Evidence Service Options

Computer forensics experts should offer various levels of service, each designed to suit individual investigative needs. For example, they should be able to offer the following services:

- Standard service
- On-site service
- Emergency service
- Priority service
- Weekend service

a) Standard Service

Computer forensics experts should be able to work on any case during normal business hours until critical electronic evidence is found. They must be able to provide clean rooms and ensure that all warranties on equipment will still be valid following their services.

b) On-Site Service

Computer forensics experts should be able to travel to different locations to perform complete computer evidence services. While on-site, the experts should quickly be able to produce exact duplicates of the data storage media in question. Their services should then be performed on the duplicate, minimizing the disruption to business and the computer system. Experts should also be able to help federal marshals seize computer data and be very familiar with the Federal Guidelines for Searching and Seizing Computers.

c) Emergency Service

After receiving the computer storage media, computer forensics experts should be able to give the highest priority to cases in their laboratories. They should be able to work on it without interruption until evidence objectives are met.

d) Priority Service

Dedicated computer forensics experts should be able to work on during normal business hours until the evidence is found. Priority service typically cuts your turn around time in half.

e) Weekend Service

Computer forensics experts should be able to work from on weekends, to locate the needed electronic evidence and will continue working on a case until evidence objectives are met. Weekend service depends on the availability of computer forensics experts.

f) Other Miscellaneous Services

Computer forensics experts should also be able to provide extended services. These services include

- Analysis of computers and data in criminal investigations
- On-site seizure of computer data in criminal investigations
- Analysis of computers and data in civil litigation.
- On-site seizure of computer data in civil litigation
- Analysis of company computers to determine employee activity
- Assistance in preparing electronic discovery requests
- Reporting in a comprehensive and readily understandable manner

- Court-recognized computer expert witness testimony
- Computer forensics on both PC and Mac platforms
- Fast turnaround time

5.2.8 Recover Deleted Data

Computers systems may crash. Files may be accidentally deleted. Disks may accidentally be reformatted. Computer viruses may corrupt files. Files may be accidentally overwritten. Disgruntled employees may try to destroy files. All of these can lead to the loss of critical data. One might think it's lost forever, but computer forensics experts should be able to employ the latest tools and techniques to recover data. In many instances, the data cannot be found using the limited software tools available to most users. The advanced tools that computer forensics experts utilize allow them to find and locate files and restore them for use. In those instances where the files have been irreparably damaged, the experts' computer forensics expertise allows them to recover even the smallest remaining fragments.

5.2.9 Securing data, information and accidental loss

Business today relies on computers. Sensitive client records or trade secrets are vulnerable to intentional attacks from, for example, computer hackers, disgruntled employees, viruses, and corporate espionage. Equally threatening, but far less considered, are unintentional data losses caused by accidental deletion, computer hardware and software crashes, and accidental modification.

Computer forensics experts should advise you on how to safeguard data by such methods as encryption and back-up. The experts can also thoroughly clean sensitive data from any computer system that needs to be eliminated. Files, records, and conversations are just as vital to be protected. Computer forensics experts should survey business of organization and provide guidance for improving the security of organizations information. This includes possible information leaks such as cordless telephones, cellular telephones, trash, employees, and answering machines.

5.2.10 Examine a Computer to Find Out user behaviour

While looking for evidence in a criminal prosecution, looking for evidence in a civil suit, or determining exactly what an employee has been up to the concerned systems, computer forensics experts should be equipped to find and interpret the clues that have been left behind. This

includes situations where:

- files have been deleted, ‘
- disks have been reformatted, or
- other steps have been taken to conceal or destroy evidence.

As previously mentioned, computer forensics experts should provide complete forensic services. These include:

- electronic discovery consultation,
- on-site seizure of evidence,
- thorough processing of evidence,
- interpretation of the results,
- reporting the results in an understandable manner, and
- Court-recognized expert testimony.

Computer forensics experts should also be able to regularly provide training to other forensic examiners, from both the government and private sectors. When other forensic examiners run into problems, they should turn to experts for solutions.

5.2.11 High-Tech Investigations

Computer forensics experts should have high level government investigative experience and the knowledge and experience to conduct investigations involving technology, whether the technology is the focus of the investigation or is required to conduct the investigation. The experts should be uniquely qualified to conduct investigations involving

- cellular telephone cloning,
- cellular subscription fraud,
- software piracy,
- data or information theft,
- trade secrets, computer crimes,
- misuse of computers by employees

Professional Duties & Ethics in Computer Forensics

Professional Duties

Professional duty or professional ethics are very important for a Computer Forensics specialist in his job, his/her prime objective in carrying professional duties is to collect evidence, preserve evidence and must ensure that the evidence is properly handled and kept secured. The mindset of a professional must be carried by the forensic experts while:

- Going for the identification of the possible evidences at the crime scene.
- While performing seizure of computer or computing related device(s).
- Acquisition of relevant data
- Protecting the system(s) during the forensic examination from any possible alteration.
- Proper care while handling of devices to prevent damage, corruption of data.
- Analyze all possibly relevant data found and also look into inaccessible areas of a disk like unallocated spaces, Slack spaces, and Hidden file/drives/volume.
- Relate the events transpired
- Expert witness adds value to your investigation.
- Document every action taken and present very concise and reliable report of the investigation

5.3 Benefits of Professional Forensic Methodology

Unlike paper evidence, computer evidence can often exist in many forms, with earlier versions still accessible on a computer disk. Knowing the possibility of their existence, even alternate formats of the same data can be discovered. The discovery process can be served well by a knowledgeable expert identifying more possibilities that can be requested as possibly relevant evidence. In addition, during on-site premises inspections, for cases where computer disks are not actually seized or forensically copied, the forensics expert can more quickly identify places to look, signs to look for, and additional information sources for relevant evidence. These may take the form of earlier versions of data files (memos, spreadsheets) that still exist on the computer's disk or on backup media or differently formatted versions of data, either created or treated by other application programs (word processing, spreadsheet, email, timeline, scheduling, or graphic).

A knowledgeable computer forensic professional should ensure that a subject computer system is carefully handled to ensure that:

- No possible evidence is damaged, destroyed, or otherwise compromised by the procedures used to investigate the computer.
- No possible computer virus is introduced to a subject computer during the analysis process.
- Extracted and possibly relevant evidence is properly handled and protected from later mechanical or electromagnetic damage.
- A counting chain of custody is established and maintained.
- Business operation is affected for a limited amount of time, if at all.
- Any client attorney information that is inadvertently acquired during a forensic exploration is ethically and legally respected and not divulged.

5.4 Steps taken by computer forensic specialists.

The computer forensics specialist needs to complete an Evidence Identification and Retrieval Checklist. Several careful steps to identify and attempt to retrieve possible evidence that may exist on a subject's computer system should be taken by the forensic expert.

Sample Checklist form

<i>Evidence Identification and Retrieval Checklist Form</i>	
Date: _____	
The computer forensics specialist should ensure that the following provisional list of actions for retrieval and identification of evidence are adhered to (check all tasks completed):	
1. Protect the subject computer system during the forensic examination from any possible alteration, damage, data corruption, or virus introduction.	☐
2. Discover all files on the subject system. This includes existing normal files, deleted yet remaining files, hidden files, password-protected files, and encrypted files.	☐
3. Recover all (or as much as possible of) discovered deleted files	☐
4. Reveal (to the greatest extent possible) the contents of hidden files as well as temporary swap files used by both the application programs and the operating system	☐
5. Access (if possible and legally appropriate) the contents of protected or encrypted files.	☐

6. Analyze all possibly relevant data found in special (and typically inaccessible) areas of a disk. This includes but is not limited to what is called unallocated space on a disk (currently unused, but possibly the repository of previous data that is relevant evidence), as well as slack space in a file (theremnant area at the end of a file in the last assigned disk cluster that is unused by current file data, but once again, may be a possible site for previously created and relevant evidence). ☐
7. Print out an overall analysis of the subject computer system, as well as a listing [of all possibly relevant files and discovered file data. ☐
8. Provide an opinion of the system layout; the file structures discovered; any discovered data and authorship information; any attempts to hide, delete, protect, and encrypt information; and anything else that has been discovered and appears to be relevant to the overall computer system examination. ☐
9. Provide expert consultation and/or testimony, as required ☐

5.5 Who can use Computer Forensic Evidence?

Many types of criminal and civil proceedings can and do make use of evidence revealed by computer forensics specialists. These are as follows:

- Criminal prosecutors use computer evidence in a variety of crimes where incriminating documents can be found, including homicides, financial fraud, drug and embezzlement record-keeping, and child pornography.
- Civil litigations can readily make use of personal and business records found on computer systems that bear on fraud, divorce, discrimination, and harassment cases.
- Insurance companies may be able to mitigate costs by using discovered computer evidence of possible fraud in accident, arson, and workman's compensation cases.
- Corporations often hire computer forensics specialists to find evidence relating to sexual harassment, embezzlement, and theft or misappropriation of trade secrets, and other internal and confidential information.
- Law enforcement officials frequently require assistance in pre-search warrant preparations and post-seizure handling of the computer equipment.
- Individuals sometimes hire computer forensics specialists in support of possible claims of wrongful termination, sexual harassment, or age discrimination.

However, there are concerns and problems with computer forensic evidence. Let's examine some of those problems.

Problems with Computer Forensic Evidence

Computer evidence is like any other evidence. It must be

- Authentic
- Accurate
- Complete
- Convincing to juries
- In conformity with common law and legislative rules (i.e., admissible) [5]

There are also special problems:

- Computer data changes moment by moment.
- Computer data is invisible to the human eye; it can only be viewed indirectly after appropriate procedures.
- The process of collecting computer data may change it—in significant ways. The processes of opening a file or printing it out are not always neutral.
- Computer and telecommunications technologies are always changing so that forensic processes can seldom be fixed for very long.

5.6 The Forensic Technician

Contrary to what is often thought, in many cases it is possible to produce reliable computer-derived evidence without recourse to specialist tools. The general principles to be followed by the forensic technician are :

- The scene of crime has to be frozen; that is, the evidence has to be collected as early as possible and without any contamination.
- There must be continuity of evidence, sometimes known as chain of custody; that is, it must be possible to account for all that has happened to the exhibit between its original collection and its appearance in court, preferably unaltered.

- All procedures used in examination should be auditable; that is, a suitably qualified independent expert appointed by the other side in a case should be able to track all the investigations carried out by the prosecution's experts.

Good results can be obtained by using the standard disk repair, network testing, and other utilities; however, complete records need to be kept. Even so, for some purposes these may not be enough, for example, where it is hoped to recover previously deleted material or where a logic bomb or virus is suspected. In these circumstances, specialist tools are needed. Special training is also required. The tools themselves don't address all of the problems of producing evidence that will stand up in court. Thus, the key features of the forensic technician are

- Careful methodology of approach, including record keeping
- A sound knowledge of computing, particularly in any specialist areas claimed
- A sound knowledge of the law of evidence
- A sound knowledge of legal procedures
- Access to and skill in the use of appropriate

5.7 Legal Tests

The rules vary from legislation to legislation, but one can give a broad outline of what happens in those countries with a common law tradition—the U.K., U.S., and the so-called old Commonwealth. The law makes distinctions between real evidence, testimonial evidence, and hearsay. Real evidence is that which comes from an inanimate object that can be examined by the court. Testimonial evidence is that which a live witness has seen and upon which he or she can be cross-examined. The hearsay rule operates to exclude assertions made other than those made by the witness who is testifying as evidence of the truth of what is being asserted. The pure hearsay rule is extremely restrictive and has been extensively modified by various statutory provisions. Thus, there are rules about the proving of documents and business books. Bankers' books have separate legislation. Some of the rules apply explicitly to computers, but many do not, although they can be (and have been) interpreted to cover many situations in which computers are involved. For example, in the U.K. there have been situations where legal rules presumably designed to help the court may in fact hinder it. In practice, these issues may be circumvented. For instance, in a criminal case, evidence may be obtained by inadmissible methods. This evidence, however, then points investigators to admissible sources of evidence

for the same sets of circumstances. An example of this could occur during a fraud investigation. In other words, computer search methods are often used to identify allegedly fraudulent transactions, but the evidential items eventually presented in court are paper-based invoices, contract notes, dockets, or other documents. In this manner, the prosecution can demonstrate to the jury the deception or breach of the Companies Act or other specific fraudulent act. Again, in civil litigation the parties may decide to jointly accept computer-based evidence (or not to challenge it) and instead concentrate on the more substantive elements in the dispute. A defendant may prefer to have a substantive defense rather than a technical one based on inadmissibility. Or, again, the legal team may not feel sufficiently competent to embark on a technical challenge.

5.8 Subject Matter of Computer Forensics

The subject matter of computer forensics can, thus, not be solely concerned with procedures and methods of handling computers, the hardware from which they are made up, and the files they contain. The ultimate aim of forensic investigation is its use in legal proceedings. At the same time, an obsession with common law and judicial rules is likely to inhibit many investigations. It might be a mistake for inquiries not to be commenced simply because of fear of possible inadmissibility. Furthermore, a number of computer-investigatory methods may turn out not to be directly admissible but may nevertheless be useful in locating noncomputer evidence that is admissible. One may have to take a somewhat pragmatic view of the precise bounds of the subject matter, but it should still be possible to define its core activities. It might help to explore the way in which forensic science in general has developed and then see what expectations one might reasonably have of computer forensics. Although forensic science was already well established, and indeed forms a central feature of many of Conan Doyle's Sherlock Holmes stories published from 1892 onwards, up until the 1970s, each forensic scientist tended to develop his or her own methods and present them ad hoc to juries. Obviously, reliance was placed on descriptions of methods used by others, but for courts, the tests of whether to believe the forensic evidence were the manner of presentation—the supposed eminence of the forensic scientist and the skill of the opposition lawyer (or rival expert who might be called). During the 1970s, a more formal checklist-based approach was introduced. This was partly to bring about standardization as between different laboratories and partly in response to the criticism (in the U.K.) that arose over such controversial cases as the Birmingham Six. In the U.K. Home Office Forensic Service, these checklists were devised by senior staff. Obviously, such checklists are revised in the light of experience—the publication of new specialist research or adverse

experience during a trial. An increasingly used feature of modern practice is quality control, which involves work being checked by an otherwise uninvolved coworker before being offered to external scrutiny. In any event, the broad tests for evidence include

Authenticity: Does the material come from where it purports?

Reliability: Can the substance of the story the material tells be believed and is it consistent? In the case of computer-derived material, are there reasons for doubting the correct working of the computer?

Completeness: Is the story that the material purports to tell complete? Are there other stories that the material also tells that might have a bearing on the legal dispute or hearing? Freedom from interference and contamination: Are these levels acceptable as a result of forensic investigation and other post-event handling?

Any approach to computer forensics would, thus, need to include the elements of

- Well-defined procedures to address the various tasks
- An anticipation of likely criticism of each methodology on the grounds of failure to demonstrate authenticity, reliability, completeness, and possible contamination as a result of the forensic investigation
- The possibility for repeat tests to be carried out, if necessary, by experts hired by the other side
- Checklists to support each methodology
- An anticipation of any problems in formal legal tests of admissibility
- The acceptance that any methods now described would almost certainly be subject to later modification

5.9 Divergences from Conventional Forensic Investigation

There will be divergences from the expectations of more traditional areas of forensic investigation. The main reason is the rate of change of computer technology. The deviser of a test for the presence of a prohibited drug, an explosive, fabric fibers, bodily tissues, and the like, can expect that over a period of time, the test may be improved or shown to be defective, but, the need for the test and most of its essential details will probably not change. However, in computers, newness and obsolescence is the norm. For example, a key feature of computer forensics

is the examination of data media: new forms and methods of data storage occur at intervals of less than 4 years. The floppy disk of 13 years ago was in 5.25 inch format and held 360 k. The current equivalent is 3.5 inches and holds 1.44 MB, and much higher densities are expected soon. A typical hard-disk size on a PC of the same date was 20-30 MB, was in 5.25 inch form, and used modified frequency modulation (MFM) controller technology. Today most PCs have hard disks in excess of 1750 MB in 2.5 inch or even 1.5 inch form using integrated development environment (IDE) or run length limited (RLL) technology. On minis and mainframes, data may be held on redundant array of independent (or inexpensive) disks (RAID), where individual files may be split and spread over eight or more separate disk surfaces. Similar changes have taken place in tape technology and the use of erasable programmable readonly memory (EPROMs).

Computer architectures have gone through profound changes in the same short period. PCs have become much more powerful, the large central mainframe is now a rarity, and large companies are now served by a multiplicity of smaller computers that all interact via a complex network. Computer peripherals keep changing as well. Modems and network routers have become intelligent, and digitizing scanners are fairly common devices. They can be subverted, for example, for forgery. Wide-area telecom methods are being used more and more. These provide opportunities for both high-tech criminals and forensic investigators. The protocols they use also keep changing. The foregoing simply lists technological changes. Similar changes have taken place in computer applications; these, in turn, have affected the type of information one might expect to find held in a computer. For example, the following technological changes have taken place over the years:

- The growth of email, both locally within large organizations and worldwide.
- The growth of client/server applications.
- The software outcome of the more complex hardware architectures.
- The client/server situation (software on).
- The ability of a PC or small local machine to interact with software and data held on other nonlocal machines and large mainframes in a way that appears to be seamless to the user. One key effect of this is that a computer document often does not exist in some computer equivalent of a filing cabinet, but, rather, is assembled on demand by the activity of one computer drawing information from many others.

- The evidence of a transaction or event may, therefore, only be provable by the presentation of all the records from all the computers involved, plus an explanation of how the assembly of the report relied on took place.
- The greater use of Electronic Data Interchanges (EDIs) and other forms of computer-based orders, bills of lading, payment authorizations, etc. EDIs have very complex structures, with some evidence being held in computers owned by the counter-parties and some by the EDI supplier/regulator.
- Computer graphics: computer-aided design (CAD) methods, particularly those that provide an element of autocompletion or filling-in of basic design ideas.
- More extended easier-to-use databases.
- The greater use of computer-controlled procedures (sales, dispatch, and emergency services; computer-controlled processes; traffic control; and manufacturing).
- The methods of writing and developing software. There is much greater use of libraries of procedures (of new computer language models). For example, object-oriented programming environments and new, more formal methods of program development; standards and methods of testing have also changed.

5.10 Other Miscellaneous Service

Computer Forensics experts should be able to provide extended services. These services include:

- Analysis of computer and data in criminal investigations.
- On-site seizure of computer data in criminal investigation.
- Analysis of computer and data in civil litigation.
- On-site seizure of computer data in civil litigation.
- Analysis of company computers to determine employee activity.
- Assistance in preparing electronic discovery requests.
- Reporting in a comprehensive and readily understandable manner.
- Court-recognize computer experts witness testimony.
- Computer forensic on both PC and MAC platform.
- Fast turnaround time

Summary

- Computer forensics is the process of methodically examining computer media (hard disk, diskettes, taps, etc) for evidence. In other words computer forensics is the collection, preservation, analysis, and presentation of computer related evidence.
- No matter how careful they are, when people attempt to steal electronic information (everything from customer databases to blueprints), they leave behind traces of their activities.
- Federal rules of civil procedure let a party or their representative inspect and copy designated documents or data compilations that may contain evidence
- Unlike paper evidence, computer evidence can often exist in many forms, with earlier versions still accessible on a computer disk. The scene of crime has to be frozen; that is, the evidence has to be collected as early as possible and without any contamination.

Check your answers

1. What is Computer Forensics?
2. Define Data Seizure.
3. How to effectively do Document Searches?
4. Define Electronic surveillance.
5. Define On-Site Service.
6. How to Secure data, information and prevent accidental loss?
7. What is High-Tech Investigations?
8. Duties of the Forensic Technician?

LESSON – 6

STANDARDS, GUIDELINES AND BEST PRACTICES

After reading this lesson you will be able to understand the following:

- Standards, Guidelines and Best Practices
- Handling the Digital Evidence
- Digital Evidence Examination Guidelines
- The principles of forensic examination
- Proper facilities
- The duties of the investigating personnel
- Designing a quality process manual
- Recommended tools and techniques
- Chain of custody
- Organization who provide standards, best practices and guidelines
- ACPO
- IOCE
- SWGDE
- DFRWS
- IACIS
- HTCIA
- ISO 27037

Structure

- 6.1 Standards, Guidelines and Best Practices**
- 6.2 Handling the Digital Crime Scene**
- 6.3 Digital Evidence Examination Guidelines**
- 6.4 The principles of forensic examination**
- 6.5 Proper Facilities**

- 6.6 The duties of investigating personnel**
- 6.7 Designing a quality process manual**
- 6.8 Recommended tools and techniques**
- 6.9 Chain of custody**
- 6.10 Organization who provide standards, guidelines and Best Practices**
 - 6.10.1 ACPO**
 - 6.10.2 IOCE**
 - 6.10.3 SWGDE**
 - 6.10.4 DFRWS**
 - 6.10.5 IACIS**
 - 6.10.6 HTCIA**
 - 6.10.7 ISO 27037**

6.1 Standards, Guidelines and Best Practices

The standard concerns the principles behind, and the forensic processes involved in investigating incidents.

6.1.1. The standards

Provides guidelines that encapsulate idealized models, for common incident investigation processes across various incident investigation scenarios involving digital evidence. This includes processes from pre-incident preparation up to and including returning evidence for storage or dissemination as well as any general advice and caveats on such processes.

6.1.2. Guidelines

The guidelines describe processes and principles applicable to various kinds of investigations. An organization's forensics guidelines should include general methodologies for investigating an incident using forensic techniques, since it is not feasible to develop comprehensive procedures tailored to every possible situation. The goal for the guidelines and procedures is to facilitate consistent, effective, and accurate forensic actions, which is particularly important for incidents that may lead to prosecution or internal disciplinary actions. Because

electronic logs and other records can be altered or otherwise manipulated, organizations should be prepared, through their policies, guidelines, and procedures, to demonstrate the integrity of such records

The International Organization for Standardization (ISO), an international non-governmental organization, and the International Electrotechnical Commission (IEC), an international not-for-profit organization, develop and publish international standards to harmonize practices between countries. In 2012, the International Organization for Standardization (ISO) and the International Electrotechnical Commission (IEC) published international standards for digital evidence handling (ISO/IEC 27037 Guidelines for Identification, Collection, Acquisition, and Preservation of Digital Evidence). These guidelines included only the initial handling of digital evidence.

In summary, this International Standard provides a general overview of all incident investigation principles and processes without prescribing particular details.

6.1.3. Best Practices

Best practice guides are available to identify and promote valid and reliable digital forensics processes and outcomes. Best practices also serve to encourage professionals to adopt a consistent methodology to facilitate the interchange of data.

The importance of the best practices or guidelines set by the renowned organizations is to make the investigation procedure universal, a synchronization for carrying out investigation where ever it is carried out in the world. Best practices guide also help to carry out the investigation fulfilling the Legal criteria. The best practices guide helps in

- Maintaining the professional ethics,
- Ways of handling the evidences,
- Following the principles of forensic examination,
- Proper facilities,
- Discusses about the duties of the investigating personnel,
- Designing a Quality process manual,
- Use of recommended tools and related Equipment,
- Computer forensics examination environment,

- Ways of storing and
- Maintaining chain of custody.

Cases in point are the U.S. Scientific Working Group on Digital Evidence's best practices for computer forensic examination, digital evidence collection, and computer forensic acquisitions, and the European Network of Forensic Science Institutes' best practice manual for the forensic examination of digital technology.

Maintaining Professional Ethics

Computer Forensics specialist or any other person involved in a forensics examination has to play an important role by maintaining strict ethical standards. The basic objective in his/her role as forensic expert is to ensure that the evidence gathered and analyzed meets the highest quality standards possible. The investigation he/she has carried out must be complete based on proper guidelines and procedures followed along with the Integrity and Authenticity. The proper handling of evidence and following forensic examination principles is must.

Professional ethics are very important for a Computer Forensics specialist in his job, his/her prime objective in carrying professional duties is to collect evidence, preserve evidence and must ensure that the evidence is properly handled and kept secured. The mindset of a professional must be carried by the forensic experts while:

- Going for the identification of the possible evidences at the crime scene.
- While performing seizure of computer or computing related device(s).
- Acquisition of relevant data
 - Protecting the system(s) during the forensic examination from any possible alteration.
- Proper care while handling of devices to prevent damage, corruption of data.
- Analyze all possibly relevant data found and also look into inaccessible areas of a disk like unallocated spaces, Slack spaces, and Hidden file/drives/volume.
- Relate the events transpired
- Expert witness adds value to your investigation.
- Document every action taken and present very concise and reliable report of the investigation

6.2 Handling the Digital Evidence

Any procedures, processes or practices framed in digital evidence recovery falls within a defined and accepted framework for Computer Forensics investigation and we must comply with the principles stated earlier. For example in identification and recovery of digital evidences, Computer Forensics specialists must go to the crime scene himself or any crime scene investigator with experience must go himself and recover the evidences. It is important that relevant legal guidelines or constraints are known to the investigator.

In the entire phase of digital forensic process the handling of evidence is very important. While collecting the digital evidences there needs to be an extra care taken. The transportation cannot be neglected. While collecting digital storage medium such as HDD, Mobile Phones, and Computers care should be taken to prevent exposure to high temperature, water, magnetic field.

For Example: Mobile phones should be collected in Faraday bags as Radio Frequency shield which prevents incoming calls and messages. Since mobile phones have small memory compared to Hard Disk, the oldest message which might be crucial may get deleted when new incoming messages are received. Also, Mobile phones if the mobile is in ON state it shouldn't be switched off and if the battery is low then, keep it charged till it reaches the forensic workstation, and anti-static bags will prevent any signals and communication of the mobile phone with the surrounding. Also the phone can be put on an airplane mode. In order to understand some basic steps to be kept in mind

6.3. Digital Evidence Examination Guidelines

A single neglectful act can result in the loss of information (digital). An investigator must have a thorough understanding of what is to be recovered from crime scene or in other words what possible evidences can be gathered. For the process of collecting the evidences few steps are to be kept in mind:

1. Preparation
2. Searching the scene
3. Prioritization
4. Evidence collection

6.3.1.Preparation

Prior to any investigation, the Computer Forensics specialist must ensure that he/she has the proper tools including forensically sterile media. All equipment, sampling materials, storage and transportation containers should be new, preferably disposable, or cleaned thoroughly before and after use. Extra care and precautionary measures should be taken so that the digital evidence is preserved. Following steps may be useful in preparation phase:

1. Follow the Legal procedures and make sure you have a search warrant.
2. Discuss about the search with involved personnel.
3. Make sure there is someone as in-charge at the crime scene.
4. Consider the safety and comfort of search personnel - do not be caught unprepared when encountering a potentially dangerous scene or inclement weather- examples are:
 - o Clothing
 - o Communication
 - o Lighting assistance
 - o Shelter
 - o Transportation
 - o Food
 - o Medical assistance
 - o Scene security
 - o Equipment
5. Organize communication with services of an ancillary nature (e.g., medical examiner, prosecute attorney) in order that questions which surface during crime scene search may be resolved.

Also remember to reduce the risk of loss or damage to digital evidence, all areas: indoor, outdoor or while transporting should be protected using procedures that are recommended to be followed to ensure digital evidence remains valid as per the Legal system.

6.3.2 Searching the Crime Scene

The search at a crime scene is a very first step an investigator or first responder takes towards a progressive forensic investigation. Areas should be searched systematically and thoroughly for digital evidences and related material. To increase investigative efficiency, areas should be targeted and prioritized according to how likely digital evidence may be found and the amount of digital data that may be recovered. The following things must be kept in mind while doing an investigation are:

1. Approach scene and first Secure it so that nobody comes near to the scene of crime
2. Initiate preliminary survey
3. Evaluate physical evidence that could be possible and primary source of evidence
4. Prepare narrative description
5. Photography of a crime scene is very important
6. Prepare a scene of crime
7. Always conduct detailed search
8. Record everything about the evidences and investigators
9. Conduct final survey

6.3.3. Securing the Crime Scene

Securing the crime is most important within the premises where crime occurred. Check for the extent to which scene has been protected. Gather information from personnel who have entered scene and have knowledge about it. Keep on making the notes. Keep out unauthorized personnel from accessing the crime scene.

6.3.4. Detailed Search and Collection of Physical Evidence:

Perform search for all the evidences that possibly should be there. Look in every corner for evidence. The digital evidence can be very fragile which means volatile data. The most volatile must be preserved first. Also,

1. Photograph all items before collection
2. Record everything in evidence form.

3. Ensure that evidence or the container of evidence is initialed by investigator collecting the evidence.
4. Do not handle evidence excessively after recovery.
5. Seal all evidence containers at the crime scene.
6. Different types of evidence require different media to preserve.
7. Cross check everything time and again before leaving the crime scene.
8. Maintain chain of custody of that evidence.

Physical evidence could be various storage media that contains digital evidence. The sources of evidence could be the following:

Common Exhibits (Physical Evidence) to be seized:-

- Memory storage media like HD / FD / CD / DVD / Flash cards / Thumb Drive / Pen Drive / ZIP Drive...
- CPU / Laptop
- SIM cards / Mobile Phones (GSM / CDMA)
- Proper packing is important
- Antistatic covers to be used to insulate media from the electromagnetic fields
- Thermocole or Bubble covers may be used to ensure shock proof
- Mobile Phones (GSM / CDMA)
- Verify, whether the SIM card is inserted in the phone
- Collect the data cables and charger unit if feasible
- Check the mobile phone has got extendable memory slot and look for the memory card and seize it
- If a standalone SIM card presents, it is also to be seized
- Proper packing is important –
- Antistatic covers to be used to insulate media from the electromagnetic fields
- Thermocole or Bubble covers may be used to ensure shock proof
- If there is a SIM card only, it should be packed properly

Memory Devices that need to be collected at the crime scene is illustrated in figure

MEMORY DEVICES



Figure : 6.1 Memory Devices



Figure 6.2 : Memory Devices



Figure 6.3: USB Devices

6.3.5. Final Survey

Final survey is more often done to attain assurance that the search or the crime scene investigation is done perfectly and nothing is left behind. What we do is:

1. Discuss all aspects of the search.
2. Double check everything.
3. Ensure all evidence is accounted.
4. Make sure possible hiding places of difficult access areas have not been overlooked in detailed search.

5. Critical issues: have you gone far enough in the search for evidence, documented all essential things, and made no assumptions which may prove to be incorrect in the future?

6.3.6. Sizing Evidence: General Guidelines concerning the seizing of evidence are provided as follows:

1. Consult with the investigating officer to determine the necessary equipment to take to the scene.
2. Review the legal authority to seize the evidence, ensuring any restrictions are noted. If necessary during the execution of the seizure, obtain additional authority for evidence outside the scope of the search.
3. When it is impractical to remove the evidence from the scene, the evidence items should be copied or imaged according to local procedures.
4. All suspects, witnesses, and by-standers should be removed from the proximity of digital evidence, ensuring that the above individuals are not in possession of potential evidence.
5. Solicit information from potential suspects, witnesses, LAN administrators, etc. to ascertain knowledge of the system to be seized (e.g., password(s), operating system(s), screen name, email address).
6. Scene should be searched systematically and thoroughly for evidence.
7. Searchers should be able to recognize the different types of evidence.

6.3.6 Prioritization

In general, a Computer Forensics specialist conducting an investigation, must do his work effectively and as quickly as possible. Because if a case arises where live forensic investigation is needed, the digital evidence is volatile and we must respond very quickly to take dump of the machine. Another thing is digital evidence from huge stores of data – that may be terabytes in size that definitely can be time consuming, disruptive, and costly. Extracting data from computers can be tampered easily by our slightest of mistake. So let us try to understand what to do, and what to prioritize? The ISIF (Information Security and Forensic Society) of Hong Kong describes: In conducting an examination for digital evidence, prioritizing the deployment of resources is critical. Yet prioritization of activities is not easy, as the following must be considered before any activities are initiated:

- The urgencies and priorities
- The nature of the case
- Time constraints
- The other types of forensic examination that may have to be carried out on the same items (E.g. fingerprint analysis).
- The items that have the potential to provide the most information in response to the various propositions.
- The items offering the best choice of target data, in terms of evidential value. After an assessment of the relevant requirements, examination activities should be assigned to the appropriate qualified individual(s).

The investigator(s) in charge and the investigation team should prioritize evidence collection to prevent loss, destruction or contamination. Team members should:

- Conduct a careful and methodical evaluation that considers all physical evidence possibilities (e.g., biological fluids, latent prints and trace evidence).
- Focus first on the easily accessible areas in open view and then proceed to out-of-view locations.
- Select a systematic search pattern for evidence collection based on the size and location of the scene(s).
- Select a progression of processing and collection methods so that initial techniques do not compromise subsequent methods.
- Concentrate on the most transient evidence and work to the least transient forms of physical evidence.
- Move from least intrusive to most intrusive processing and collection methods.
- Regularly assess environmental and other factors that may affect the evidence.
- Remain aware of multiple scenes (e.g., victims, suspects, vehicles and locations).
- Recognize other methods that are available to locate, document, and collect evidence (e.g., alternate light sources, and blood pattern documentation).
- Team members should carefully collect, preserve, package and transport evidence.

- Maintain scene security throughout processing and until the scene is released.
- Document evidence collection by recording its location at the scene, the date of collection and who collected it.
- Collect each item identified as evidence.
- Establish a chain of custody.
- Obtain standard and reference samples from the scene.
- Obtain control samples.
- Consider obtaining elimination samples.
- Immediately secure electronically recorded evidence (e.g., answering machine tapes, surveillance camera videotapes and computers) from the vicinity.
- Identify and secure evidence in containers (e.g., label, date and initial container) at the crime scene. Different types of evidence require different containers (e.g., porous/nonporous and crushproof).
- Package items to avoid contamination and cross-contamination.
- Document the condition of firearms/weapons prior to rendering them safe for transportation and submission.
- Avoid excessively handling the evidence after it has been collected.
- Maintain the evidence at the scene in a manner designed to diminish degradation or loss.
- Transport and submit evidence items for secure storage.

6.4. The principles of forensic examination

There are Cardinal Rules of Evidence which must be followed by the experts as a part of best practice and these principles include:

1. Never mishandle evidence
2. Never work on original evidence
3. Use proper software utilities
4. Never trust the suspect operating system
5. Document everything

The reason behind following these principles are:

1. Maintenance of data integrity as well as data authenticity without which the evidence is not admissible in court
2. Prevention of contamination of data
3. Proper and comprehensive documentation, along with chain of custody etc.
4. Following proper Digital Forensic Scientific Methodology

6.5. Proper facilities

The creation of a digital forensic laboratory is not a trivial issue and will be dependent upon a number of factors and be constrained by number of regulations and rules. It involves the provisioning of proper facilities. Facilities Include working space, trained personnel, Hardware, software and so on. In a typical laboratory space required for following functions include:

- Reception area with storage
- Waiting area/Meeting area
- Storage for personal possessions and equipment
- Viewing room
- Dismantling area
- Secure Storage
- Sensitive investigation area
- Imaging area
- Analysis area
- Management Office
- A significant number of power points will be required in each area. A number of network ports will also be needed for the laboratory network, but these should be carefully considered and controlled.
- The laboratory facilities should have adequate security.
- Depending on the configuration of the working areas within the laboratory, consideration should be given to the separation of the workstations into cubicles

separated by privacy screens to prevent inadvertent visibility of material on the screen of workstation by anyone using the other workstation.

- Three distinct types of storage are required. A significant volume of storage is needed for the evidence and there is also a need for storage of images created and any product of investigation, both images, products for onsite storage, and a backup system of offsite storage should also be facilitated. Also there needs to be a space to store all of the cables, connectors, attachments and tools required in the process of digital forensics investigations.
- The laboratory will also require a storage area for documents, sterile disks, and other disposable items as well as an area for the short term storage of equipment.
- The laboratory will need to have facilities such as intrusion detection, fire detection and motion sensor alarm system installed.

6.6. The duties of the investigating personnel

Investigating Personnel:

Computer Forensics examination cannot be possible without trained personnel. A computer forensic investigation cannot be complete without having the required set of:

1. **Skilled personnel:** who are very well trained to either conduct or assist in an examination.
2. **Quality Assurance:** An important element is a computer forensics Quality Process Manual that will help to ensure consistency of process and provide a valuable reference resource for persons who may be involved in an investigation
3. **Proper tools and facilities:** without the use of proper standard, forensic tools we cannot simply carry away the investigation. These tools or facilities help to keep the evidence secure and safe. Also, helps to save our time during investigation and definitely is very handy.

The person who has an expertise in the subject of computer forensics is usually called as Computer Forensics Examiner/Specialist etc. In simple words he is an expert in this area. The Computer Forensics Examiner/Specialist cannot be a person who is not well trained in technical stuff. He/she must understand the various technologies and techniques used by the criminals universally. The Computer Forensics Examiner/Specialist must also have an idea of Legal system,

jurisdiction and IT related Laws. He/she must have a deep knowledge about the subject and thorough understanding of what transpires so that He/she can interpret rightly what exactly has happened. Few points regarding the areas He/she must be familiar with:

1. Understanding about the Hardware of the devices
2. Software and tools being used in Digital Forensics Investigation like EnCase, FTK etc.
3. Knowledge of different Operating Systems (MOBILE/COMPUTERS)
4. Knowledge of Internet/web technology
5. Understanding of Networks and Databases
6. Business, conceptual and/or operational knowledge
7. Well versed in Programming Languages
8. Familiar with latest Gadgets (functionality)
9. Knowledge of Procedures for gathering and preserving digital evidences
10. Knows the professional or ethical standards.

6.7 Designing a Quality process manual

The Digital forensic methodology is to be followed with accuracy and once the procedures followed and the quality of the work done, there should be a documented computer forensics Quality Process Manual covering all systems, processes and methods used in the examination and reporting of forensic digital evidence. The Information Security and Forensics Society, Hong Kong (ISFS) describes the process documentation and use of Quality Process Manual.

It provides detailed steps and procedures to be followed during an examination. Such a Quality Process Manual would not only help ensure consistency of process, it could additionally serve as an important reference resource for personnel who may find themselves in a first responder situation. This information can be particularly critical as first responders who are unaware of proper forensics procedures may inadvertently destroy valuable evidence.

6.7.1. Process Documentation

Process Documentation includes the following:

- Standard operating procedures
- Contacts
- Evidence handling and tracking
- Corrective action policy/procedures
- Complaint and conflict resolution

Among other things, this Quality Process Manual should document organizational policy outlining the following:

- Steps to be taken in the event a forensic examination is necessary
- Procedures to follow
- Persons to contact

Whenever appropriate the Quality Process Manual should provide standardized form templates. It should be organized in an easily accessible, reference able and comprehensive format, be regularly reviewed and updated if necessary. Readers should be aware that such standards documentation is common among firms that value quality, for instance organizations that are ISO 9000 certified. A Quality Process Manual also a good indicator to staff, investors and others of the importance of quality to the organization.

6.7.2. For organizations conducting investigations:

For organizations that are conducting forensics examinations, their Quality Process Manual should also include procedures and information on:

- Standard Operating Procedures
- Complaints and conflict resolution
- Corrective Action Policy/Procedures

6.7.3. Standard Operating Procedures

The Standard Operating Procedures should define everything starting from:

- **Incident identification**– Where an event is recognized as an incident it should be classified into a type of incident.
- **Preparation** – Detailed procedures for the proper preparation of tools, techniques, monitoring authorizations and management support.
- **Approach strategy** – Provide details about the best practices and guidelines on how approach for search and seizure of evidences so that positive result is guaranteed.
- **Preservation** – Isolating, securing and preserving the state of physical and digital evidence.
- **Collection** – Procedures and processes for recording the physical scene and duplication of digital evidence using standardized and accepted practices.
- **Documentation** - Maintaining record of actions taken during an investigation.
- **Analysis** – Look for evidences and draw conclusions based on evidence found.
- **Presentation** – Provide a structured and clear report which explains the conclusions reached. It shouldn't be more technical, non-technical person should be able to understand easily.

6.8. Use of recommended tools and related Equipment

6.8.1. Forensic Hardware:

Forensic hardware includes workstation, write blockers, and forensic devices.

(a) Forensic Workstations

Forensic Recovery of Evidence Device (**FRED, Digital Intelligence**) family of workstations consists of integrated forensic processing platforms capable of handling the most challenging computer case. FRED is available in Mobile, Stationary and laboratory configurations. These systems are designed for both the acquisition and examination of Digital evidence. FRED professional forensic systems and the Digital Intelligence Ultrabay universal write protected, imaging bay, deliver the ability to easily duplicate evidence directly from IDE/SCSI/SATA hard

drives, USB Devices, Firewire devices, floppies, CDs, DVDs, ZIP cartridges, DLT-V4 tapes and PC Card/smart card/SD –MMC/Memory stick/Compact Flash Media in a forensically sound environment. Various categories of FRED workstations are:

- FRED – Forensic Recovery of Evidence Device an ideal tool for laboratory for imaging and processing
- FREDDIE – a highly portable solution which meets both imaging and processing requirements FRED interrogation Equipment.
- FRED- L – The Laptop of FRED family of tools
- FRED- microFRED the smallest, full powered Forensic Workstation
- FRED Systems: is the complete forensic hardware and hardware solution. The 19" LCD monitor is included. The available OSs on this system is MS 6.22, Wins 98, Wins XP Pro, and Linux 9.1 Pro. Some software such as Norton GHOST 10.0 & 2003, Nero DVD/CD
- Authoring Software, DriveSpy, Image, PDWipe, PDBlock, and PARTare included.

Other open source forensic workstation with pre-installed software with collection of tools includes:

- SIFT Workstation – Ubuntu flavor with collection of forensic tools pre-installed
- Kali Linux- Linux based workstation with collection of forensic tools

(b) Forensic Network

Forensic Network is a series of processing and imaging computers connected and integrated directly with a high speed, high capacity server to share resources. The file server operates as the core of the Forensic Network and can be used as a central storage facility for Forensic Images as well as the applications software for use by the client processing and imaging stations.

(c) Forensic Write Blockers

Forensic write blockers are **Write blockers** are devices that allow acquisition of information on a drive without creating the possibility of accidentally damaging the drive contents. They do this by allowing read commands to pass but by blocking write commands, hence their name.

There are two ways to build a write-blocker: the blocker can allow all commands to pass from the computer to the drive except for those that are on a particular list. Alternatively, the blocker can specifically block the write commands and let everything else through.

Write blockers may also include drive protection which will limit the speed of a drive attached to the blocker. Drives that run at higher speed work harder (the head moves back and forth more often due to read errors). This added protection could allow drives that can not be read at high speed (UDMA modes) to be read at the slower modes (PIO).

There are two types of write blockers, Native and Tailgate. A Native device uses the same interface on for both in and out, for example a IDE to IDE write block. A Tailgate device uses one interface for one side and a different one for the other, for example a Firewire to SATA write block. Write blockers are also called as bridges.

There are both hardware and software write blockers. Some software write blockers are designed for a specific operating system. One designed for Windows will not work on Linux. Most hardware write blockers are software independent.

ATA and SCSI hardware write blockers, as well as other custom solutions, to effectively address specific write blocking requirements. UltraKit, FireFly, FireBlock, SCSIBlock and FireChief devices can maintain the integrity of evidence. The following figure illustrates various parts of write blockers.

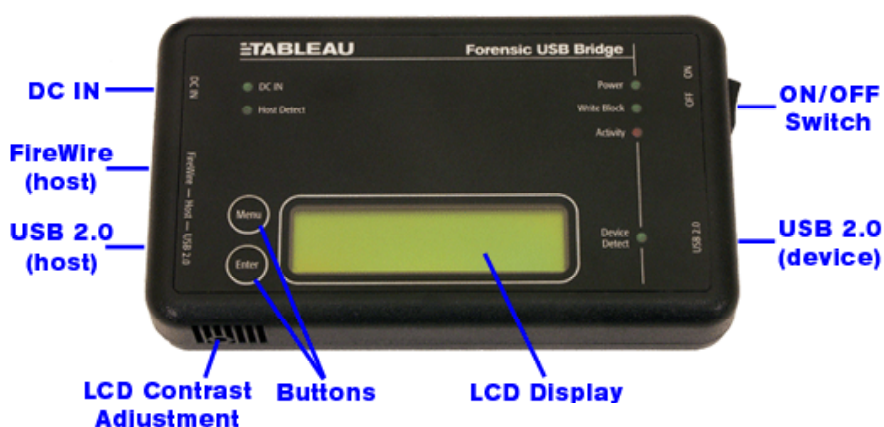


Figure 6.4: Write Blocker



Figure 6.5 : Classic Bridge/Write Blockers Family

The Laboratory should have a collection of classic bridges/write blocker family and be used according to the type of hard disc dismounted from CPU of the suspect computer.

(d) Forensic Devices

The following are various forensic devices:

- Fannie - forensic area network numerous imaging enclosure
- Rack-a-Tacc password decryption
- Tacc1441 Hardware Accelerator
- Modular Accessories
- Forensic Duplicator
- Hardcopy 3 & Hardcopy 2
- Shadow 2

(e) Forensic Software Tools

The forensic software tools frequently used for cyber forensics include imaging tools, examination and analysis tools, visualization tools and the like.

(i) Digital Intelligence Software

Digital Intelligence has created several forensic software tools in-house specifically for Forensic use. These tools include DriveSpy, Image, Part, PDBlock and PDWipe.

(ii) Accessdata Forensic toolkit

The forensic tools available are Ultimate Toolkit, Forensic Toolkit, and specialized password recovery based on applied cryptography – Password Recovery Toolkit and Registry Viewer.

(iii) Encase from Guidance Software

Encase Forensic Edition, by Guidance Software, is the world's leading solution for computer investigations and forensics. It is the oldest of GUI Based IT Forensic Tools, and it includes other useful features including the ability to preview and acquire disks through many types of connections.

(iv) The Sleuth Kit and Autopsy Browser powerful forensic tools to work in UNIX or Linux Environments.

(v) The Coroner's Toolkit (TCT) is an open source set of forensic tools for performing post mortem analysis on UNIX system.

(vi) Mandiant First Response as a first response tool for gathering snapshot of the network with very limited intrusiveness prior to a detailed forensic examination.

(vii) ProDiscover is a complete IT forensic tool that can access over the network to enable media analysis and network behaviour analysis.

(viii) i2 Analyst Notebook

This is a very different type of analysis tool from those information security professionals are used to perform Link Analysis, a crucial aspect of incident response is usually done manually or by trying to use log correlators. This is a true link analyser in analysing complex crimes and security incidents. Link analysis is applied to incident response and it is used as a visualization software to link inter relationship between displayed.

(ix) Nuix Email Investigation Software is a powerful email investigation software tool that performs link analysis, email information visualization, hierarchical relationship between chain of emails.

(x) Paraben Forensic Tools is a Forensic Software for PDA, MobilePhone, text searching, data acquisition and email examination.

(xi) Netwitness is a network traffic security analyzer (securityintelligence) is used as a forensic incident response tool to gatherinformation from connected computers.

(xii) Forensic Network is a series of processing and imaging computersconnected and integrated directly with a high capacity server to shareresources. The file server operates as the core of the ForensicNetwork and can be used as a central storage facility for ForensicImages as well as applications software for use by the clientprocessing and imaging stations. Workstation clients on the networkperform the actual imaging and processing tasks, while the centralfile server stores the images and case work.

(xiii) CyberCheck Suite is a suite of forensic software tools to performdata analysis which has a capability of analysing storage media suchas hard disks, and optical media images and analyses images forevidences developed by C-DAC Thiruvananthapuram.

(xiv) Hot Pepper Technology

Authors of EMAIL Detective, a dedicated software solution for recovering and reconstructing AOL email. EMD is the most comprehensive AOExtraction tool available to forensic agencies.

(xv) Recommended Tools

The Collection of tools that is recommended is the “best of breed” of forensic tools has been listed in the Table. The computer forensic tools sets are categorized as good, the better and best.

S.No	Product	Supplier	Unix/ Linux	Windows	Analysis W=Windows U=Unix/Linux	Remote Capture	GUI	Requires Remote agents	Pre Forensic Audit
1	Coroners Toolkit	Open Source	Yes	No	U	No	No	No	No
2	Encase	Guidance Software/ Opentext	No	Yes	W,U	No	Yes	No	Yes
3	Forensic toolkit	Access data	No	Yes	W,U	No	Yes	No	Yes
4	i2 analyst notebook	i2 Inc	No	Yes	-	-	Yes	No	No
5	LogLogic	LogLogic	Yes	No	-	Yes	Yes	No	No
6	Mandiant First Response	Mandiant	No	Yes	W	Yes	Yes	Yes	Yes
7	Netwitness	Man Tech Intl.	Yes	No	-	-	No	No	No
8	ProDiscover Incident Response	Technology Partners	No	Yes	W, U	Yes	Yes	N	Yes
9	Sleuthkit and Autopsy Browser	Open source	Yes	No	W,U	No	Yes	No	No

Table 6.1 : Summary of Computer Forensic Tools

Choice of the tools should take into consideration the performance, reliability and repeatability and also the caveats. Having considered all these factors, examination should be performed utilizing these tools in a forensically sound manner following the best practices so as to enable admissibility of the evidence in the court of law.

6.9. Computer forensics examination environment

All equipment and procedures used during forensic procedures should be appropriate for that particular purpose and be properly maintained for operational considerations. Few basic points for the tools and other procedures followed:

- All media used in making copies must be forensically sterile.
- Equipment used must be accurate and suitable for the work.
- Properly validation of equipment's
- Proper Working Space/Forensics workstation should be in place.
- Regular audits
- Review of technique or procedure
- Following recommended guidelines for forensic procedures
- The methods, materials and equipment used have been demonstrated to be fit for their respective purpose
- There are appropriate quality control and quality assurance procedures in place for monitoring performance
- The technique or procedure is documented
- The results obtained are reliable and reproducible.

6.9.1. Forensic Environment

Laboratories and/or forensic workspaces for the examination of digital evidence items should be designed in proper manner where all the conditions like environmental, electrical, security related etc. are in place without any missing factors. The Forensic Workstation should be equipped with all tools required to carry out the investigation. The workstation should allow the investigator to make an efficient and effective use of the resources available. The area should also be very secure and safe.

6.9.2. Audit

Audits and update check of the quality system, procedures and practices should be conducted in a timely manner to ensure that quality and competency standards are maintained.

6.10. Chain of custody

Chain of custody refers to the chronological documentation or paper trail, showing the seizure, custody, control, transfer, analysis, and disposition of physical or electronic evidence. Chain of Custody is very important in criminal cases, it is vital that a chain of custody is maintained

for protection of potential evidence along with a detailed action performed onto that particular evidence, with whom it was lying etc. Chain of custody maintains a track of evidence from its original source (at the time of seizure) to what is offered as evidence in court. With digital evidence, a chain of custody is critical because the data can be altered or destroyed with relative ease.

Chain of custody is very CRITICAL and it relates to that documentation regarding how evidence is handled.

It establishes continuity of who/what/where about the digital evidence.

- Who collected the evidence?
- What comprises the evidence?
- When evidence was collected?
- If hardware (take a photo)
- Make, model, serial #
- Description of the evidence, technical information
- Name and signature of individual receiving evidence
- Case number & tag (bag & tag)
- If electronic, cryptographic hashes

Requirements for a chain of custody

The basic idea of chain of custody is to prove what all happened with this evidence, so the requirement from a chain of custody is to be a support, proof document related to the condition of evidence from the day it was seized to the day it is being presented in the court. The Chain of custody clears the facts related to the criminal case proceeding like:

- No information/data has been added or changed in the digital evidence and it gets clear when we use write blocker.
- Tells about working on the duplicate copy of the evidence by doing imaging process on the original. No changes have occurred can be later verified by comparing hash values of both original and the imaged copy.
- Tells about the condition of the evidence and how it was preserved.

- If any third party or some other person has handled the device that is also maintained and verified from chain of custody.

The following Table shows the various requirements for chain of custody forms as suggested by National Institute of Standards and Technology(NIST).

Anywhere Police Department

EVIDENCE CHAIN OF CUSTODY TRACKING FORM

Case Number: _____ Offense: _____

Submitting Officer: (Name/ID#) _____

Victim: _____

Suspect: _____

Date/Time Seized: _____ Location of Seizure: _____

Description of Evidence		
Item #	Quantity	Description of Item(Model, Serial #, Condition, Marks, Scratches)

Chain of Custody				
Item #	Date/Time	Released by (Signature & ID#)	Received by (Signature & ID#)	Comments/Location

EVIDENCE CHAIN-OF-CUSTODY TRACKING FORM (Continued)

Chain of Custody				
Item #	Date/Time	Released by (Signature & ID#)	Received by (Signature & ID#)	Comments/Location

Final Disposal Authority
Authorization for Disposal Item(s) #: _____ on this document pertaining to (suspect): _____ is(are) no longer needed as evidence and is/are authorized for disposal by (check appropriate disposal method) ☐ Return to Owner ☐ Auction/Destroy/Divert Name & ID# of Authorizing Officer: _____ Signature: _____ Date: _____
Witness to Destruction of Evidence Item(s) #: _____ on this document were destroyed by Evidence Custodian _____ ID#: _____ in my presence on (date) _____. Name & ID# of Witness to destruction: _____ Signature: _____ Date: _____
Release to Lawful Owner Item(s) #: _____ on this document was/were released by Evidence Custodian _____ ID#: _____ to _____ Name _____ Address: _____ City: _____ State: _____ Zip Code: _____ Telephone Number: (____) _____ Under penalty of law, I certify that I am the lawful owner of the above item(s). Signature: _____ Date: _____ Copy of Government-issued photo identification is attached. ☐ Yes ☐ No
This Evidence Chain-of-Custody form is to be retained as a permanent record by the Anywhere Police Department.

6.11. Organization who provide standards, best practices and guidelines

6.11.1.Association of Chief Police Officers (ACPO)

Association of Chief Police Officers functioning from Northern Ireland have designed and published a guide with procedures to be followed while collecting computer based evidence to ensure good field practice.

The Association of Chief Police Officers (ACPO), officially The Association of Chief Police Officers of England, Wales and Northern Ireland, was a not-for-profit private limited company that for many years led the development of policing practices in England, Wales, and Northern Ireland. Established in 1948, ACPO provided a forum for chief police officers to share ideas and coordinate their strategic operational responses, and advised government in matters such as terrorist attacks and civil emergencies. ACPO coordinated national police operations, major investigations, cross-border policing, and joint law enforcement. ACPO designated Senior Investigative Officers for major investigations and appointed officers to head ACPO units specialising in various areas of policing and crime reduction. ACPO was funded by Home Office grants, profits from commercial activities and contributions from the 44 police authorities in England, Wales, and Northern Ireland.

6.11.2.International Organization on Computer Evidence (IOCE)

The discipline continued to grow in the 1990s, with the first conference on collecting evidence from computers held in 1993. Two years later, the International Organization on Computer Evidence (IOCE) was established.

6.11.3.Scientific Working Group on Digital Evidence (SWGDE)

The Federal Crime Laboratory Directors group formed SWGDE in 1998. It was noted that the traditional audio and video examination and processing was becoming digital and, along with digital still photography, was converging with computer forensics. As a result, they formed a group to explore digital evidence as a forensic discipline.

The initial members were the forensic laboratories of the

- ATF,
- DEA,

- FBI,
- IRS-CID,
- US Customs,
- US Postal Inspection Service, and
- US Secret Service.

In addition,

- NASA and
- Department of Defense Computer Forensics Laboratory participated from the beginning.

In an effort to widen the participation, representatives from North Carolina, Pennsylvania, and Illinois State Crime Laboratories were invited to participate along with the Florida Department of Law Enforcement. Later on, representatives of other state and local agencies (including Ocean City, Maryland and Lakewood, Colorado) were accepted for membership. Many of those state and local agencies discovered and joined SWGDE by attending a presentation given at IACIS, HTCIA, GMU, as well as announcements sent to IACP and various e-mail lists. It should be noted that the members of SWGDE individually have many years of digital forensic experience. Most of the members of SWGDE are members of prominent organizations such as

- American Academy of Forensic Science,
- IACIS,
- HTCIA, and
- IOCE, and many are officers of these organizations.

Early discussions revealed that, in fact, there were many similarities and overlap in the disciplines of audio, video, image analysis and computer forensics and that it was desirable to develop standards and harmonize operations among these disciplines. As a result, the Scientific Working Group on Digital Evidence was formed, under the same umbrella that covers the

- Scientific Working Groups on DNA (SWGDAM),
- Questioned Documents (SWGDOC),
- Trace Evidence (SWGMEAT), Drugs (SWGDRUG),

- Fire and Explosives (SWGEX),
- Fingerprints (SWGFAST), and
- Imaging (SWGIT).

6.11.4.Digital Forensic Research Workshop (DFRWS)

Digital Forensic Research Workshop is a non-profit, volunteer organization dedicated to bringing together everyone with a legitimate interest in digital forensics to address the emerging challenges of our field. DFRWS organizes digital forensic conferences, challenges, and international collaboration to help drive the direction of research and development.

DFRWS conferences provide a friendly atmosphere to share research papers, practitioner presentations, and works in progress. Every gathering includes technical workshops, demos, panels, and other “breakout sessions” covering various issues related to digital forensics.

Ultimately, it is the goal of DFRWS to cultivate trans-disciplinary coproduction of knowledge that stimulates healthy growth in this rapidly evolving field. It is notable that many novel developments in the field have their roots in works and breakout sessions at DFRWS conferences. As such, DFRWS conferences are not only a snapshot of the state of the research in the field, but are also a useful pointer towards the future.

Fostering trans-disciplinary approaches, to address emerging challenges in digital forensics by bringing together innovative researchers, developers, and practitioners from around the globe.

The first Digital Forensic Research Workshop was held in August 2001. The vision was to bring together researchers, practitioners, industry, academics, law enforcement, and military with the goal of opening lines of communication. The result of this inaugural event was the technical report “A Road Map for Digital Forensic Research” that is frequently referenced as a foundational framework for the domain.

Since then, an annual conference has been held in various cities around the United States and Europe to present and discuss current and emerging challenges in digital forensic science. The scope of DFRWS has increased to include annual challenges and technical working groups. In 2005, Digital Forensic Research Workshop, Inc. became a 501(c)(3) non-profit organization.

DFRWS is about

- INCLUSION: Reaching out to everyone with a legitimate interest in digital forensic research.
- INNOVATION: Pushing the envelope of what is currently possible in digital forensics
- USABILITY: Practical solutions for real-world problems.
- SCIENTIFIC PRACTICE: Systematic processes and reliable results

6.11.5 International association of computer investigative specialist (IACIS)

On October 23, 1989, the International Association of Computer Investigative Specialists (IACIS) was created. On this auspicious day in history, twenty like-minded law enforcement professionals assembled to hammer out the framework that was destined to become arguably one of the most influential computer forensic organizations in the world. Although that initial meeting lasted less than two hours, the organization's name, objective/goals, manner of incorporation, organizational structure, director terms, means of selecting directors, authority and responsibilities of directors, membership type and qualification, conditions of membership status, and membership dues were all decided in a meeting that lasted less than two hours.

One of the key issues being decided during that meeting was what to call this new fledgling organization. Ultimately, the choice fell between the International Association of Computer Investigative Specialists (IACIS) and the International Fraternity of Computer Investigative Specialists (IFCIS). It was also decided that membership in IACIS would be open to sworn law enforcement personnel that had successfully completed the SCER training.

In February 1991, IACIS held its first training class at held in Portland, Oregon. Some of the topical courses taught during this first training event included: preparing search warrants, undeleting files, file compression, and ethical use of software. It is interesting to note that many years later; these foundational courses are still an integral part of the IACIS training curriculum being taught today.

From its very inception, IACIS was built upon two foundational pillars – training and certification. Aside from training and foundational to its core purpose, IACIS has been instrumental in the development and implementation of a carefully constructed certification process for its membership. In those earlier days, those successfully completing the IACIS training were granted

the DOS Seizure Certificate (DSC) and offered the opportunity to complete an additional set of post-training problems and testing. Those successfully completing the problem set and rigorous testing were granted the DOS Processing Certificate (DPC).

6.11.6.High Technology Investigation Association (HTCIA)

The High Technology Crime Investigation Association (HTCIA) was formed to provide education and collaboration to our global members for the prevention and investigation of high tech crimes. HTCIA aspires to help all those in the high technology field by providing extensive information, education, collective partnerships, mutual member benefits, astute board leadership and professional management. The HTCIA is a registered not-for-profit association and its mission is to provide education and collaboration to our global members for the prevention and investigation of high tech crimes. Their core Purpose is to promote collaboration, education and training.

6.11.7 National Institute of Standards and Technology (NIST)

NIST's mission was to promote U.S. innovation and industrial competitiveness by advancing measurement science, standards, and technology in ways that enhances economic security and improves quality of life.

NIST's vision was to be the world's leader in creating critical measurement solutions and promoting equitable standards. Their efforts was to stimulate innovation, foster industrial competitiveness, and improve the quality of life.

NIST's core competencies include:

- Measurement science
- Rigorous traceability
- Development and use of standards

NIST's core values

NIST is an organization with strong values, reflected both in history and current work. NIST leadership and staff will uphold these values to ensure a high performing environment that is safe and respectful of all. Core values include:

- **Perseverance:** Planning the future with scientific knowledge and imagination to ensure continued impact and relevance for stakeholders.

- **Integrity:** Ethical, honest, independent, and provide an objective perspective.
- **Inclusivity:** Collaborative work to harness the diversity of people and ideas, both inside and outside of NIST, to attain the best solutions to multidisciplinary challenges.
- **Excellence:** Rigorous and critical thinking to achieve world-class results and continuous improvement in everything.
- NIST implements practical cybersecurity, cyber forensics and privacy through outreach and effective application of standards and best practices necessary for the U.S. to adopt cybersecurity capabilities.
- With a world-class measurement and testing laboratory encompassing a wide range of areas of computer science, mathematics, statistics, and systems engineering, NIST's cybersecurity program supports its overall mission to promote U.S. innovation and industrial competitiveness by advancing measurement science, standards, and related technology through research and development in ways that enhance economic security and improve our quality of life.
- The need for cybersecurity standards and best practices that address interoperability, usability and privacy continues to be critical for the nation. NIST's cybersecurity programs seek to enable greater development and application of practical, innovative security technologies and methodologies that enhance the country's ability to address current and future computer and information security challenges.

6.11.8.ISO 27037

ISO 27037 is the first of a developing family of international standards that seek to create a common baseline for the practice of digital forensics. It is not intended to replace local laws or usurp local and national governments' authority to regulate the practice of digital forensics. Rather, its intent is to facilitate the usability of evidence obtained in one jurisdiction by a legal process operating in another jurisdiction.

As its title suggests, ISO 27037 only addresses the initial steps of the forensics process: identifying, obtaining and preserving potential digital evidence. Other steps in the forensics process are the subject of additional standards currently under development. The same is illustrated in the figure

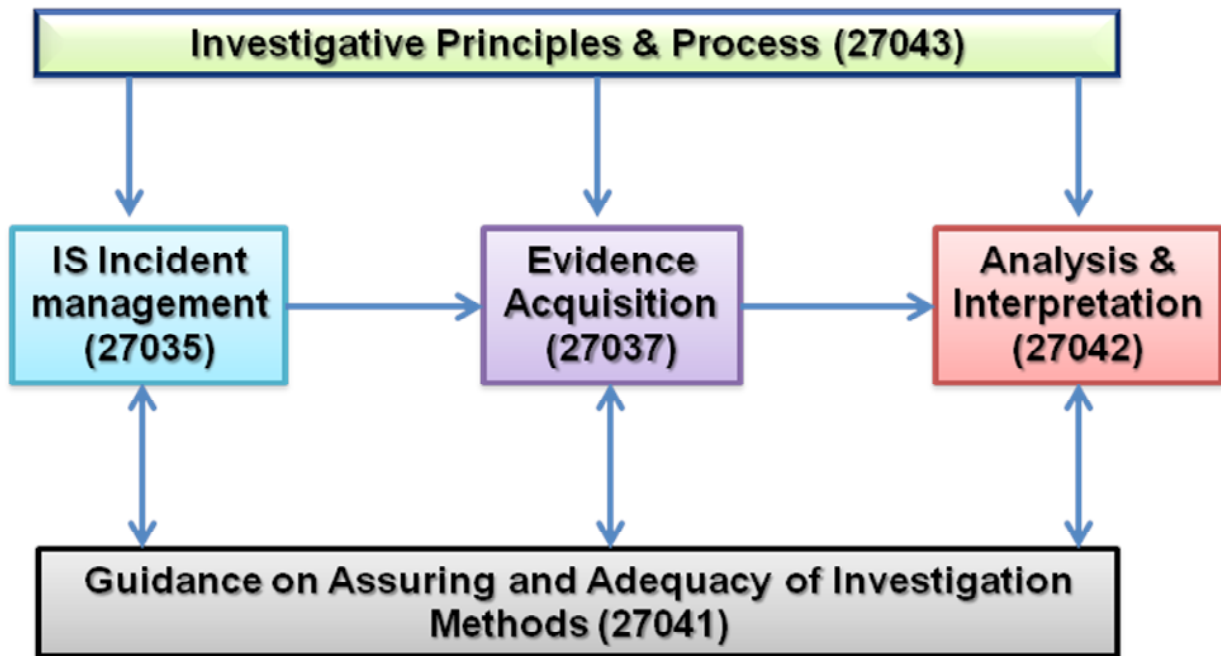


Figure 6.6: Developing International Standards

Evidence handling process according to ISO 27037 is illustrated in the figure 4.2.

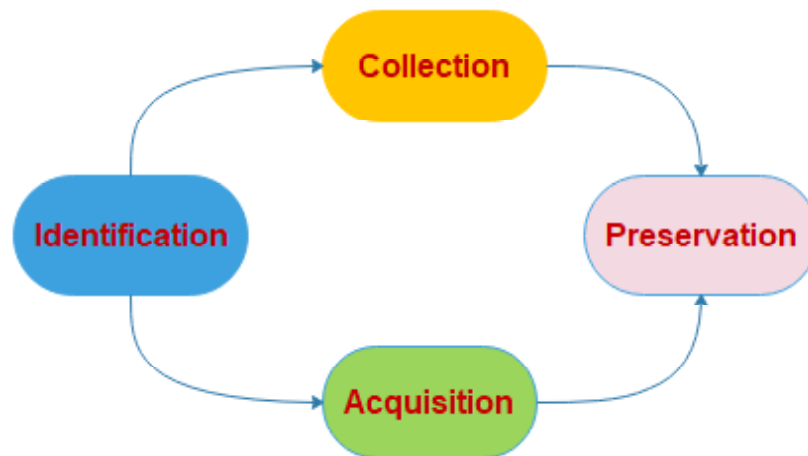


Figure 6.7: Evidence handling process (27037)

Identification

The forensics process begins with the identification of items that may be or may contain potential digital evidence. Formally, identification is the *“process involving the search for, recognition and documentation of potential digital evidence”*.

Although the identification of potential digital evidence sounds simple in principle, there are subtle complexities. For example, digital evidence has both a physical and virtual representation. Consider a hard drive containing potential digital evidence. The physical location of the evidence is the hard drive, but the evidence itself is the data contained within the drive. Furthermore, it also may not be at all obvious where potential digital evidence is housed. A server may have very few directly attached disks and have a significant part of its storage within a SAN or NAS. As will be discussed later, these aspects of the cloud environment compound the difficulties in identifying relevant evidence.

Collection and Acquisition

After potential digital evidence is identified, it must either be collected or acquired:

- Collection – *“Process of gathering items that contain potential digital evidence.”*
- Acquisition – *“Process of creating a copy of data within a defined set.”*

Collection is roughly equivalent to the standard law enforcement practice of seizing items containing potential digital evidence under authority of a legal order (i.e., search warrant) and removing them to a forensics lab or other facility for processing and analysis. Acquisition is more common in the private sector due to the need to minimize business impact of an ongoing investigation. Similar concerns with reducing the impact on other applications and customers will make acquisition the more likely process in the cloud environment as well.

It should be noted that the copy created during acquisition can range from the forensic image of a hard drive to a copy of the contents of a server's memory to the logical contents of an individual user's email box depending on the purpose and scope of the investigation. In all cases, the requirements for the copy are very similar: it must be made using a well-understood, defensible, well-documented process. Furthermore, the process must include integrity measures to ensure that the copy has not been modified since acquisition. The wide variety of potential digital evidence to be copied, and the requirements on the copying process, make acquisition a more complex and challenging process than collection.

Preservation

Once potential digital evidence has been collected or acquired, it must be preserved. ISO 27037 defines preservation as the *“process to maintain and safeguard the integrity and/or original condition of the potential digital evidence”*. The preservation of potential digital evidence is a complex and important process. Evidence preservation helps assure admissibility in a court of law. However, digital evidence is notoriously fragile, and is easily changed or destroyed. Given that the backlog in many forensic laboratories ranges from six months to a year (and that delays in the legal system might create further delays), potential digital evidence may spend a significant period of time in storage before it is analyzed or used in a legal proceeding. Storage requires strict access controls to protect the items from accidental or deliberate modification, as well as appropriate environment controls.

Summary

- Standards provides guidelines that encapsulate idealized models, for common incident investigation processes across various incident investigation scenarios involving digital evidence. This includes processes from pre-incident preparation up to and including returning evidence for storage or dissemination as well as any general advice and caveats on such processes.
- -The guidelines describe processes and principles applicable to various kinds of investigations. An organization's forensics guidelines should include general methodologies for investigating an incident using forensic techniques, since it is not feasible to develop comprehensive procedures tailored to every possible situation.
- Best practice guides are available to identify and promote valid and reliable digital forensics processes and outcomes. Best practices also serve to encourage professionals to adopt a consistent methodology to facilitate the interchange of data. The importance of the best practices or guidelines set by the renowned organizations is to make the investigation procedure universal, a synchronization for carrying out investigation where ever it is carried out in the world. Best practices guide also help to carry out the investigation fulfilling the Legal criteria.
- A single neglectful act can result in the loss of information (digital). An investigator must have a thorough understanding of what is to be recovered from crime scene or in other words what possible evidences can be gathered.

- The search at a crime scene is a very first step an investigator or first responder takes towards a progressive forensic investigation. Areas should be searched systematically and thoroughly for digital evidences and related material. To increase investigative efficiency, areas should be targeted and prioritized according to how likely digital evidence may be found and the amount of digital data that may be recovered.
- Securing the crime is most important within the premises where crime occurred. Check for the extent to which scene has been protected. Gather information from personnel who have entered scene and have knowledge about it. Keep on making the notes. Keep out unauthorized personnel from accessing the crime scene.
- Antistatic covers to be used to insulate media from the electromagnetic fields.
- There are Cardinal Rules of Evidence which must be followed by the experts as a part of best practice and these principles include: Never mishandle evidence; never work on original evidence; Use proper software utilities; never trust the suspect operating system; Document everything.
- Computer Forensics examination cannot be possible without trained personnel. A computer forensic investigation cannot be complete without having the required set of: Skilled personnel; quality assurance and proper tools & facilities.
- Computer forensics Quality Process Manual covering all systems, processes and methods used in the examination and reporting of forensic digital evidence should be made available at all forensic laboratory.
- The Standard Operating Procedures should define everything starting from **Incident identification**– Where an event is recognized as an incident it should be classified into a type of incident; **Preparation** – Detailed procedures for the proper preparation of tools, techniques, monitoring authorizations and management support; **Approach strategy** – Provide details about the best practices and guidelines on how approach for search and seizure of evidences so that positive result is guaranteed; **Preservation** – Isolating, securing and preserving the state of physical and digital evidence; **Collection** – Procedures and processes for recording the physical scene and duplication of digital evidence using standardized and accepted practices; **Documentation** - Maintaining record of actions taken during an investigation; **Analysis** – Look for evidences and draw conclusions based on evidence found;

Presentation – Provide a structured and clear report which explains the conclusions reached. It shouldn't be more technical, non-technical person should be able to understand easily.

- Forensic laboratory should have the facilities such as forensic hardware, software, devices, write blockers, UPS
- Chain of custody refers to the chronological documentation or paper trail, showing the seizure, custody, control, transfer, analysis, and disposition of physical or electronic evidence. Chain of Custody is very important in criminal cases, it is vital that a chain of custody is maintained for protection of potential evidence along with a detailed action performed onto that particular evidence, with whom it was lying etc. Chain of custody maintains a track of evidence from its original source (at the time of seizure) to what is offered as evidence in court
- Association of Chief Police Officers functioning from Northern Ireland have designed and published a guide with procedures to be followed while collecting computer based evidence to ensure good field practice.
- The Federal Crime Laboratory Directors group formed SWGDE in 1998. It was noted that the traditional audio and video examination and processing was becoming digital and, along with digital still photography, was converging with computer forensics. As a result, they formed a group to explore digital evidence as a forensic discipline.
- Digital Forensic Research Workshop is a non-profit, volunteer organization dedicated to bringing together everyone with a legitimate interest in digital forensics to address the emerging challenges of our field. DFRWS organizes digital forensic conferences, challenges, and international collaboration to help drive the direction of research and development.
- International Association of Computer Investigative Specialists (IACIS) was created and it held its first training class at held in Portland, Oregon. Some of the topical courses taught during this first training event included: preparing search warrants, undeleting files, file compression, and ethical use of software. From its very inception, IACIS was built upon two foundational pillars – training and certification. Aside from training and foundational to its core purpose, IACIS has been instrumental in the development and implementation of a carefully constructed certification process for its membership. In

- The High Technology Crime Investigation Association (HTCIA) was formed to provide education and collaboration to our global members for the prevention and investigation of high tech crimes.
- NIST's mission was to promote U.S. innovation and industrial competitiveness by advancing measurement science, standards, and technology in ways that enhances economic security and improves quality of life.

Check your answers

Explain the following

1. Standards, Guidelines and Best Practices
2. Handling the Digital Evidence
3. Digital Evidence Examination Guidelines
4. The principles of forensic examination
5. Proper facilities

LESSON - 7

COMPUTER FORENSICS

Learning Objectives

After reading this unit, you will be able to

- Ø Define Cyber Forensics and its Importance
- Ø Understand the Computer Forensic Life Cycle
- Ø Knowledge about best practices to be followed
- Ø Issues in Computer Forensics

Structure

- 7.1 Cyber Forensics Definition, Importance and Objective**
- 7.2 Computer Forensic Lifecycle**
- 7.3 Best Practices to be followed**
- 7.4 Where to look for Evidence**
- 7.5 Retrieving and reviewing logs**
- 7.6 Imaging of HDD**
- 7.7 Registry Analysis**
- 7.8 RAM Analysis**
- 7.9 Tools used in Cyber Forensics**
- 7.10 Issues in Computer Forensics (Legal/Technical/Anti-Forensics)**

7.1 Cyber Forensics

In the very first unit of Cyber Forensics we will be learning about the goal of cyber forensics, and the ways of finding the digital evidences along with the legal procedures to be followed. Cyber forensics follows the investigative procedure for cyber-crimes. Crime crimes are the latest in the series of crime that occur in our society with the increase and dependency on technology and internet. We will start with looking into what cyber forensics is and what procedures to be followed.

7.1.1 Definition

Cyber forensics is a systematic procedure for collection of data from computers, networks and other storage devices for analysis of the digital evidence and its presentation so that the evidence is admissible in the court of law.

Other definitions related to cyber forensics:

Computer forensics uses computer investigation and analysis techniques to collect evidence regarding what happened on a computer that is admissible in a court of law.

“The application of computer investigation and analysis techniques to gather evidence suitable for presentation in a court of law”

CERT-US about computer forensics: “We define computer forensics as the discipline that combines elements of law and computer science to collect and analyze data from computer systems, networks, wireless communications, and storage devices in a way that is admissible as evidence in a court of law”.

7.1.2 Importance

Let's understand it in this way, we are now seeing a change in the world with technology becoming a part of our life and emergence of a new type of crime which is more related to technology or we can say a bit more technical. In the case like frauds, harassment, child pornography many evidences and other relevant document files are easily found on the system which suspect has used. It is not about just opening the computer and simply looking into files and draws a conclusion. It is much more than that; there is a scientific investigation procedure to be followed during the process otherwise our evidence won't be admissible in the court of law. So in order to safeguard the evidence cyber forensics plays a very important role.

7.1.3 Objective

The objective of a Computer Forensic is to help in identifying computer systems, HDD, media or other devices contains potential evidence and secure that storage media, and it will be followed by creating a duplicate image of the original device in order to preserve the integrity of the evidence. The use of special digital forensic tools in order to simplify the extraction of digital evidence from the suspect devices so, that it can be produced in the court of law and are accepted. The aim of cyber forensics is to find the digital evidences that are supposed to be the

clue in solving a criminal case. The cyber forensics gives a whole new dimension to the ways of dealing with cyber-crimes.

7.2 Cyber Forensics Life Cycle

The Digital forensic Investigation needs to follow a systematic procedure for carrying out its investigation. Mainly there are six stages in a cyber-forensics life cycle and starts from:

- Identification
- Seizure
- Acquisition
- Authentication
- Analysis
- Presentation

Identification

The very basic step involved in cyber forensic life cycle is the Identification of the digital media that can be helpful in getting the evidence, what do we mean, whenever a forensics expert reaches a crime scene the most important thing is to search for the material that is present at crime scene and look for whatever digital device are present so that later we can do an investigation onto that. Along with identifying the evidences we must label them, which simply means to mark whatever we found.

Seizure

The next step after identifying the evidence is to take that from the crime scene to the forensic workstation for further analysis. We need to follow best practices recommended by the experts and law enforcement agencies. Different procedures for seizure are to be done if we have found a mobile phones, laptops, computers etc. for example if we found a laptop which is an ON state we should never turn it off, and vice versa. All these best practices will be discussed later. The forensics expert must carry specialized hardware and software based tools to the crime scene for seizure of the device or in some cases live forensic investigation must be carried out if required. Gather all the evidences, protect them and carry them to the forensic workstation, most important is maintain chain of custody.

Acquisition

This step is also known as IMAGING process, there is something called as cardinal rule of forensics in which one of the key points is never work on the original. Same applies to the Digital forensics investigation, we can never work on the original media chances are we may tamper the evidence, that way data integrity will be lost and our case obviously is also gone. So we must create a duplicate media of the original so that even if we might corrupt the duplicate data accidentally, still we have preserved an original copy. Disk imaging differs from just copying, all the files on a disk are same in copying but in imaging the disk structure and relative location of data on the disk are preserved. Also will performing imaging its must that we use write blockers which will help to prevent the data get written onto original disk, it becomes read-only thus helping in maintaining data integrity. Also before doing imaging or alongside the process hash value is to be taken of the original device. Hashing is a one way function that creates a unique value for a particular file; the hashing helps us to ensure data integrity is maintained by comparing the hash values of both the original and the duplicate media files.

Authentication

As already described above what hashing means in this stage after creating duplicate image hash value of the already generated Original device is compared with just now created image file of the same media. Both the values should match. If the hash values don't match then the data will be considered as tampered. There is no point of continuing with the investigation then. This step authenticates that the data integrity is maintained and no manipulation is done onto the original disk.

Analysis

Now after the image is made, Forensic expert has to start the examination of the device, extract all the information possible from the device as well as if necessary perform recovery of the deleted data. As per his expertise he/she knows where to look for evidences inside the hard disk or other storage devices. Also it is recommended to use registered forensic tools and not any open source tools available in the market.

Presentation

This stage is also known as documentation phase, some mention it as reporting and review phase. What is to be done starts after completing the analysis and interpretation, forensic

expert now has to generate a report in which he/she has to write down all the findings during her analysis of the suspect media. It is must that the report written should be clear for understanding and it should have a proper structured format.

7.3 Best Practices in Evidence Collection

One of the most important phase of cyber forensics investigation process is collection of evidence. Whenever we reach to a crime scene as a first responder we have to be very cautious with the finding of evidence and its collection because a single negligence can result in tampering of evidence and basically we are dealing with electronic devices where the data can be volatile. So, we have best practices for the collection of evidences, the most widely followed best practice guide is from the Association of Chief Police Officers (ACPO) UK Police, there website is www.acpo.police.uk, they released a “Good Practice Guide for Computer-Based Electronic Evidence”. Its available on internet for free download.

7.4 Various Components where Evidence may be Hidden

Where data can be kept hidden within HDD,like in;

- hidden or deleted logical volumes on the hard drives
- Slack spaces
- Unallocated spaces
- deleted volumes, files and emails

7.6 Imaging of HDD

So, imaging is the process of making an exact copy of a disk. Imaging is sometimes also called disk cloning or ghosting, but the latter terms usually refer to images created for purposes other than evidence preservation.

Disk imaging differs from just copying, all the files on a disk are same in copying but in imaging the disk structure and relative location of data on the disk are preserved.

When you create a disk image (a bit stream copy), each physical sector of the disk is copied so that the data is distributed in the same way, and then the image is compressed into a file called an image file, This image is exactly like the original, both physically and logically. We usually perform bit stream imaging because in bit stream imaging exact copy of every data bit

by bit is created from the original storage device, including all hidden files, temp files, corrupted files, file fragments and erased files that have not yet been overwritten.

Need for Imaging

- As per cardinal rule never work on the Original.
- To Preserve the Evidence.
- Later it is required to prove the Integrity of the evidence.
- If by mistake or during the operation any error occurs by which our evidence is tampered we can still acquire the Image from the Original Evidence.

Types of Imaging Processes

The imaging process can be of two types:

1. DEAD Imaging
2. LIVE Imaging

Whenever we power off the system and remove the hard disk and perform acquisition at forensic workstation using hardware/software device is known as Dead Imaging.

Whenever imaging process is to be done on a live system is known as Live Imaging.

There are both Hardware and Software based tools available for the Imaging process, some of most widely used tools are;

Hardware based tools:

- True Imager
- DIBS (RAID/PERU)
- LOGICUBE

Software based tools:

- FTK
- EnCase
- Dd (Linux utility)
- Safe Back

HDD

Hard disk drive is a most widely used data storage device for keeping the information permanent and is accessible. HDD are used because of being more reliable and has got very high performance. The best part of HDD is its storage space which is increasing with the technology nowadays we have HDD storage in Terabytes. The picture below is of the HDD and its various components.

RAM

RAM is also a type of data storage device but it's Volatile, which means it doesn't store information permanently; the information is lost once the system is turned off. Also the process of reading and accessing the file is stored temporarily in RAM so that next time if the access is needed to that particular file it should be very fast. RAM is used to optimize the system performance but there is also a significant importance of RAM in cyber forensics mostly in the case of LIVE Forensics where we are supposed to take a dump of the RAM, which can be later analyzed and surely helps in providing the details about the systems running activities.

Register Analysis

Introduction to Registry:

The Windows registry serves as a primary storage location for system configurations and other information. System, users, applications and hardware in Windows make use of the registry to store their configuration and it is constantly accessed for reference during their operation. Windows 9x/ME, Windows CE, Windows NT/2000/XP/2003 store configuration data in registry. The registry is introduced to replace most text-based configuration files used in Windows 3.x and MS-DOS, such as .ini files, autoexec.bat and config.sys.

Registry Structure:

Registry consists of two basic elements KEYS and VALUES

Keys

- Keys are similar to folder; each key can contain sub keys, which may contain further sub keys, and so on.
- Keys are referenced with syntax similar to Windows' path names, using backslashes to indicate levels of hierarchy.

- E.g. HKEY_LOCAL_MACHINE\Software\Microsoft\Windows refers to the subkey “Windows” of the subkey “Microsoft” of the subkey “Software” of the HKEY_LOCAL_MACHINE key.

Values

- Name/data pairs stored within keys.
- Each key has one or more values.
- There are 3 parts in a value, which are Name, Type and Data.

There are 5 Root Keys (also called Hives) in the registry.

- HKEY_CLASSES_ROOT(HKCR)
- HKEY_CURRENT_USER(HKCU)
- HKEY_LOCAL_MACHINE(HKLM)
- HKEY_USERS(HKU)
- HKEY_CURRENT_CONFIG(HKCC)

We can check out details like Most Recently used documents, Storage Devices connected to the system, Networks to which system was connected, system and user related information, virtual machine and much more.

E.g. how to get details about mounted devices onto the system and recently opened documents.

- Computer\HKEY_LOCAL_MACHINE\SYSTEM\Mounted Devices
- Computer\HKEY_CURRENT_USER\Software\Microsoft\Windows\Current Version\Explorer\RecentDocs

7.8 RAM Analysis

The type of data inside the system is either persistent or volatile, the persistent data is the one which resides onto the system even if power is turned off like HDD, and the volatile data never resides on the system once the power is turned off all the information is lost, like in the case of RAM memory. So, in order to get the information from the RAM while system is ON we need to take a RAM dump which can be analyzed later on. As per the order of volatility described

by different areas of research in the subject, first information to be captures from the system is from the RAM. RAM Analysis is must when performing LIVE Analysis of the system.

Tools used for RAM Analysis:

- COFEE tool
- Windd
- WindowsSCOPE
- Volatility

Also the study proves that RAM can be analyzed after power loss, because the electrical charge stored in the memory cells takes time to dissipate, an effect exploited by the cold boot attack. The length of time that data is recoverable is increased by low temperatures and higher cell voltages. But in most of the circumstances it is not at all possible; we must have that controlled temperature in room to continue the operation.

7.9 Computer Forensic Tools available

IMAGING: Both Hardware and Software based tools are available

- True Back Imager (Hardware)
- LOGCUBE (Hardware)
- DIBS RAID-PERU (Hardware)
- FTK Imager (Software)
- EnCase (Software)
- ProDiscover (Software)
- Safe Back (Software)

ANALYSIS: for analysis of the device image most widely used tool are:

- EnCase
- FTK
- Cyber Check Suite
- Autopsy/Sleuth kit

RAM ANALYSIS/LIVE FORENSIC ANALYSIS:

- Volatility
- COFEE
- Live Response tool
- Live RAM Capture
- Xplico

DATA CARVING TOOLS:

- Bulk Extractor
- Foremost

MOBILE FORENSIC TOOLS:

- Paraben Device seizure
- Cellbrite Mobile Forensics
- Oxygen Forensic suite

7.10 Issues in Computer Forensics

The cyber forensic or we can say cyber forensic investigation is facing a lot of issues or I would say challenges regarding many new techniques coming to evade and hinder the forensic investigation process. For example let us just say about data recovery, while examining the data we use sophisticated tools to recover deleted data in order to deter the forensic process many software tool are available which criminal can use in order to over write the data onto the disk thus making it almost impossible to recover data. Similarly steganography makes it easy for criminals to hide the message within a different file. So, every other domain whether its memory, network, mobile etc. have their own issues and challenges for cyber forensics. Some basic issues related are:-

Encryption

Encryption topic is related to Cryptography; Encryption is the technique to encode messages. The plaintext message is encrypted by an encryption algorithm using randomly generated key, so the message changes from plaintext to cipher text, now the only way to retrieve back this message is when we decrypt the cipher text using that particular key. The

person who encrypted this data can easily get access to the message using the key but we as an investigator don't know the key so, it gets hard for us to decrypt the message. These techniques are used by the criminals often to hamper the investigation process.

Increasing Amount of Data

Previously we use to devices with minimum data capacity only in Gigabytes, but now we have data capacity in Terabytes, so the question is how to deal with the devices having more storage capacity, how much time will it take us to complete the imaging process and also analyzing the entire disk. There is a time constraint in forensic processes to crack the case and that too if we get the data drive of more than Terabyte of capacity it is really going to take more time. So, the ways to manage and investigate more data storage devices is getting difficult and is going to be a serious issue in the future.

Evolving Dynamic Technologies

As we have seen how technology is changing and we have seen a boom in technology from last ten years, and are advancing at a very rapid rate. Few of the emerging technologies like Virtualization, Cloud computing will make it more difficult for forensic investigators to look for forensic artifacts. With the coming of new devices and operating systems, up to date knowledge alongside the growing technology is required in dealing with these devices and Technology.

Legal Issue

Forensic is actually meant to provide a scientific procedure to the crime occurred and is related to the court of Law. E.g. let us talk about the Indian Penal code (jurisdiction of our own country) we can't simply go and perform Forensic investigation at some place, there are Laws and rules which deal with criminal procedures whether it is FIR, Seizure, search and other procedures, all of them are compulsorily meant to be followed. Cyber Law (IT ACT 2000) deals with the Computer related crimes and punishments.

International cooperation is very important in case of cybercrimes, since we are connected with internet we can use computer from any part of the world and criminal can be simply sitting in one country and does illegally hacks into the system of other country. If countries don't have cyber laws enacted in their states and no cooperation is seen between the countries how can there be a trial for the criminal, how will he be arrested? And many other issues remain unclear if international cooperation does not exist.

Another important and mostly relevant issue with the cyber-crimes is the offences committed by the juveniles. What law says about the juvenile if he commits a crime? College going students usually try to do hacking attempts which finally land them in jail but the question is what law has to say about the juvenile. Is it right to charge him on this crime or not and many other questions always revolve in case of Juveniles.

Anti Forensics

Anti-Forensics are different methods used by a person to hamper/clog forensic investigation process. The prime objective of using such techniques is to make it difficult for investigators to find the traces of the evidence.

The Anti-Forensics Techniques are classified into four types:

- Data Hiding
- Artifact Wiping
- Trial Obfuscation
- Attack on Forensic itself

The Data hiding is about looking out for ways to hide data inside the system. Some of the common techniques of data hiding are Steganography and Encryption.

In case of artifact wiping the use tools like disk clean which wipe the entire data from the hard disk are used to delay the investigation procedure.

Trial Obfuscation is mainly used to create confusion to the investigator so that the path of investigation gets diverted. The use of secure protocols, removing of log files from the systems, being anonymous while using the services and many other methods make it very hard for the investigating person to get the details of the event in depth.

Nowadays different ways have been identified where the vulnerabilities in the forensic tools is exploited, there is something called timestamp in the system which is related to the every file created onto the disk, the tool have been made which change the time stamp of the files. Or one more example if we want to delay the process of investigation we simply break the device into pieces, this simple step also has an impact and cause delay in the investigation process.

Summary

So, till now we have understood about the cyber forensics and how much important role it plays in tackling cyber-crimes, then, we learned about the Forensic Life cycle that needs to be adopted in cyber forensic investigation along with best practices that need to be carried out. Then we had an overview about where evidences can be kept hidden. We also started Imaging process and different tools used in cyber forensics, then we looked into what is meant by registry analysis and RAM analysis. In the end we had a look into different issues and challenges that are coming in the way of cyber forensic investigation.

Check your answers

1. Define Cyber Forensic.
 2. Describe Computer Forensic Life Cycle.
 3. What is imaging and what tools are used for imaging.
 4. What is a Write Blocker?
 5. What is DEAD and LIVE Imaging?
 6. What do you mean by Live Forensics? Explain RAM analysis.
 7. What is the purpose of Registry from Forensic prospective?
 6. Discuss in detail the various issues and challenges to cyber forensics.
 7. Discuss in detail importance of cyber forensics.
-
1. Volatility tool can be used for imaging. True / False
 2. Autopsy/Sleuth Kit is a data imaging tool. True / False
 3. Steganography is not an anti-forensic technique. True / False
 4. Steps involved in Digital Forensic Life Cycle _____,
_____, _____, _____,
_____, and _____.
 5. To prove data integrity we use _____ for authentication.

LESSON - 8

LIVE FORENSICS

Learning Objectives

After reading this unit, you will be able to

- Ø Understand about the Live Acquisition of the Evidences
- Ø Best practices for live forensics
- Ø Knowledge about Imaging process
- Ø Understanding the importance of the Documentation process

Structure

- 8.1 Live Acquisition of Evidence**
- 8.2 Live Network Analysis**
- 8.3 RAM Analysis**
- 8.4 Photography at the Crime scene**
- 8.5 Imaging of HDD using Write blockers**
- 8.6 Documentation process**

8.1 LIVE ACQUISITION

The Acquisition techniques are of two types, one in which the image of the device is taken at the forensic workstation and not directly on site while as the live acquisition is taking of image on-site while the system is still running. During Live acquisition all the details of the processes running onto the system, network related details and hard disk image are taken without seizing the system. The emphasis is given on collecting volatile data.

The live acquisition process can be bit complicated to perform, all the running programs information is to be collected. The live acquisition is very beneficial also when we talk about the current status of the system and we can also retrieve the details of deleted data which can be present as a temporary files. These details are very crucial and also we see day by day the data storage capacity is increasing and it is getting very complex to do imaging process of the entire disk, so in such cases live acquisition proves to be very crucial. In order to make our task quite

simple and manageable in lesser time we use software tools like COFEE from Microsoft which helps in taking RAM dump and sys log files for the later analysis.

8.2 LIVE NETWORK ANALYSIS

The cyber forensics is divided into many sub parts like Media forensics, Device forensics, and Operating system forensic etc. Similarly one such part is network forensics, in which the details about the techniques and ways to be adopted for collection of evidence in networks is discussed.

8.2.1 Objective

The objective of network forensic is to do analysis of the network connections and network traffic in order to prove the incident. The network forensic also can be done offline as well as online. The offline depends on log files analysis etc. while as live network forensics is about capturing the real time traffic coming into the network, monitoring of the network and IDS systems. The network forensics helps to know who the source of attack is and learn about the attacker's behavior.

8.2.2 Steps

The steps involved in live network forensics are:

- Capture
- Fetch
- Analysis
- Reporting

CAPTURE: In this step the network traffic is to be captured, using packet sniffing tools. Network tapes can also be used to access the data going in the network. Some of the most common tools used are Wireshark and tcpdump.

FETCH: In this step all the log details must be collected whether it is from routers, systems, firewall, IDS/IPS etc. so, that we can correlate the stuff while doing investigation.

ANALYSIS: In this step the search is made about a particular type of data we are looking for. Search can be made just for looking into protocols, or the entire captured packet can be

inspected. Also reconstruction of TCP and UDP stream is done to understand the communication taking place. The most common tool used for analysis is Wireshark.

REPORTING:All the findings, evidences related to the case are to be noted down, as we have seen in the digital forensic life cycle already. The reporting should be done in a way so that it is understandable and not much complex.

8.3 RAM ANALYSIS

RAM is also a type of data storage device but it's Volatile, which means it doesn't store information permanently; the information is lost once the system is turned off. Also the process of reading and accessing the file is stored temporarily in RAM so that next time if the access is needed to that particular file it should be very fast. RAM is used to optimize the system performance but there is also a significant importance of RAM in cyber forensics mostly in the case of LIVE Forensics where we are supposed to take a dump of the RAM, which can be later analyzed and surely helps in providing the details about the systems running activities.

The artifacts collected from RAM are:

- Current running processes
- Network configuration
- User logins
- Encryption keys

Objective:The purpose of RAM analysis is to get the information regarding what all processes are running at that very particular stage in the system. The RAM analysis provides the very crucial information which cannot be found out once the system is turned off.

Steps:Once we reach a crime scene and while performing the seizure of the devices if we found out a system is in ON-STATE, never turn it off. Always try to plug in the power cable if possible till it is been taken to the forensics workstation. If necessary then simply do live forensic operation on site. Run the tools like volatility or COFEE which comes as a live pen drive, simply insert that into the system and take RAM (memory) dump of that devices. The dump file gets saved in the pen drive and records the current running state of that very system, now analyze the data at forensic workstation.

8.4 PHOTOGRAPHY AT THE CRIME SCENE

One of the most important phase of cyber forensics investigation process is collection of evidence. Whenever we reach to a crime scene our first role is to identify and collect the digital evidence, another very important thing to be followed which doing crime scene investigation is Photography, it is very compulsory that we take the pictures of the crime scene and also of the evidences we have collected from there. The photography are additional proof in support of the evidence. Also it is recommended if the device is in the Power On state, take a picture of it. So we find a similarity in the digital and traditional forensics.

The photography of a crime scene is also termed as forensic photography or crime scene photography. The photography acts as an aid to our investigation and helps to prove the evidences recovered from the crime scene. The pictures must be very clear and explain what is being captured. The captured photography also become a type of authentic record of evidences. Photographs are to be taken of both the crime scene and the evidences recovered from the crime scene. There will always be specialized people in his area who will be taking the photographs.

8.5 IMAGING OF HDD

So, imaging is the process of making an exact copy of a disk. Imaging is sometimes also called disk cloning or ghosting, but the latter terms usually refer to images created for purposes other than evidence preservation.

Disk imaging differs from just copying, all the files on a disk are same in copying but in imaging the disk structure and relative location of data on the disk are preserved.

When you create a disk image (a bit stream copy), each physical sector of the disk is copied so that the data is distributed in the same way, and then the image is compressed into a file called an image file, This image is exactly like the original, both physically and logically. We usually perform bit stream imaging because in bit stream imaging exact copy of every data bit by bit is created from the original storage device, including all hidden files, temp files, corrupted files, file fragments and erased files that have not yet been overwritten.

Need for Imaging

- As per cardinal rule never work on the Original.
- To Preserve the Evidence.

- Later it is required to prove the Integrity of the evidence.
- If by mistake or during the operation any error occurs by which our evidence is tampered we can still acquire the Image from the Original Evidence.

IMAGING USING WRITE BLOCKERS

The question which generally will come to our mind whether imaging process is different from imaging using write blocker. Yes it is. The imaging process is not considered legitimate and unaltered if the write blockers are being used during the course. What do we mean? The write blocker is a device that makes the original hard disk read only devices thus preventing anything to get written onto that disk while imaging process is on. Which clearly proves that the data integrity has been maintained. In case a person doesn't use the write blocker chances are that the data onto the disk can get altered and our evidence will be considered as tampered thus, not accepted in the court of law.

Hardware based tools:

- True Imager
- DIBS (RAID/PERU)
- LOGICUBE

Software based tools:

- FTK
- EnCase
- Dd (Linux utility)
- Safe Back

LOGICAL IMAGE OF ENCRYPTED HDD

This is one more very important part on imaging process taking a logical image. What it exactly means and why we need it? The imaging process can be done in 2 ways.

- Physical imaging
- Logical Imaging

In physical imaging we simply take the bit by bit image of the entire hard disk drive. While as in a logical acquisition only a part within the hard disk drive is being imaged. It can be a complete one partition or a part of it where we are sure that the evidence will be residing. Now the question why we need logical imaging, reason is encryption of HDD. If we see in today's time we mostly come across with the HDD which are fully encrypted, so in that case if we take a physical image of the drive we will not be able to boot the image and an error message will be shown so it gets hard to look for evidence. In such cases what we do is take a logical image which we help us to at least read that particular part of the hard disk drive and we don't need to take a complete image of entire disk and also results in saving our time.

8.6 DOCUMENTATION

The last phase of Digital forensic life cycle is documentation, this step is about the detailed writing of every action we have performed onto the evidence from seizure to the analysis phase. One of the prime elements of documentation is to write down the details about the devices seized from the crime scene, which comprises of Serial number of that device, Details about the device model, any other unique distinction about that device. Whatever evidences we have collected must be written as the documentation.

As per Scientific working group on Digital evidences (SWGDE) has written down in their best practices for cyber forensics what all must be present in the Evidence handling documentation:

- copy of legal authority
- Chain of custody
- The initial count of evidence to be examined
- Information regarding the packaging and condition of the evidence upon receipt by the examiner
- A description of the evidence, and
- Communications regarding the case.

Examination documentation should be case specific, and contain sufficient details to allow another forensic examiner, competent in the same area of expertise, to be able to identify what has been done and to access the findings independently. Documentation should be preserved according to the examiner's agency policy.

ANTI STATIC BAGS

During the crime scene investigation one of the important things to be done in a perfect manner is seizure procedure. The evidences collected are to be transported safely to the forensic workstation without getting damaged. One of the key thing is anti-static bags, which are being used to carry the digital evidences from the crime scene. The antistatic bags are made of polyethylene material and are designed to protect digital evidences which are sensitive to static electricity, such as hard drives and other electronic components. Similarly for the mobile phones faraday's bag is being used so that the devices don't receive any kind of signals and no further communication takes place through this device after seizure.

Summary

So far we have discussed about:

- Importance of Live Acquisition of evidences.
- Packet sniffing for live network analysis.
- Importance of Photography
- Imaging using write blockers
- And documentation process

Check your answers

1. Why we do live acquisition?
2. What is DEAD and LIVE Imaging?
3. What do you mean by Logical Imaging and why is it beneficial?
4. What is meant by packet sniffing, and describe various steps involved in it?
5. What role does photography play in crime scene investigation?
6. What are Anti-static bags?
7. Discuss in detail importance of documentation?

Do you know?

1. Volatility tool can be used for imaging. True / False
2. Logical imaging can be done on devices which are not encrypted. True / False

3. Wireshark can't be used to do packet analysis. True / False
4. The chronological documentation showing the seizure, custody, control, transfer, analysis, and disposition of evidence, physical or electronic is known as _____.

LESSON - 9

MOBILE FORENSICS

Learning Objectives

After reading this unit, you will be able to

- Ø Understand the importance mobile devices as an evidence
- Ø Different operating systems used in mobile phones
- Ø Evidences found mobile devices
- Ø Issues pertaining to the mobile phone forensics

Structure

- 9.1 Overview of mobile forensics**
- 9.2 Role of mobile devices in investigation**
- 9.3 Operating systems available in mobile phones**
- 9.4 Communication Technologies**
- 9.5 Storage media available**
- 9.6 Evidences possible to extract from mobile phones**
- 9.7 Tools used in Mobile Forensics**
- 9.8 Issues/Limitations in mobile phone investigation**

9.1 Mobile Forensics

After learning and understanding the Digital forensic and its various sub parts now we will be discussing about the mobile forensic, this sub part of digital forensics is also sometimes called as device forensics. It deals with the forensic analysis (investigation, extraction) of the evidences from mobile phones and other media devices related to it and even the personal assistant devices. Nowadays what we call personal gadgets also come under device forensics.

9.2 Role of Mobile phones evidences in digital forensics

Mobile phones more or less have become the part of our day to day life activities, what's more interesting is that there is a huge wealth of information stored onto these devices. Everyone

just for their convenience keep their data in mobile phones since it's the device which always remains close to the people. The important site is that majority of the criminal cases are solved with the help evidences related to the mobile devices, like call data records, geo location etc. what makes mobile devices more interesting is they go along with us where ever we go so, one important point is where this mobile phone was recovered its location and not just what it is.

9.3 TYPES OF MOBILE OPERATING SYSTEMS

Most of the manufacturers come up with their own operating systems for their devices. So, there is not just one operating system across all mobile phones but a very wide range of operating systems and that too every manufacturer has their own features added to the devices. Thus making the forensic investigation more complex across various platforms to deal with. Let's discuss the most widely used operating systems in mobile phones.

- Android
- IOS
- Windows
- BlackBerry

ANDROID:

The android has become the most widely used operating system in mobile phones. The android is based on Linux kernel and is currently developed by Google. The android is widely used in smartphones and tablets, technology has also introduced android with TV's and write watches and other gadgets. The android has got a user friendly interface. Nowadays the storage capacity of the mobile phones is also increasing. The new firmware and latest operating systems updates are giving a new challenge for forensic investigation.

IOS:

The IOS is an operating system for iPhone developed by Apple. This is also a most widely used operating system in the world with different operating system, features and security related challenges. Every operating system in itself is a vast subject of study.

Windows:

The windows operating system is designed and developed by Microsoft. It is new in the market in terms of mobile phone operating systems. It has not been that much successful with

its previous version but from windows 8 update it has definitely gained popularity. Also Microsoft is giving windows phones under their own name.

BLACKBERRY:

The Blackberry Company has been making smartphone devices from a long time and is definitely not a new comer in the market, the blackberry phones have got their own operating system by the name of company. It also most widely used phone in the world and known for its security features.

These were some of the operating systems that we see around but other Java based and Symbian phones are also there. The question is in order to perform forensic operations on these devices we must have an in-depth knowledge of the functioning of these operating systems. So it has been a challenge to deal with all types of operating systems and mobile devices that we come across.

COMMUNICATION PROTOCOL

The various communication protocols used by mobile devices are:

- Cellular
- Wi-Fi
- Bluetooth
- NFC

CELLULAR:

A cellular network also known as mobile network is a wireless network distributed over land areas, each area has got one transceiver, known as a base station. This wireless network provides radio coverage over a wide geographic area. This enables a large number of portable transceivers (e.g., mobile phones, pagers, etc.) to communicate with each. The cellular network can be used for both voice and data facilities. The GSM architecture that is explained later in the chapter gives a clearer picture how it functions.

Wi-Fi:

The Wi-Fi Alliance defines Wi-Fi as any “wireless local area network” (WLAN) product based on the Institute of Electrical and Electronics Engineers’ (IEEE) 802.11 standards”. Wi-Fi

is a WLAN technology used by many devices e.g. personal computers, smartphones, digital cameras, tablet computers and digital audio players. These can connect to a network resource such as the Internet via a wireless network access point. Such an access point (or hotspot) has a range of about 20 meters (66 feet) indoors and a greater range outdoors. Hotspot coverage can be as small as a single room with walls that block radio waves, or as large as many square kilometers achieved by using multiple overlapping access points. Wi-Fi can be less secure than wired connections, such as Ethernet, precisely because an intruder does not need a physical connection. Wi-Fi has adopted various encryption technologies. The early encryption WEP proved easy to break. Higher quality protocols (WPA, WPA2) were added later. Wi-Fi is widely used in all the organizations even at universities/colleges and also at homes. It simply requires a router that becomes an access point and send signal up to a certain range. All the devices in that coverage can make use of that facility. Nowadays Wi-Fi connectivity is given across many major cities in the country.

9.4 Telecommunication Technologies

The technology has been drastically changing, when we just look at the mobile technology how telecommunication has changed over a period of time. Sometime back we were on 2G, then 3G now we have moved on to 4G. Some of the telecommunication protocols are:

- GSM
- CDMA
- GPRS
- EV-DO
- EDGE
- DECT
- TDMA
- iDEN

GSM:

GSM known as Global System for Mobile Communications is a standard developed by the European Telecommunications Standards Institute (ETSI) to describe protocols for second-

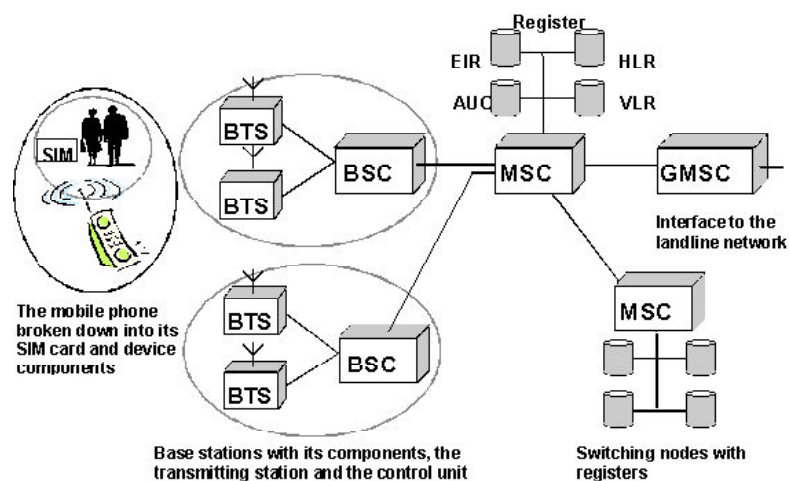
generation (2G) digital cellular networks used by mobile phones. It has become the default global standard for mobile communications - with over 90% market share, operating in over 219 countries and territories. The technology has seen a drastic change from 1G to 2G then 3G and now onto 4G and continues. 2G networks were developed as a replacement for first generation (1G) analog cellular networks. After analog we saw use of circuit-switching, then we moved onto packet data switching and we came to see GPRS (General Packet Radio Services) and EDGE (Enhanced Data rates for GSM Evolution or EGPRS). Later, the 3GPP developed third-generation (3G) UMTS standards followed by fourth-generation (4G) LTE Advanced standards.

Network Structure of GSM:

The main components of GSM are:

- Base Transceiver Station – used for receiving signals from the devices.
- Basic Switching Center –core network part
- Main Switching Center – the optional part which allows packet-based Internet connections

Below is the diagram of GSM Architecture and its main components:



GPRS:

Mostly used in GSM, General packet radio service is a packet oriented mobile data service on the 2G and 3G. GPRS was originally standardized by European Telecommunications

Standards Institute (ETSI) in response to the earlier CDPD and I-mode packet-switched cellular technologies. GPRS usage is typically charged based on volume of data transferred. GPRS is a best-effort service, implying variable throughput and latency that depend on the number of other users sharing the service concurrently. In 2G systems, GPRS provides data rates of 56–114 Kbit/second. It Provides moderate-speed data transfer, by using unused time division multiple access (TDMA). GPRS supports the following protocols; Internet protocol (IP), Point-to-point protocol (PPP), X.25 connections.

CDMA:

Code division multiple access (CDMA) is a channel access method used by various radio communication technologies. CDMA is an example of multiple access, where several transmitters can send information simultaneously over a single communication channel. This allows several users to share a band of frequencies. CDMA uses spread-spectrum technology. CDMA is used as the access method in many mobile phone standards such as cdmaOne, CDMA2000 (the 3G evolution of cdmaOne), and WCDMA (the 3G standard used by GSM carriers), which are often referred to as simply CDMA.

EV-DO:

Evolution-Data Optimized is a telecommunications standard for the wireless transmission of data through radio signals, typically for broadband Internet access. EV-DO is an evolution of the CDMA2000 standard that supports high data rates and can be deployed alongside a wireless carrier's voice services. It uses advanced multiplexing techniques including code division multiple access (CDMA) as well as time division multiplexing (TDM) to maximize throughput.

EDGE:

Often known as Enhanced GPRS or Enhanced Data rates for Global Evolution is a digital mobile phone technology that allows improved data transmission rates as a backward-compatible extension of GSM. EDGE is considered a pre-3G radio technology. EDGE is standardized also by 3GPP as part of the GSM family. EDGE delivers higher bit-rates per radio channel, resulting in a threefold increase in capacity and performance compared with an ordinary GSM/GPRS connection. EDGE can be used for any packet switched application, such as an Internet connection.

TDMA:

Time division multiple access (TDMA) allows several users to share the same frequency channel by dividing the signal into different time slots. The users transmit in rapid succession, one after the other, each using its own time slot. This allows multiple stations to share the same transmission medium (e.g. radio frequency channel) while using only a part of its channel capacity. TDMA is used in the digital 2G cellular systems such as Global System for Mobile Communications (GSM), IS-136, Personal Digital Cellular (PDC) and iDEN, and in the Digital Enhanced Cordless Telecommunications (DECT) standard for portable phones. It is also used extensively in satellite systems, combat-net radio systems, and PON networks for upstream traffic from premises to the operator. TDMA is a type of time-division multiplexing, with the special point that instead of having one transmitter connected to one receiver, there are multiple transmitters. In the case of the uplink from a mobile phone to a base station this becomes particularly difficult because the mobile phone can move around and vary the timing advance required to make its transmission match the gap in transmission from its peers.

iDEN:

Integrated Digital Enhanced Network (iDEN) is a telecommunications technology, developed by Motorola, which provides its users the benefits of a trunked radio and a cellular telephone. iDEN places more users in a given spectral space, compared to analog cellular and two-way radio systems, by using speech compression and time division multiple access (TDMA).

9.5 Data Storage Medium

The mobile phone has been a rich source of information store, data or the information is being stored on various storage medium:

- Internal RAM
- ROM
- External Memory Cards
- SIM cards
- Email Servers
- Servers of service providers and
- Personal computers

Internal RAM and ROM:

The RAM is used by the device to accelerate and have fast processing of the tasks. The use is same as we know in PC's but with something extra what is that? In android phones Apps that you download and install are first loaded in the RAM and then executed. Those Apps remain in the RAM after you are no longer using them and they have been shifted to background. The next time you use those apps then will be retrieved back from the RAM. Our most frequently used Apps remain on the RAM.

ROM or Read Only Memory is a permanent by nature and can't be changed. The ROM or read only memory forms a part of your internal storage and that part is not Accessible for users to write on. The rest of your internal storage is where you store Apps and can store media files and other documents. The memory type used is basically Flash memory or technically EEPROM (electrically erasable and programmable read only memory). It is ROM of your internal storage where your operating system resides. Also OEM's add some preinstalled apps in this memory sections which cannot be deleted on users end either.

External Memory cards:

The external memory cards are storage devices used by mobile phones to expand their storage capacities. There is limited internal storage for mobile phones which is changing nowadays so, to save extra data external memory cards are used. The file system used in memory cards is FAT. The information retrieved from the external memory cards while doing forensic investigation is huge and is comparatively easier to recover than internal storage of the phone. The same digital forensic methodology can be adopted on them as well. The storage capacity of memory cards ranges from 512mb – 64GB and when more.

SIM cards:

SIM known as Subscriber Identity Module is a kind of smartcard. The SIM card is used in GSM Mobile phones and is given to all the subscribers who want to use their service. The SIM card has some details printed on them like Name of the service provider and a unique number called ICCID (Integrated Circuit Card Identification). SIM has an EEPROM, the file system is hierarchical having Master File, Dedicated file and Elementary files.

There is possibility of finding digital evidences from SIM cards as well like:

- Last number dialled
- Contacts (varies from 100 – 200)
- SMS (some of the messages can be saved in SIM card)
- Location Information

All these details are very important when we do mobile phone analysis, there are bundle of tools available to look into SIM cards for the information present inside. Either some of the commercial tools have inbuilt feature for analysing SIM or we have SIM card analysis tools available: SIMCon, SIMSeizure etc.

Email Servers:

Every email that is sent passes through a series of mail servers along its way to its intended recipient. When we click on sent button it simply doesn't just reach to the recipient but the mail has to pass through the mail servers. Without this series of mail servers, you would only be able to send emails to people. Mail servers can be broken down into two main categories: outgoing mail servers and incoming mail servers. Outgoing mail servers are known as SMTP, or Simple Mail Transfer Protocol, servers. Incoming mail servers come in two main varieties. POP3, or Post Office Protocol, version 3, servers are best known for storing sent and received messages on PCs' local hard drives. IMAP, or Internet Message Access Protocol, servers always store copies of messages on servers. Most POP3 servers can store messages on servers, too, which is a lot more convenient.

Servers of service providers:

The service provider has got servers on which the details about the subscriber is stored. The service provider can easily provide details about the subscriber's identity, his address and other relevant details. Besides that we get call data records of the person which is very valuable, also geolocation and roaming details can be fetch from these servers.

Personal computers:

The personal computer we have already discussed but why we have mentioned it in mobile forensics is because the mobile phone is always synced with the person's personal computer. It is important when we transfer the data from the system to mobile most mobile

operating systems have their own software's which are used to create backup for saving data and transfer/managing of mobile data.

9.6 Evidences possible to extract from Mobile Phones

First we will generally look at what evidences are possible to be extracted from mobile phones. Below is the list of important data that can be extracted from the mobile devices:

- Audio Files (Music and Voice)
- Calendar Entries
- Call History (Inbound and Outbound)
- Contacts/Phonebook
- Email
- Internet History
- Instant Messaging (IM) chat
- Memos
- Multimedia Messages (MMS)
- Pictures
- Short Message Service (SMS) or Text Messages
- System Firmware Information
- Telecommunication Settings
- Videos
- Voice Mail

9.7 Tools used in Mobile Forensics

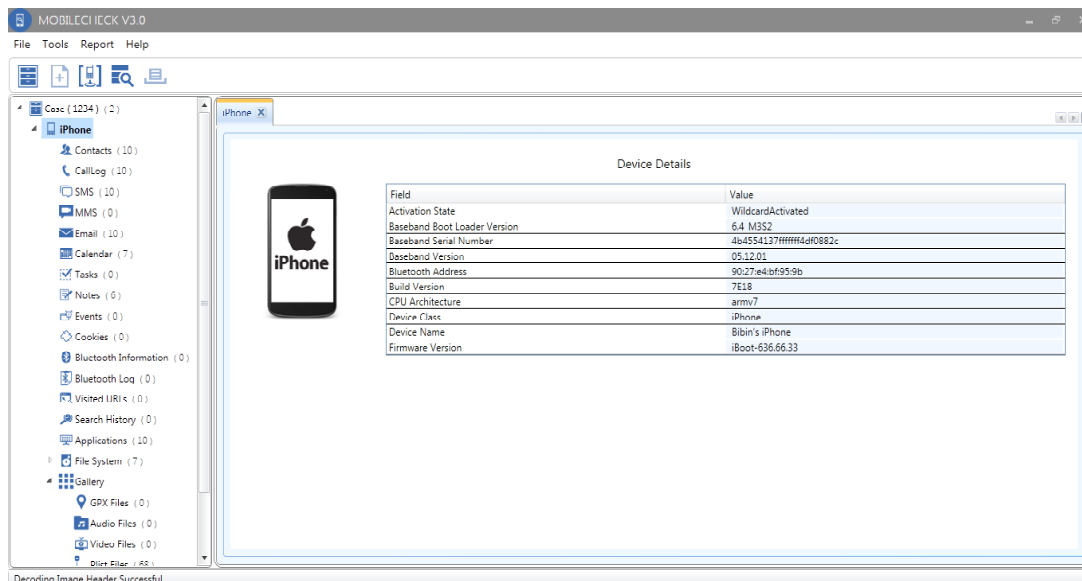
There are some very sophisticated and specialized tools available that can be used for investigation of the mobile phones. The Devices have a wide range of drivers available which provide support to the various mobile devices attached for acquisition and analysis.

Some of the mobile forensic tools are:

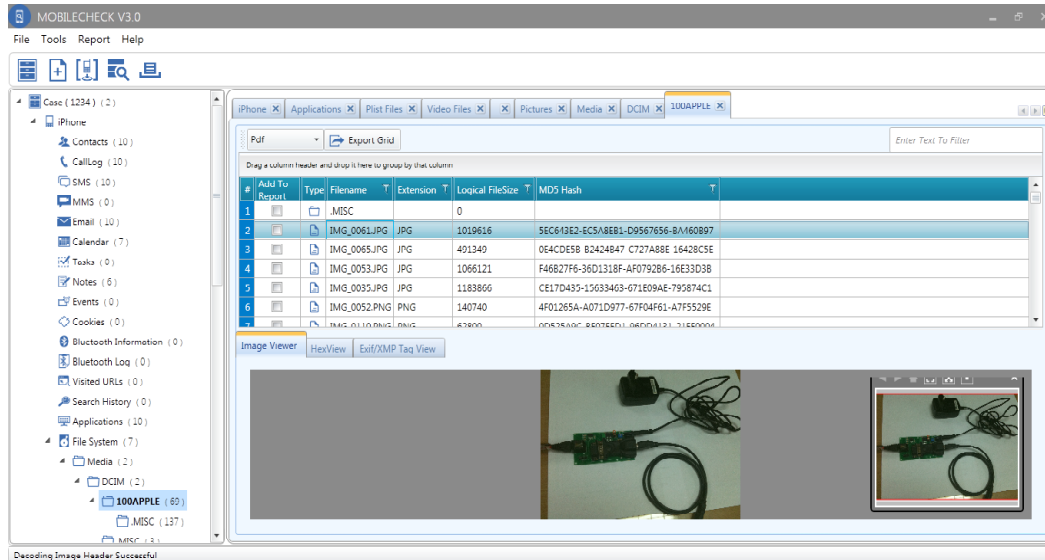
- Paraben Device Seizure
- CellIDEK from Logicube
- Oxygen Forensic Suite
- MOBILedit Forensic
- UFED from Cellbrite
- .XRY

Example of Mobile Check Tool

The mobile checker is tool for mobile phone acquisition and analysis, this tool can be download from CDAC's website. The tool is just for academic purpose and can't be used investigation procedures. The basic idea behind the tool is to understand of how the stuff works and educate the students of this subject. Below I have pasted some screen shots of this particular tool in which analysis is done of an I Phone. You can use this tool at your place and understand the operations.



The above image is the interface of the tool and the image of I Phone taken for analysis is opened in its interface.



This image above is of the analysis phase where we are looking into the gallery of the phone and we are able to see some pictures stored onto the device.

Best Practices

Every organization has described the best practices to be followed for the mobile phone forensics. In this particular topic we will be looking at the best practices defined by the Scientific Working Group on Digital Evidence (SWGDE), I will just be writing about the limitations with the mobile phones and evidence collection as described by them rest of the information can be gathered from their website:

9.8 Issues/Limitation in Mobile Forensic Investigation

- Incoming and Outgoing Signals – Attempts should be made to block incoming and outgoing signals of a mobile phone.
- Cables – Data Cables are often unique to a particular phone.
- COM Ports – Some tools may require the use of specific ports.
- Condition of the Evidence – Commercially available tools may not provide solutions to deal with physically damaged mobile phones.

- Destruction of Data – There are methods to destroy data locally and remotely on a mobile phone.
- Drivers – Conflicts may occur due to existing operating system drivers, proprietary drivers, driver version inconsistencies, and vendor specific drivers.
- Dynamic Nature of the Data – Data on active (powered-on) mobile phones is constantly changing. There are no conventional write-blocking methods for mobile phones.
- Encryption – Data may be stored in an encrypted state preventing analysis.
- Equipment – Equipment used during examinations may not be the most recent version due to agency verification requirements of hardware, firmware and/or software.
- Field analysis – First responders should be aware of the risks associated with triaging mobile phones. Triaging mobile phones is not considered a full examination. The phone should be protected for further examination.
- Hash Values – Individual data objects (e.g., graphics, audio, and video files) will often maintain consistency between the forensic workstation and the hash value reported by the mobile phone application.
- Industry Standards – Manufacturers and carriers lack standardized methods of storing data (e.g., closed operating systems and proprietary data connections).
- Legal Issues – Unopened emails, unread text messages, and incoming phone calls of seized phones may present non-consensual eavesdropping issues.
- Loss of Power – Many mobile phones may lose data or initiate additional security measures once discharged or shut down.
- Passwords – Authentication mechanisms can restrict access to a phone and/or data. Traditional password cracking methods can lead to permanent inaccessibility or destruction of data. There are different methods to protect a phone (e.g., Personal Identification Number (PIN), Phone Unlock Key (PUK), and handset protection). ?
- Removable Media Cards – Processing these cards inside the phone poses risks (e.g., not obtaining all data including the deleted data, and altering date/time stamps).
- Identity Module e.g., SIM, USIM, CSIM, RUIM Cards – Lack of or removal of an identity module may prevent the examiner from accessing data stored on the internal

memory of a handset. Inserting an identity module from another phone may cause the loss of data.

- Training – The individual copying data from a mobile phone should be trained to ensure the integrity of the data.
- Unallocated Data / Deleted Data – Many mobile phone forensic tools may only provide the logical acquisition of data. Deleted data may only be recoverable from a physical acquisition.

Evidence Collection General Guidelines concerning the collection of evidence are provided as follows:

- Consult with the investigator to determine the necessary equipment to take to the scene.
- Review the legal authority to collect the evidence, ensuring any restrictions are noted. If necessary during the collection, obtain additional authority for evidence outside the original scope.
- All individuals not involved in the collection process should be removed from the proximity of the mobile phone to prevent modifications to the data.
- Solicit information from mobile phone user to determine the phone number, pass codes, pattern locks or PINs.
- If the phone is unable to be processed immediately, turn off phone, remove battery if practical, and do not turn it back on.
- The benefits of turning off the phone include:
 - Preserving call logs and last cell tower location information (LOCI). Preventing overwriting deleted data.
 - Preventing data destruction signals from reaching the mobile phone. Preventing improper mobile phone handling (i.e., placing calls, sending messages, taking photos or deleting files).
- The risks of turning off the mobile phone include possibly engaging authentication mechanisms (e.g., passwords, PINs, etc). Exigency may dictate that the mobile phone remains on for immediate processing. If the mobile phone must be left on, isolate it from its network while maintaining power.

- Radio Frequency (RF) shielding – Mobile phones communicate with cell towers. Allowing this communication will change data on the phone.
- Many mobile phones can be placed in “Airplane” mode limiting access to cell towers (e.g., 911 calls still available). This requires user input on the handset.
- Disable Wi-Fi, Bluetooth, RFID and IrDA communications if practical. Searchers should be able to recognize different types of mobile phone evidence.
- The scene should be searched systematically and thoroughly for evidence. Document the scene according to policy. Collect associated chargers, cables, peripherals, and manuals.

9.9 Summary

- The discipline continued to grow in the 1990s, with the first conference on collecting evidence from computers held in 1993. Two years later, the International Organization on Computer Evidence (IOCE) was established.
- Digital Forensic Research Workshop (DFRWS)
- Association of Chief Police Officers functioning from Northern Ireland have designed and published a guide with procedures to be followed while collecting computer based evidence to ensure good field practice.
- International Association of Computer Investigative specialist (IACIS)

Digital Forensic Research Workshop (DFRWS)

- DFRWS is a non-profit, volunteer organization dedicated to bringing together everyone with a legitimate interest in digital forensics to address the emerging challenges of our field.

DFRWS is About

- INCLUSION: Reaching out to everyone with a legitimate interest in digital forensic research.
- INNOVATION: Pushing the envelope of what is currently possible in digital forensics
- USABILITY: Practical solutions for real-world problems.
- SCIENTIFIC PRACTICE: Systematic processes and reliable results

So, till now we looked into Mobile Phones and how the related technologies has come up, then we discussed the importance of mobile devices as a forensic rich source of evidence. Then we made an overview of the telecommunication protocols used. Also we saw the places from where evidences can be extracted like RAM, external memory cards, SIM cards, service providers and personal computers which are all related to the mobile technology. Finally we looked for all the possible evidences that can be extracted from a mobile phone along with the functioning of Mobile Checker Tool from CDAC for Academic purpose.

9.10 Check your answers

1. Define Device Forensic?
2. Describe the role played by mobile device in Investigation?
3. Explain GSM and GPRS Technologies?
4. What are the possible evidences that can be extracted from mobile devices?
5. What kind of information can be extracted from SIM Card?
6. Write down the different tools used in Mobile Forensics?
7. What is the purpose of Registry from Forensic prospective?
6. Discuss Limitations in Mobile Forensics Investigation?

LESSON - 10

TYPES OF COMPUTER FORENSICS TOOLS AND TECHNOLOGY

After reading this lesson you will be able to understand the following:

- Tools and Types of Military Computer Forensics Technology
- Tools and Types of Law Enforcement Computer Forensic Technology
- Tools and Types of Business Computer Forensic Technology

Structure

- 10.1. Tools and Types of Military Computer Forensics Technology**
- 10.2. Tools and Types of Law Enforcement Computer Forensic Technology**
- 10.3. Tools and Types of Business Computer Forensic Technology**

10.1. Tools and Types of Military Computer Forensics Technology

Key objectives of cyber forensics include rapid discovery of evidence, estimation of potential impact of the malicious activity on the victim, assessment of the intent and identity of the perpetrator. Difficulty arises with real time tracking of potentially malicious activity is especially difficult when the pertinent information has been intentionally hidden, destroyed or modified in order to elude discovery. US – DOD cyber forensics includes evaluation and indepthexamination of data related to both the trans- and post-cyberattack periods.

- National Law Enforcement and Corrections Technology Center (NLECTC). They partnered with NIJ. The computer forensics experiment (CFX-2000) resulted from this partnership. They developed military Research and Development to demonstrate New Methodology and new prototypes.
- National Institute of Justice (NIJ) sponsors research and development or identifies best practices to address those needs.
- The experiment used a realistic cyber crime scenario specifically designed to exercise and show the value added of the directorate-developed cyber forensic technology.
- The CFX-2000 central proposition is to determine accurately the following:

- o The motives
- o Intent
- o Targets
- o Sophistication
- Identity and location of cyber criminals and cyber terrorists
- Develop integrated forensic analysis framework and R&D prototype and cyber forensic tools.
- The NLECTC assembled a diverse group of computer crime investigators from DoD and federal, state, and local law enforcement to participate in the CFX-2000 exercise hosted by the New York State Police Forensic Investigative Center in Albany, New York. Officials divided the participants into three teams. Each team received an identical set of software tools and was presented with identical initial evidence of suspicious activity. The objective of each team was to uncover several linked criminal activities from a maze of about 30 milestones that culminated in an information warfare crime.
- The cyber forensics tools involved in CFX – 2000 software and R&D prototypes.
- Synthesizing Information from Forensic Investigations (SI-FI) integration environment supports the collection, examination, and analysis processes employed during a cyber-forensic investigation.
- SI-FI prototype uses digital evidence bags (DEBs) – investigators can seal & Authorized user can reopen the DEBs.

10.2. Tools and Types of Law Enforcement Computer Forensic Technology

Computer forensics involves the preservation, identification, extraction and documentation of computer evidence stored in form of magnetically encoded information (data). Often the computer evidence was created transparently by the computers operating system and without the knowledge of the computer operator. Such information may actually be hidden from view and special forensic software tools and techniques are required to preserve, identify, extract and document the related evidence.

Computer forensics tools and techniques have become important resources for use in internal investigation of leads, civil lawsuits and computer security risk management. Computer forensics tools and techniques have proven to be a valuable resource for law enforcement in identification of leads and in processing computer related evidence.

Forensic software tools and methods can be used to identify passwords, logons, and their information that is automatically dumped from the computer memory as a transparent operation of today's popular personal computer operating systems. Such computer forensic software tools can also be used to identify backdated files and to tie a diskette to the computer that created it.

10.2.1. Computer Evidence Processing Procedures

Computer Preservation of Evidence

MIRROR IMAGE BACKUP SOFTWARE

SafeBack is used to create mirror-image (bit-stream) backup files of hard disks or to make a mirror-image copy of an entire hard disk drive or partition. The process is analogous to photography and the creation of a photo negative. Once the photo negative has been made, several exact reproductions can be made of the original. Unlike a photo, SafeBack image files cannot be altered or modified to alter the reproduction. This is because SafeBack is an industry standard self-authenticating computer forensics tool that is used to create evidence-grade backups of hard drives. With the release of SafeBack version 3.0 or higher, the integrity of SafeBack files is maintained through the use of two separate mathematical hashing processes that rely upon the National Institute of Standards and Technology (NIST)-tested Secure Hash Algorithm 256 (SHA256). Users of prior versions of SafeBack are encouraged to upgrade to take advantage of the greater levels of accuracy achieved with version 3.0. Information about upgrades can be found on the Internet at <http://www.forensicsintl.com/>. The upgrade of SafeBack has new and added features and it takes into account the last sector error finding by NIST concerning the older SafeBack version 2.0.

Backup image files created with SafeBack can be written to any writable magnetic storage device, including small computer system interface (SCSI) tape backup units. SafeBack preserves all the data on a backed-up or copied hard disk, including inactive or deleted data. Backup image files can be restored to another system's hard disk. Remote operation via a parallel port connection allows the hard disk on a remote PC to be read or written by the master system. A

date- and time-stamped audit trail maintains a record of SafeBack operations during a session, and when the default is used an SHA256 hash is recorded in the output audit file. This hash can be used to cross-validate the accuracy of the process with any other software utility that relies upon the NIST-tested SHA256 algorithm. To avoid possible claims that the SafeBack image file may have been altered after the fact, SafeBack now safeguards the internally stored SHA256 values. Any alterations of computer data are quickly brought to the attention of the operator of the program when the SafeBack image file is restored. Simply put, SafeBack is a DOS-based utility used to back up and restore hard disks. SafeBack picks up every last bit of data—unused and erased data included—on the original disk and stores it in a tape or disk file (or series of files). SafeBack can take that same backup file and recreate the original disk on your own system. SafeBack does not write or otherwise modify the original system and can (and should) be started from a boot diskette. SafeBack also has a couple of derivative operating modes. The first is Verify mode, where restoring from a backup disk is done, but the data is thrown away. This is more useful than it first appears to be because it allows the operator of the program to scan his or her backups to make sure that they will read back without errors, without having to go through the setup required by a standard SafeBack restore procedure. The other derivative operation is Copy, which feeds the Restore function directly with the output of the Backup function, with no intermediate files. This is less useful than it first appears to be. If the operator of SafeBack is considering making a copy, he might as well make a backup image file and then restore it as needed.

PRIMARY USES

The primary uses of SafeBack are as follows:

Used to create evidence-grade backups of hard disk drives on Intel-based computer systems
 Used to exactly restore archived SafeBack images to another computer hard disk drive of equal or larger storage capacity
 Used as an evidence preservation tool in law enforcement and civil litigation matters
 Used as an intelligence gathering tool by military agencies

10.2.2.PROGRAM FEATURES AND BENEFITS

The program features and benefits of SafeBack are as follows:

- DOS based for ease of operation, for speed, and to eliminate the problems created by a separate operating system concerning the potential alteration of data.

- No software dongle. Software dongles get in the way and they restrict your ability to process several computers at the same time.
- Incorporates two separate implementations of the NIST-tested SHA256 algorithm to ensure the integrity of all data contained on the target computer storage device.
- Provides a detailed audit trail of the backup process for evidence documentation purposes, and the SafeBack default outputs an SHA256 hash value that can be compared with other utilities when cross-validation of findings are deemed to be important.
- Checks for possible data hiding when sector cyclic redundancy checks (CRCs) do not match on the target hard disk drive. These findings are automatically recorded in the SafeBack audit log file.
- Accurately copies all areas of the hard disk drive.
- Allows the archive of non-DOS and non-Windows hard disk drives (Unix on an Intel-based computer system).
- Allows for the backup process to be made via the printer port.
- Duplicate copies of hard disk drives can be made from hard disk to hard disk in direct mode.
- SafeBack image files can be stored as one large file or separate files of fixed sizes. This feature is helpful in making copies for archive on CDs.
- Uses tried and proven evidence preservation technology with a long-term legacy of success in government agencies.
- Does not compress relevant data to avoid legal arguments that the original computer evidence was altered through data compression or software translation.
- It is fast and efficient. In spite of the extensive mathematical validation, the latest version of SafeBack runs as fast or faster than prior versions. Processing speeds are much faster when state-of-the-art computer systems are used to make the backup.
- Makes copies in either physical or logical mode at the option of the user.
- Copies and restores multiple partitions containing one or more operating systems.

- Can be used to accurately copy and restore most hard disk drives including Windows NT, Windows 2000, and Windows XP configured drives.
- Accuracy is guaranteed in the backup process through the combination of mathematical CRCs that provides a level of accuracy that far exceeds the accuracy provided by 128-bit CRCs (RSA MD5).
- Writes to SCSI tape backup units or hard disk drives at the option of the user.
- The current version of SafeBack compresses unused and unformatted sections of the hard disk drive to increase processing speeds and to conserve storage space concerning the writing of the SafeBack image file [3].

Trojan Horse Programs

The need to preserve the computer evidence before processing a computer should be clearly demonstrated by the computer forensic instructor through the use of programs designed to destroy data and modify the operating systems. The participant should be able to demonstrate his or her ability to avoid destructive programs and traps that can be planted by computer users bent on destroying data and evidence. Such programs can also be used to covertly capture sensitive information, passwords, and network logons.

Computer Forensics Documentation

The documentation of forensic processing methodologies and findings is important. This is even true concerning computer security risk assessments and internal audits, because without proper documentation, it is difficult to present findings. If the security or audit findings become the object of a lawsuit or a criminal investigation, then documentation becomes even more important. Thus, the computer forensic instructor should also teach the participant the ins and outs of computer evidence processing methodology (which facilitates good evidence-processing documentation and good evidence chain of custody procedures). The benefits will be obvious to investigators, but they will also become clear to internal auditors and computer security specialists.

File Slack

The occurrence of random memory dumps in hidden storage areas should be discussed and covered in detail during workshops. Techniques and automated tools that are used to capture and evaluate file slack should be demonstrated in a training course. Such data is the

source of potential security leaks regarding passwords, network logons, email, database entries, and word processing documents. These security and evidence issues should also be discussed and demonstrated during the training course. The participants should be able to demonstrate their ability to deal with slack and should demonstrate proficiency in searching file slack, documenting their findings, and eliminating the security risk.

Data-Hiding Techniques

Trade secret information and other sensitive data can easily be secreted using any number of techniques. It is possible to hide diskettes within diskettes and to hide entire computer hard disk drive partitions. These issues should be discussed in any computer forensics training course from a detection standpoint, as well as from a security risk standpoint. Tools that help in the identification of such anomalies should be demonstrated and discussed (like AnaDisk™ [see sidebar, “AnaDisk Diskette Analysis Tool”]) in the training course. Participants should be required to demonstrate their understanding of such issues. This aspect of the training becomes especially important during the last day of the course when the participants are called on to extract their Certificate of Completion from a special floppy diskette. Data-hiding courses are only open to classified government agencies and businesses that have a demonstrated need to know about this kind of information as outlined in a company’s training policies. This is because the information covered in a data-hiding course can be used to defeat government computer security review processes and techniques.

COPYQM: DISKETTE DUPLICATION SOFTWARE

CopyQM Plus essentially turns a personal computer into a diskette duplicator. In a single pass, diskettes are formatted, copied, and verified. This capability is useful for computer forensics specialists and computer security specialists who need to preconfigure floppy diskettes for specific uses and duplicate them. Classified government agencies and government contractors are required to perform periodic examinations of government computer systems to determine if classified data may reside on nonclassified computer systems. Programs like New Technology Inc.’s TextSearch Plus (see sidebar, “Text Search Plus”) and TextSearch NT were designed specifically for this purpose and they are both certified by the U.S. DoD for use in classified government risk assessments. CopyQM is also certified by the U.S. DoD for use in the duplication of “search disks” used in classified U.S. government computer risk reviews. CopyQM Plus can also create self-extracting executable programs that can be used to duplicate specific diskettes. This feature makes CopyQM an ideal tool for use in security reviews because once a CopyQM

disk-creation program has been created, it can be used by anyone to create pre-configured security risk assessment diskettes. When the resulting program is run, the diskette image of the original diskette will be restored on multiple diskettes automatically. This process requires little technical knowledge and it allows computer specialists to delegate more of the security risk assessment responsibilities to employees with minimal technical knowledge. The diskette images can also be password protected when the diskette images are converted to self-extracting programs. This is helpful when you want to keep computer forensic and computer security software tools away from curious hands.

Techniques:

Disk Structure

Participants should be able to leave a training course with a good understanding of how computer hard disks and floppy diskettes are structured and how computer evidence can reside at various levels within the structure of the disk. They should also demonstrate their knowledge of how to modify the structure and hide data in obscure places on floppy diskettes and hard disk drives.

Data Encryption

A computer forensics course should cover, in general, how data is encrypted; it should also illustrate the differences between good encryption and bad encryption. Furthermore, demonstrations of password-recovery software should be given regarding encrypted WordPerfect, Excel, Lotus, Microsoft Word, and PKZIP files. The participant should become familiar with the use of software to crack security associated with these different file structures.

Matching a Diskette to a Computer

New Technology Inc. has also developed specialized techniques and tools that make it possible to conclusively tie a diskette to a computer that was used to create or edit files stored on it. Unlike some specialgovernment agencies, New Technology Inc. relies on logical rather than physical data storage areas to demonstrate this technique. Each participant is taught how to use special software tools to complete this process.

Data Compression

The participant should be shown how compression works and how compression programs can be used to hide and disguise sensitive data. Furthermore, the participants should learn how

password-protected compressed files can be broken; this should be covered in hands-on workshops during the training course.

Erased Files

The training participant should be shown how previously erased files can be recovered by using DOS programs and by manually using data-recovery techniques. These techniques should also be demonstrated by the participant, and cluster chaining will become familiar to the participant.

Internet Abuse Identification and Detection

The participant should be shown how to use specialized software to identify how a targeted computer has been used on the Internet. This process will focus on computer forensics issues tied to data that the computer user probably doesn't realize exists (file slack, unallocated file space, and Windows swap files).

The Boot Process and Memory Resident Programs

The participant should be able to take part in a graphic demonstration of how the operating system can be modified to change data and destroy data at the whim of the person who configured the system. Such a technique could be used to covertly capture keyboard activity from corporate executives, for example. For this reason, it is important that the participants understand these potential risks and how to identify them.

10.3. Tools and Types of Business Computer Forensic Technology

Finally, let's briefly look at the following types of business computer forensics technology:

- Remote monitoring of target computers
- Creating trackable electronic documents
- Theft recovery software for laptops and PCs
- Basic forensic tools and techniques
- Forensic services available

10.3.1 Remote Monitoring of Target Computers

Data Interception by Remote Transmission (DIRT) from Codex Data Systems (CDS), Inc. [7] is a powerful remote control monitoring tool that allows stealth monitoring of all activity on one or more target computers simultaneously from a remote command center. No physical access is necessary. Application also allows agents to remotely seize and secure digital evidence prior to physically entering suspect premises.

10.3.2 Creating Trackable Electronic Documents

There are so many powerful intrusion detection tools that allow the user to create trackable electronic documents that it is beyond the scope of this chapter to mention them all. See “Intrusion Detection Systems” in Chapter 3 for a detailed explanation of some of these tools. In general, most of these tools identify (including their location) unauthorized intruders who access, download, and view these tagged documents. The tools also allow security personnel to trace the chain of custody and chain of command of all who possess the stolen electronic documents.

10.3.3. Theft Recovery Software for Laptops and PCs

If your PC or laptop is stolen, is it smart enough to tell you where it is? According to a recent FBI report, 98% of stolen computers are never recovered. According to Safeware Insurance, 1,201,000 PCs and laptops were stolen in 2002 and 2003, costing owners \$7.8 billion dollars [9]. According to a recent joint Computer Security Institute/FBI survey, 72% of the Fortune 1000 companies experienced laptop theft [9]. Nationwide losses to computer component theft cost corporate America over \$11 billion a year. So if your company experiences computer-related thefts and you do nothing to correct the problem, there is a 92% chance you will be hit again.

What Is the Real Cost of a Stolen Laptop or PC?

When you lose your wallet, the last thing you think of is how much it is going to cost to replace your wallet. The same is true when equipment (especially a computer) is stolen. Our mothers always told us, an ounce of prevention is worth a pound of cure. They were right. Think about what it really costs to replace a stolen computer.

- The price of the replacement hardware.
- The price of replacing the software.

- The cost of recreating data. If possible at all, do you keep perfect back-ups?
- The cost of lost production time or instruction time.
- The loss of customer goodwill (lost faxes, delayed correspondence or billings, problems answering questions and accessing data).
- The cost of reporting and investigating the theft, filing police reports and insurance claims.
- The cost of increased insurance.
- The cost of processing and ordering replacements, cutting a check, and the like.
- If a thief is ever caught, the cost of time involved in prosecution [7].

So, doesn't it make sense to use an ounce of prevention? You don't have to be a victim.

With that in mind, SecurityKit.com has a solution: PC PhoneHome™ [9] is a software application that will track and locate a lost or stolen PC or laptop anywhere in the world. It is easy to install. It is also completely transparent to the user. If your PC PhoneHome-protected computer is lost or stolen, all you need to do is make a report to the local police. In other words, PC PhoneHome is a transparent theft protection and recovery software system that you install on your laptop or PC. Once installed, it sends an stealth email message to your address every time the computer connects to the Internet. Let's look at the following scenario [9].

PC PhoneHome Scenario

Imagine you're a small business owner and you just went out and invested several thousand dollars in the latest laptop computer, fully loaded with all the bells and whistles. On your first business trip with your new computer you leave it (you think) safely hidden in your hotel room while you entertain a client. Later that night you return to your room and your laptop, with all your work in it, has disappeared without a trace. The financial loss is bad enough, but the hours of work you've lost is worse, and the sensitivity of the information in your laptop, if it gets into the hands of the wrong people, could be a disaster [9]. Now imagine a simple, inexpensive software system that offers real hope of tracking your laptop and pinpointing its location anywhere in the world. Is this really possible or is it just another fanciful hi-tech gimmick from the imagination of

the writers of the latest James Bond movie? It's no gimmick. It's PC PhoneHome, the latest in computer theft recovery software [9]. PC PhoneHome is a software application that, when installed in your laptop or desktop computer, secretly transmits an electronic message to an email address of your choice. This allows you to track and locate your computer, thus providing the potential for its ultimate recovery as well as apprehension of the thief [9].

10.3.4 Basic Forensic Tools and Techniques

Today, many computer forensics workshops have been created to familiarize investigators and security personnel with the basic techniques and tools necessary for a successful investigation of Internet and computer-related crimes. So many workshops have been created that it is beyond the scope of this chapter to mention them all. However, throughout the book, a number of them will be mentioned in detail. Workshop topics normally include: types of computer crime, cyber law basics, tracing email to its source, digital evidence acquisition, cracking passwords, monitoring computers remotely, tracking online activity, finding and recovering hidden and deleted data, locating stolen computers, creating trackable files, identifying software pirates, and so on.

10.3.5. Forensic Services Available

Through computer forensic evidence acquisition services, forensic experts for companies like Capitol Digital Document Solutions [10] can provide management with a potent arsenal of digital tools at its disposal. They have the necessary software and hardware to travel to designated sites throughout the world to acquire an exact image of hard drives, tapes, etc. This image is an exact duplication of the source media and allows evaluation within their laboratories with minimal disruption to others. Services include but are not limited to

- Lost password and file recovery
- Location and retrieval of deleted and hidden files
- File and email decryption
- Email supervision and authentication
- Threatening email traced to source
- Identification of Internet activity
- Computer usage policy and supervision

- Remote PC and network monitoring
- Tracking and location of stolen electronic files
- Honeypot sting operations
- Location and identity of unauthorized software users
- Theft recovery software for laptops and PCs
- Investigative and security software creation
- Protection from hackers and viruses (see sidebar, "Virus/Trojan/Worm Protection")

Specialized forensic techniques:

Threats to the strategic value of your business almost always involve a computer or network because that is where your company's proprietary information and business processes are located. A simple and virtually undetectable fraud that posts a few cents to a phony account can reap a perpetrator thousands of dollars flowing through accounts payable. A malicious change to an individual's personnel records could cost the person a job and a career. Divulging a company's financial records could damage it on Wall Street, in the marketplace, and before shareholders. Corporate espionage can steal trade secrets. Posting libelous information on the Internet about a company or individual can damage a reputation beyond recovery. Employees of a company might be stealing from it or using company resources to work for themselves, or they can be using excessive work time to surf pornographic sites and play. Computer forensics investigators examine computer hardware and software using legal procedures to obtain evidence that proves or disproves allegations. Gathering legal evidence is difficult and requires trained specialists who know computers, the rules of evidence gathering, and how to work with law enforcement authorities. Computer forensics examiners should be called in when a threat to a company's business and reputation is serious. Any organization that does not have a way to detect and stop malicious behavior can be victimized with no legal recourse. Preserving evidence according to Federal Rules of Evidence gives choices that otherwise would not exist. When an intruder attacks or steals from an organization, the ability or threat to get law enforcement involved may be the only way to reduce the damage or prevent future occurrences. Gathering computer evidence is also useful for confirming or dispelling concerns about whether an illegal incident has occurred and for documenting computer and network vulnerabilities after an incident. Companies employ computer forensics when there is serious risk of information being compromised, a potential loss of competitive capability, a threat of lawsuits, or potential damage

to reputation and brand. Some companies regularly use forensic investigations to check employee computers. In theory, employees are less tempted to stray when they know they are being watched. On the other hand, when the cost of a forensic investigation exceeds potential gain, there is little reason to use it. Companies have used legal evidence gathering to drive home points with employees and external intruders even though the cost of investigations exceeded recovery. Usually, however, a full-scale investigation is not needed to stop an inappropriate action such as surfing that wastes time. Computer forensics also may not be needed when computers and networks play a minor role in an incident or threat, but this may not always be clear. The relationship between the computer and an event under inquiry is critical, and sometimes until a forensics examination has been done, one cannot know whether a computer was a significant part of an event or not.

Legal Evidence

A computer forensics examiner always should gather and preserve evidence according to Federal Rules of Evidence. The examiner has three basic tasks: finding, preserving, and preparing evidence. Finding and isolating evidence to prove or disprove allegations is as difficult as preserving it. Investigators can plow through thousands of active files and fragments of deleted files to find just one that makes a case. Computer forensics has been described as looking for one needle in a mountain of needles. Preserving computer evidence is important because data can be destroyed easily. The 1s and 0s that make up data can be hidden and vanish instantly with a push of a button. As a result, forensics examiners assume every computer has been rigged to destroy evidence, and they proceed with care in handling computers and storage media. Preparing evidence requires patience and thorough documentation so it can withstand judicial scrutiny. For example, a hacking incident at a Web music store was thrown out of court because examiners who prepared the case failed to follow rules of evidence that documented where evidence had come from and that it had not been changed. Preserving computer evidence requires pre-incident planning and training of employees in incident discovery procedures. System administrators sometimes think they are helping a forensics examiner when they are actually destroying evidence. Managers should make sure that there's minimal disturbance of the computer, peripherals, and area surrounding the machine. If a computer is turned on, leave it on; if turned off, leave it off. Moreover, never run programs on a computer in question. For example, running Windows to examine files destroys evidence in the swap file. Finally, never let a suspect help open or turn on a machine. Gathering computer evidence goes beyond normal data recovery. Unfortunately, there are no certified procedures for safe evidence

gathering, nor is there a single approach for every type of case. Examiners work in secure laboratories where they check for viruses in suspect machines and isolate data to avoid contamination. Examiners will, for example, photograph equipment in place before removing it and label wires and sockets so computers and peripherals can be reassembled exactly in a laboratory. They transport computers, peripherals, and media carefully to avoid heat damage or jostling. They never touch original computer hard disks and floppies. They make exact bit-by-bit copies and they store the copies on a medium that cannot be altered, such as a CD-ROM. When suspects attempt to destroy media, such as cutting up a floppy disk, investigators reassemble the pieces to read the data from it. Nor do examiners trust a computer's internal clock or activity logs. The internal clock might be wrong, a suspect might have tampered with logs, or the mere act of turning on the computer might change a log irrevocably. Before logs disappear, investigators are trained to capture the time a document was created, the last time it was opened, and the last time it was changed. They then calibrate or recalibrate evidence based on a time standard or work around log tampering, if possible. Investigators always assume the worst. It is a rule in computer forensics that only the physical level of magnetic materials where the 1s and 0s of data are recorded is real, and everything else is untrustworthy. A suspect might have corrupted all of the software operating systems, applications, and communications in a computer, or the software itself might erase evidence while operating, so forensic examiners avoid it. Examiners search at the bit level of 1s and 0s across a wide range of areas inside a computer, including email, temporary files in the Windows operating system and in databases, swap fields that hold data temporarily, logical file structures, slack and free space on the hard drive, software settings, script files that perform preset activities, Web browser data caches, bookmarks, and history and session logs that record patterns of usage. They then correlate evidence to activities and sources. Investigators have many tricks that help them get around the clever suspect. For example, they often do not attempt to decode encrypted files. Rather, they look for evidence in a computer that tells them what is in the encrypted file. Frequently, this evidence has been erased, but unencrypted traces remain to make a case. For data concealed within other files or buried inside the 1s and 0s of a picture, an investigator can tell the data is there even though it is inaccessible. Nearly identical files can be compared to see their minute differences. When forensic examiners find computer evidence, they must present it in a logical, compelling, and persuasive manner that a jury will understand and a defense counsel cannot rebut. This requires step-by-step reconstructions of actions with documented dates and times, charts and graphs that explain what was done and how, testimony that explains simply and clearly what a suspect did or did not do, and exhibits that can withstand scrutiny. Case presentation requires experience, which only can be gained through courtroom

appearances. This is why lawyers and managers should retain computer forensics examiners who have a record of successful expert testimony on computer evidence. An experienced examiner knows the questions that opposing attorneys will ask and the ways to provide answers that withstand challenge. A skilled litigator can defeat an inexperienced examiner for failing to collect evidence in a proper manner and failing to show that evidence supports allegations. Not long ago, attorneys knew little about computers and how they operated, but today they do and they are increasingly skilled at challenging examiners' methods .

A Growing Service

With the growth of computers and networks comes the growth of crime committed through or with computers and networks. It is a fast-growing field because computers and networks have moved to the heart of business and societal operations. However, it is not a service that most corporations will or should establish internally. Because investigations are so specialized, few organizations have the human or technical resources to gather and compile evidence that withstands court challenges. Large multinational corporations have or may develop the capability, but most organizations will purchase computer forensics as needed or keep a computer forensics firm on retainer. It's important that managers and lawyers remember that computer evidence is fragile and that the best way to handle an incident is to isolate it until examiners take over .

HIDDEN DATA AND HOW TO FIND IT

As if you didn't have enough to worry about, today's technology presents your business with as many problems as it does solutions. Computers that work miracles in your day-to-day operations often malfunction—and you lose valuable data. The email that makes communicating so simple, carries deadly viruses that infect your machines and spread, causing massive data losses throughout your network. Hackers, both inside and outside your company, can access your information, manipulate it, hide it, steal it, and cause huge losses of data [14]. In many cases, documents and files deleted from a computer can be found and recovered using the methods of computer forensics. When files or documents are deleted from a computer, the majority of the actual information is typically left behind. Although the user may think the deleted document has been eradicated, this is usually not the case [14]. Documents and files deleted or hidden even years ago may be recovered through a computer investigation. Deleted or hidden files are one of the prime targets of the computer forensic technician searching for evidence

[14]. What can you do about it right away? You should turn to computer forensic technicians or specialists (like Kessler International) for hard drive data recovery and other data recovery services [14]. These technicians specialize in professional data recovery and will restore your data quickly—right when you need it. These teams of data recovery experts know how to retrieve your lost data from damaged and corrupt storage media including hard drives, back-up systems, temporary storage units, and more. They can also restore individual corrupt files back to their original condition [14].

Spyware and Adware

Spyware is Internet jargon for advertising supported software (adware). It is a way for shareware authors to make money from a product, other than by selling it to the users. There are several large media companies that approach shareware authors to place banner ads in their products in exchange for a portion of the revenue from banner sales. This way, you don't have to pay for the software, and the developers are still getting paid. If you find the banners annoying, there is usually an option to remove them by paying the regular licensing fee.

Why Is It Called Spyware?

While this may be a great concept, the downside is that the advertising companies also install additional tracking software on your system, which is continuously calling home, using your Internet connection to report statistical data to the “mothership.” While according to the privacy policies of the companies, there will be no sensitive or identifying data collected from your system and you shall remain anonymous, the fact still remains that you have a liveserver sitting on your PC that is sending information about you and your surfing habits to a remote location.

Are All Adware Products Spyware?

No, but the majority are. There are also products that display advertising but do not install any tracking mechanism on your system.

Is Spyware Illegal?

Even though the name may indicate so, spyware is not an illegal type of software in any way. However, there are certain issues that a privacy-oriented user may object to and therefore prefer not to use the product. This usually involves the tracking and sending of data and statistics via a server installed on the user's PC and the use of your Internet connection in the background.

What's the Hype About?

While legitimate adware companies will disclose the nature of data that is collected and transmitted in their privacy statement (linked from their database, there is almost no way for the user to actually control what data is being sent). The technology is in theory capable of sending much more than just banner statistics and this is why many people feel uncomfortable with the idea.

On the Other Hand

Millions of people use advertising supported spyware products and could not care less about the privacy hype. In fact, some spyware programs are among the most popular downloads on the Internet.

Real Spyware

There are also many PC surveillance tools that allow a user to monitor all kinds of activity on a computer, ranging from keystroke capture, snapshots, email logging, chat logging, and just about everything else. These tools are often designed for parents, businesses, and similar environments but can be easily abused if they are installed on your computer without your knowledge. Furthermore, these tools are perfectly legal in most places, but, just like an ordinary tape recorder, if they are abused, they can seriously violate your privacy.

ENCRYPTION METHODS AND VULNERABILITIES

The use of encryption provides a different kind of challenge for the forensic investigator. Here, data recovery is only half the story, with the task of decryption providing a potentially greater obstacle to be overcome. Encryption, whether built into an application or provided by a separate software package, comes in different types and strengths. Some of the most commonly used applications provide encryption protected by passwords that can be readily defeated by investigators with the right tools and the time to use them. Other types of encryption, readily available to the general public, can be configured and used to create encrypted data that goes beyond the ability of the professional investigator to decrypt it using software. Nevertheless, in these cases it may still be possible to decrypt data by widening the scope of the investigation to include intelligence sources beyond the computer under investigation. For example, public key encryption can be used to create highly secure, encrypted data. To decrypt data encrypted in this fashion, a private key and passphrase is needed. The private key may be found on the suspect's machine or backed up to removable media. Similarly, the passphrase may be recorded

somewhere on the computer in case it is forgotten or may be written down somewhere and kept in a nearby location. For example, three recent developments have added to the remarkable insecurity of the Net. One affects email. The other two affect network systems. First, a weakness has been discovered in the world's most popular encryption program that, in some circumstances, allows the encryption program to be completely bypassed. People using this program to encrypt email to protect its privacy and confidentiality may be thwarted despite their efforts. Second, hackers have recently discovered a cloaking program that allows them to blow past firewalls on servers and networks without being detected. Third, a flaw has been announced that affects networks around the globe regarding the file transfer protocol (FTP) used on the Internet. These three revelations taken together are seriously bad news for Internet privacy, confidentiality, and security.

Security and wireless technologies

There are two types of RIM devices within each model class. The Exchange Edition is meant for use in a corporate environment, while the Internet Edition works with standard POP email accounts. The Exchange Edition employs Triple-DES encryption to send and receive, but the Internet Edition communicates in clear text. Neither employs an encrypted files system.

Relevance of RIM Computer Forensics A RIM device shares the same evidentiary value as any other personal digital assistant (PDA). As the computer forensics investigator may suspect of most file systems, a delete is by no means a total removal of data on the device. However, the RIM's always-on, wireless push technology adds a unique dimension to forensic examination. Changing and updating data no longer requires a desktop synchronization. In fact, a RIM device does not need a cradle or desktop connection to be useful. The more time a PDA spends with its owner, the greater the chance is that it will more accurately reflect and tell a story about that person. Thus, the RIM's currently unsurpassed portability is the examiner's greatest ally.

Evidence Collection: Gathering Logs

The first procedure of evidence collection violates the computer forensic method by requiring the investigator to record logs kept on the unit that will be wiped after an image is taken. The logs must be accessed on the original unit before the programmer software development kit (SDK) tool is applied. The logs are not accessed via the standard user interface. Rather, they are reviewed using the following hidden control functions.

Imaging and Profiling

An image should be taken of the file system as the very first step as long as the logs are not required or a method of extracting the logs from the image is developed. An image or bit-by-bit backup is acquired using an SDK utility that dumps the contents of the Flash RAM into a file easily examined with a hex editor. The Program Loader, which is used to perform most of the inspection in addition to taking the image, will cause a reset each time it is run. A reset can mean a file system cleanup. This means that to get a partition table, you risk changing the file system and spoiling the data.

Evidence Review

Two options are available for information review using the hex dump: manual review of the Hex file using a hex editor and loading of the hex file into the BlackBerry SDK simulator for review. The hex editor will provide access to the entire file system including deleted or dirty records indicated by byte 3 of the file header. Using the SDK will assist in decoding dates on extant records.

The File System

The RIM file system is abstracted to appear as a database to most available Application Programming Interfaces (APIs) in order to simplify programming. This abstraction qualifies as a file translation layer (FTL), hiding what is really a quite complicated system of file management. Under the hood of your RIM device is a standard Flash RAM used to store all nonvolatile data.

Flash is organized similarly to dynamic random access memory (DRAM) and has the 65 ns read performance to match. However, write performance is slower by a factor of nearly 1000 times. Writing flash is a binary AND or NAND, meaning each 1 in memory can be toggled to 0 but not back again without an erasure. However, an erasure can only occur in blocks of 64 KB, while writing can occur in any interval. An erasure costs more in terms of time than a write because of conditioning. Conditioning means forcing every 1 to a 0 before resetting (erasing) all bits to 1 again in order to prolong the life of the Flash RAM

Data Hiding Data hiding is accomplished in several ways on a RIM device. Hidden databases, partition gaps, and obfuscated data are but a few. Presented here are a few ideas to get the computer forensics investigator thinking in the right direction. Custom databases with no icon in the Ribbon graphical user interface (GUI) are capable of providing hidden data transport. A hacker may write a program that utilizes a database accessible only through device

synchronization. The average user or uninformed investigator will never have knowledge of the hidden database. For example, a database reader can thwart such an attempt, as it will provide access to all databases on a unit. Unfortunately, it will need to be installed on the unit investigated for it to function. Now let's look at what the computer forensics specialist sees in firewall logs, especially what port numbers mean. The computer forensics specialist can use this information to help figure out what hackers or worms are up to.

SUMMARY

- National Law Enforcement and Corrections Technology Center (NLECTC).
- Forensic software tools and methods can be used to identify passwords, logons, and their information that is automatically dumped from the computer memory.
- SafeBack is used to create mirror-image (bit-stream) backup files of hard disks or to make a mirror-image copy of an entire hard disk drive or partition. The process is analogous to photography and the creation of a photo negative.
- Trade secret information and other sensitive data can easily be secreted using any number of techniques. It is possible to hide diskettes within diskettes and to hide entire computer hard disk drive partitions.
- CopyQM Plus essentially turns a personal computer into a diskette duplicator. In a single pass, diskettes are formatted, copied, and verified.
- **Business computer forensics technology:**
 1. Remote monitoring of target computers
 2. Creating trackable electronic documents
 3. Theft recovery software for laptops and PCs
 4. Basic forensic tools and techniques
 5. Forensic services available
- A computer forensics examiner always should gather and preserve evidence according to Federal Rules of Evidence. The examiner has three basic tasks: finding, preserving, and preparing evidence

Check your Answer

Define Safeback and What are their benefits.

Different types of data hiding techniques?

Define file Slack?

What is data encryption and Data compression?

What is Business computer forensics technology and its type?

Define spyware and adware?

Model Question Paper
M.Sc., Cyber Forensics and Information Security
First Year - Second Semester
Core Paper - VI
Basics of Cyber Forensics

Time : 3 Hours

Max Marks: 80

Section - A

Answer the following in 50 words in each

1. Define Forensic Dictum.
2. What is data carving ?
3. Define Malware Forensics?
4. What is digital Signature ?
5. Describe the Locard's Exchange principle ?
6. Explain different types of reconstruction.
7. What is primary and secondary storage?
8. Explain Ordered Vs. Unordered record structures.
9. What is Magic Number?
10. What is Phishing ?

Section - B

Answer any Five of the following in 250 Words in Each

(5 x 6 = 30)

1. What digital forensics spectrum ?
2. What are the various perspectives that needs to be considered while setting up a cyber forensics laboratory ?
3. What are the challenging aspects of digital evidence ?

4. What is a cyber trail ? Explain with a suitable example.
5. What are the steps involved in presenting the evidence in a court room ?
6. What is cyber forensics ? What are the goals of cyber forensics ? Explain its stages with example.

Section - C

Answer the following in 500 words

(3 x 10 = 30)

1. What is event reconstruction ? What are the different types of reconstruction ? Explain with suitable examples.
2. What are standards, guidelines and best practices? Name any three organizations which provides best practices.
3. What is the forensics significance of file system ? What happens when a suspect deletes a file in windows operating system ?